WORKSHEET 1.1: BASICS OF RINGS

A *ring* $(R,+,\cdot)$ is an abelian group $(R,+)$ together with a multiplication $(\cdot)$ and a multiplicative identity element $1_R \in R$ satisfying the following axioms:

(i) (Multiplicative Identity): $1_R \cdot a = a \cdot 1_R = a$ for all $a \in R$,

(ii) (Associativity): $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c \in R$, and

(iii) (Distributivity): $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$ and $(b + c) \cdot a = (b \cdot a) + (c \cdot a)$ for all $a, b, c \in R$.

As a warning some people will not include condition (i) in the definition of a ring, but this is rare to unheard of in modern commutative algebra. (See the historical remark at the end of this worksheet.) As is often the case once we are familiar with these operations we will normally just drop some of the notational baggage and just write ab for $a \cdot b$ and 1 for 1_R.

A ring R is *commutative* if and only if $ab = ba$ for all $a, b \in R$. As the name of the course suggests we will primarily be interested in commutative rings in this course. To the point we shall adopt the following convention: Throughout this course, "ring" means *commutative* ring with unity. Commutative algebra broadly can be thought of as the study of rings and their associated objects. At a high level we will study rings from four perspectives: 1) element-wise, 2) homomorphisms, 3) ideals, and 4) modules. You hopefully have some familiarity with 1, 2, and 3, but today we will review and remember what it is like working with these objects.

An element $u \in R$ is a *unit* if there exists $v \in R$ such that $uv = vu = 1$, in which case we will often denote v by u^{-1}. A nonzero element $a \in R$ is a *zero divisor* if there exists a nonzero element $b \in R$ such that $ab = 0$. A non-zero element, which is not a zero divisor is a *non-zero divisor*.

(1) Let R be a ring and consider $R[[t]]$ the ring of formal power series with coefficients in R. The goal of this exercise is to prove that:

$$(R[[t]])^\times = \left\{ u + tf(t) \;\middle|\; u \in R^\times,\ f(t) \in R[[t]] \right\}.$$

(a) Elements of $R[[t]]$ can be written as $f = \sum_{k=0}^{\infty} f_k t^k$ for $f_k \in R$. We call f_0 the constant term of f. If $f = \sum_{k=0}^{\infty} f_k t^k$ and $g = \sum_{k=0}^{\infty} g_k t^k$ then find a formula for the constant term of fg.

(b) Using the formula from 1a prove that if f is a unit then $f_0 \in R^\times$.

(c) Find a formula for the coefficient of t^k appearing in fg.

(d) Assume that $f_0 \in R^\times$. Prove that f is a unit by explicitly constructing an element $g \in R[[t]]$ such that $fg = 1$. (Hint: The condition that $fg = 1$ places constraints on each coefficient of fg. Use the constraints together with part 1c to recursively solve for the coefficients of g.)

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A non-empty subset $I \subset R$ is an *ideal* if it is closed by outside multiplication by elements of R and is closed under addition. Precisely, $I \subset R$ is a non-empty subset satisfying:

(i) if $a \in I$ then $ra \in I$ for all $r \in R$ and

(ii) if $a, b \in I$ then $a + b \in I$.

Note this immediately implies that an ideal always contains 0 and is an abelian group with respect to addition. Given elements $a_1, \ldots, a_t \in R$ we write $\langle a_1, \ldots, a_t \rangle \subset R$ for the the smallest ideal containing $a_1, \ldots, a_t$. Concretely, we may describe this as the set of all R-linear combinations of $a_1, \ldots, a_t$:

$$\langle a_1, \ldots, a_t \rangle = \{ r_1 a_1 + \cdots + r_t a_t \mid r_1, \ldots, r_t \in R \}.$$

An ideal $I \subset R$ is *principal* if there exists an element $a \in R$ such that $I = \langle a \rangle$.

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(1) (a) Prove that an element $u \in R$ is a unit if and only if the principal ideal $\langle u \rangle = R$.

(b) Prove that if ab and b are units, then a is a unit.

(c) Prove that if u is a unit and $a \in R$, then a and ua generate the same principal ideal.

(d) In the ring $\mathbb{Z}/\langle 12 \rangle$, determine which elements are units.

Let R and S be rings. A ring homomorphism from R to S is a function $\phi : R \to S$ such that:

(i) (Multiplicative): $\phi(ab) = \phi(a)\phi(b)$ for all $a, b \in R$,

(ii) (Additive): $\phi(a + b) = \phi(a) + \phi(b)$ for all $a, b \in R$, and

(iii) (Preserves Unit): $\phi(1_R) = 1_S$.

Notice that condition (ii) implies that $\phi(0_R) = 0_S$, so ϕ is a homomorphism of the underlying abelian groups. It will turn out that condition (iii) is very useful as the following exercise demonstrates. There are two important sets associated to any ring homomorphism $\phi : R \to S$ the kernel and image of ϕ:

$$\ker(\phi) := \{ a \in R \mid \phi(a) = 0 \} \subset R \qquad \text{and} \qquad \operatorname{img}(\phi) := \{ \phi(a) \in S \mid a \in R \} \subset S.$$

More succinctly, $\ker(\phi) = \phi^{-1}(\{0\})$ and $\operatorname{img}(\phi) = \phi(R)$. One checks that $\ker(\phi)$ is an ideal of R, and that $\operatorname{img}(\phi)$ is a subring of S; however, it need not be an ideal of S.

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(1) Given two rings R and S we let $\mathrm{Hom}_{\mathbf{comRing}}(R, S)$ be the set of ring homomorphisms from R to S. When there is little room for confusion we will often drop the subscript **comRing** and simply write $\mathrm{Hom}(R, S)$.

(a) Prove or disprove: If R and S are rings, then $\mathrm{Hom}(R,S)$ is a ring under point-wise multiplication and point-wise addition.

(b) Give an explicit description of the hom-sets below:

(i) $\mathrm{Hom}(\mathbb{Z},\mathbb{Z})$

(ii) $\mathrm{Hom}(\mathbb{Z}/\langle 2\rangle,\mathbb{Z})$

(iii) $\mathrm{Hom}(\mathbb{Z},\mathbb{Z}/\langle 2\rangle)$

(iv) $\mathrm{Hom}(\mathbb{Z},\mathbb{C}[x])$

(c) Based on part 1b state and prove a characterization of $\mathrm{Hom}(\mathbb{Z},R)$ for any ring R.

(2) Let $\phi : R \to S$ be a ring homomorphism.

(a) Prove that if $J \subset S$ is an ideal then $\phi^{-1}(J) \subset R$ is an ideal.

(b) Assuming ϕ is injective, give a nice description of $\phi^{-1}(J)$ for an ideal $J \subset S$.

(c) Find a counterexample showing that if $I \subset R$ is an ideal, then $\phi(I)$ need not be an ideal.

(d) Prove that if ϕ is surjective then $\phi(I)$ is an ideal of S.

(e) Let $I \subset R$ be an ideal. Prove that if $\ker(\phi) \subset I$, then $\phi^{-1}(\phi(I)) = I$.

(f) Is the assumption that $\ker(\phi) \subset I$ needed in part 2e?

As a small historical remark, despite it being common practice to include the existence of a multiplicative identity in the definition of a ring, in her original definition of a ring Emmy Noether did not do this. In E. Noether's 1921, article *Idealtheorie in Ringbereichen* – which can reasonably be viewed as the birth of commutative algebra – she says, "*Aus diesen Eigenschaften folgt die Existenz der Null; ein Ring braucht aber keine Einheit zu besitzen...*"; roughly translating to, "From these properties the existence of the zero element follows; however a ring is not required to possess a unit".

§ 1.

Ringbereich, Ideal, Endlichkeitsbedingung.

1. Der zugrunde gelegte Bereich Σ sei ein (kommutativer) *Ring* in abstrakter Definition[7]; d. h. Σ bestehe aus einem System von Elementen $a, b, c, \ldots, f, g, h, \ldots$, in dem eine den üblichen Bedingungen genügende Relation als *Gleichheit* definiert ist; und in dem durch zwei Operationen (Verknüpfungsarten), *Addition* und *Multiplikation*, aus je zwei Ringelementen a und b stets eindeutig je ein drittes als Summe $a+b$ und als Produkt $a \cdot b$ gewonnen wird. Der Ring und die sonst ganz willkürlichen Operationen müssen dabei den folgenden Gesetzen genügen:

1. *Dem assoziativen Gesetz der Addition*: $(a+b)+c = a+(b+c)$.
2. *Dem kommutativen Gesetz der Addition*: $a+b = b+a$.
3. *Dem assoziativen Gesetz der Multiplikation*: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
4. *Dem kommutativen Gesetz der Multiplikation*: $a \cdot b = b \cdot a$.
5. *Dem distributiven Gesetz*: $a \cdot (b+c) = a \cdot b + a \cdot c$.
6. *Dem Gesetz der unbeschränkten und eindeutigen Subtraktion.*

Es gibt in Σ ein einziges Element x, das die Gleichung $a + x = b$ befriedigt. (Man bezeichnet $x = b - a$.)

Aus diesen Eigenschaften folgt die Existenz der Null; ein Ring braucht aber keine Einheit zu besitzen; und es kann das Produkt zweier Elemente verschwinden, ohne daß ein Faktor verschwindet. Ringe, für die aus dem Verschwinden eines Produktes stets das Verschwinden eines

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WORKSHEET 1.2: QUOTIENTS AND PRIME AND MAXIMAL IDEALS

Throughout this course, "ring" means *commutative* ring with unity. A general theme throughout commutative algebra is that when given a special type of ideals, you should often ask whether it can be classified by morphisms in some way. In this worksheet we study quotient rings, which gives us a way to classify all ideals in terms of homomorphisms, i.e., ideals are precisely the subsets of rings that arise as kernels. We saw one direction of this claim on a previous exercise, the other is the following very important theorem.

Theorem 1. *Let R be a ring and $I \subset R$ an ideal.*

(1) There exists a ring, which we denote R/I, together with a surjective ring homomorphism $\pi : R \to R/I$ such that $\ker(\pi) = I$. Moreover, R/I and π are unique up to isomorphism.

(2) There is a inclusion-preserving bijection:

$$\{ \textit{ ideals in } R/I \} \xrightarrow{\sim} \left\{ \begin{array}{c} \textit{ideals in } R \\ \textit{containing } I \end{array} \right\}$$

$$J \longmapsto \pi^{-1}(J)$$

You have hopefully seen a proof of this theorem where you give an explicit construction of R/I as a set of equivalence classes determined by I. Later in this worksheet we will see how we can also determine R/I by its universal property.

Two important types of ideals that will appear throughout this course are prime and maximal ideals. A proper ideal $P \subset R$ is *prime* if whenever $ab \in P$ we have $a \in P$ or $b \in P$. A proper ideal $I \subset R$ is *maximal* if it is maximal among proper ideals of R with respect to inclusion. Put differently, a proper ideal $I \subset R$ is maximal if and only if given a proper ideal $J \subset R$ such that $I \subset J$ then $J = I$.

Theorem 2. *Let $I \subset R$ be an ideal. Then I is prime if and only if R/I is an integral domain, and I is maximal if and only if R/I is a field.*

Recall a non-zero ring R is an *integral domain*, often shortened to just *domain*, if R has no zero-divisors. A field is a ring R such that every non-zero element of R is a unit. An important equivalent characterization is that a ring R is a field if and only if the only ideals of R are $\langle 0 \rangle$ and R. If $\mathfrak{m} \subset R$ is a maximal ideal the *residue field* of $\mathfrak{m}$ is the field $R/\mathfrak{m}$.

Note the above theorem says that both prime and maximal ideals can be characterized by morphisms in the sense: i) An ideal $I \subset R$ is prime if and only if there exists a surjective morphism $\phi : R \to S$ where S is an integral domain and $I = \ker(\phi)$; and ii) An ideal $I \subset R$ is maximal if and only if there exists a surjective morphism $\phi : R \to \mathbb{K}$ where $\mathbb{K}$ is a field and $I = \ker(\phi)$. Since every domain can be viewed as a subring of a field, a fact we shall prove later in the course, we can also rephrase (i) and (ii) above as: An ideal $I \subset R$

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is prime if and only if there exists a morphism $\phi : R \to \mathbb{K}$ where $\mathbb{K}$ is a field and $I = \ker(\phi)$. Moreover, I is maximal if and only if one can find ϕ which is surjective.

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(1) **Operations on Ideals.** Throughout this problem let I and J be ideals in a ring R.

(a) Prove or disprove that $I \cap J$ is an ideal.

(b) Prove or disprove that $I \cup J$ is an ideal.

(c) Prove or disprove that the product of ideals, defined below, is an ideal:

$$IJ := \{a_1 b_1 + \cdots + a_t b_t \mid a_i \in I \ \ b_i \in J, \text{ for all } t\}.$$

(d) Prove or disprove that the sum of ideals, defined below, is an ideal:

$$I + J := \{a + b \mid a \in I \ \ b \in J\}.$$

(e) Prove or disprove that the radical of an ideal, defined below, is an ideal:

$$\sqrt{I} := \left\{a \in R \mid a^k \in I \text{ for some } k \in \mathbb{Z}_{\geq 1}\right\}.$$

(2) Let $\mathbb{K}$ be a field. The rings $\mathbb{Z}$, $\mathbb{K}[x]$, and $\mathbb{K}[[t]]$ are all principal ideal domains (PIDs) meaning domains where every ideal is generated by a single element. If $I = \langle a \rangle$ and $J = \langle b \rangle$ give, with proof, a careful concrete description of generators for IJ, $I \cap J$, $I + J$, and $\sqrt{I}$ in each of these rings.

Throughout this course we will see numerous constructions and operations (e.g., quotients, tensor products, localizations, completions, etc.) each of these will have an explicit construction where we describe the objects precisely as sets. However, there is an alternative perspective, which is just as useful, describing these objects via their *universal properties*, which roughly asks what makes each object special amongst all other such objects. The standard framework for a universal property is, "Object X is unique up to isomorphism among all objects having property P."

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(3) **Universal Property of Quotients.** Let R be a ring and $I \subseteq R$ an ideal. Let $\pi : R \to R/I$ be the quotient map. Suppose $\varphi : R \to S$ is a ring homomorphism such that $I \subseteq \ker(\varphi)$. Define a function

$$\widetilde{\varphi} : R/I \to S \qquad \text{by} \qquad \widetilde{\varphi}(r + I) = \varphi(r).$$

(a) Prove that $\widetilde{\varphi}$ is well-defined ring homomorphism.

(b) Prove that $\widetilde{\varphi} \circ \pi = \varphi$.

(c) Prove that $\widetilde{\varphi}$ is the unique ring homomorphism $R/I \to S$ with the property that

$$\widetilde{\varphi} \circ \pi = \varphi.$$

(d) Explain in words what this says about the quotient ring R/I.

(4) In this problem, we wish to consider the evaluation map:

$$\mathrm{ev}_i : \mathbb{Z}[x] \to \mathbb{C}, \qquad f(x) \mapsto f(i)$$

(a) Prove that ev_i is a ring homomorphism.

(b) Show that $\langle x^2+1 \rangle \subset \ker(\mathrm{ev}_i)$.

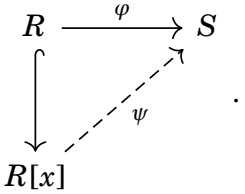
(c) Use Question 3 to produce a ring homomorphism

$$\widetilde{\mathrm{ev}_i} : \mathbb{Z}[x]/\langle x^2+1 \rangle \to \mathbb{C}.$$

(d) What does $\widetilde{\mathrm{ev}_i}$ do to the class of x in $\mathbb{Z}[x]/\langle x^2+1 \rangle$?

(e) Prove that $\mathbb{Z}[x]/\langle x^2+1 \rangle$ is isomorphic to $\mathbb{Z}[i]$. Is $\langle x^2+1 \rangle$ prime and/or maximal?

(5) **Universal Property of Polynomial Rings.** The goal of this exercise is to prove the following characterization of polynomial rings. Let $\varphi : R \to S$ be a ring homomorphism. For any element $s \in S$ there is a unique ring homomorphism $\psi : R[x] \to S$ such that $\psi(x) = s$ and the following diagram commutes:



(a) Prove there is at most one ring homomorphism $\psi : R[x] \to S$ with the above properties.

(b) Construct such a homomorphism explicitly by setting

$$\psi\left(\sum_{i=0}^{n} a_i x^i\right) = \sum_{i=0}^{n} \varphi(a_i) s^i.$$

(c) Conclude that to give a ring homomorphism $R[x] \to S$ is equivalent to giving:

- a ring homomorphism $R \to S$, and
- an element of S.

(6) **Evaluation maps.** Let R be a ring and let $a \in R$. Define the evaluation map

$$\mathrm{ev}_a : R[x] \to R, \qquad f(x) \mapsto f(a).$$

(a) Prove that ev_a is a surjective ring homomorphism and $\ker(\mathrm{ev}_a) = \langle x - a \rangle$.

(b) Use Question 3 to show that ev_a factors through a ring homomorphism $R[x]/(x-a) \to R$.

(c) Explain what this means if R is a field or R is an integral domain.

(d) Give an example of a ring R and prime/maximal ideal $I \subset R[x]$ such that $I \neq \ker(\mathrm{ev}_a)$ for all $a \in R$.

WORKSHEET 1.3: GRÖBNER BASES PT. I

Polynomial Division. Recall the following classical theorem:

Theorem 1. *Let $g \in \mathbb{K}[x]$ be a non-zero polynomial. Every polynomial $f \in \mathbb{K}[x]$ can be written uniquely as $f = qg + r$ where $q, r \in \mathbb{K}[x]$, and either $r = 0$ or $\deg(r) < \deg(g)$.*

While this theorem might seem innocuous when first learned as a teenager, it turns out to be very powerful when studying ideals and polynomials in $\mathbb{K}[x]$. The proof of this theorem also provides an algorithm for computing q and r, which when presented in pseudocode looks something like

Algorithm 1 Division Algorithm

Require: g, f
Ensure: q, r
  $q \leftarrow 0$; $r \leftarrow f$
  while $r \neq 0$ **and** $\mathrm{LT}(g)$ divides $\mathrm{LT}(r)$ **do**
    $q \leftarrow q + \mathrm{LT}(r)/\mathrm{LT}(g)$
    $r \leftarrow r - (\mathrm{LT}(r)/\mathrm{LT}(g))g$
  end while

The goal of this worksheet is to begin exploring how we might extend this theorem to polynomial rings in more variables. Along the way we will see how these explorations will lead to both computational tools for studying the commutative algebra of polynomial rings over fields, but also gives us useful tools for proving theorems in this setting as well.

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(1) Use Theorem 1 to prove the following:

  (a) If $f \in \mathbb{K}[x]$ is a non-zero polynomial then f has at most $\deg(f)$ roots in $\mathbb{K}$.

  (b) If $g \in \mathbb{K}[x]$ is a non-zero polynomial then $f \in \langle g \rangle$ if and only if $r = 0$.

(2) Implement the algorithm above in Macaulay2, and use your code to compute q and r for the following polynomials. (If you want to test your code versus the built-in functions use `time f//g` and `time f%g` to compute q and r respectively and time the computation.)

  (a) $f = x^3 + x + 1, \quad g = x + 1.$

  (b) $f = x^5 - 1, \quad g = x - 1.$

  (c) $f = x^2 + 1, \quad g = x^3 + x.$

  (d) $f = x^{100} - 1, \quad g = x^2 - 1.$

  (e) $f = x^{1000} - 1, \quad g = x - 1.$

  (f) $f = \sum_{i=0}^{500} x^i, \quad g = x^2 + x + 1.$

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The computations in Question (2) are deliberately familiar: in one variable division works because there is a canonical notion of "largest term," namely highest degree. In several variables there is no such natural choice. To imitate the division algorithm we must first choose an ordering on monomials, and the next questions explain exactly what properties that ordering must satisfy.

Definition 2. A monomial ordering on $\mathbb{K}[x_1,\dots,x_n]$ is a relation $<$ on $\mathbb{Z}^n_{\geq 0}$ satisfying the following:

(i) $<$ is a total ordering on $\mathbb{Z}^n_{\geq 0}$,

(ii) if $\alpha < \beta$ and $\gamma \in \mathbb{Z}^n_{\geq 0}$ then $\alpha+\gamma < \beta+\gamma$, and

(iii) $<$ is a well-ordering on $\mathbb{Z}^n_{\geq 0}$.

Recall that a *total ordering* means that of any two elements $\alpha,\beta \in \mathbb{Z}^n_{\geq 0}$ exactly one of the following statements is true: $\alpha<\beta$, $\alpha=\beta$, or $\beta<\alpha$. That $<$ is a well-ordering on $\mathbb{Z}^n_{\geq 0}$ means that every non-empty subset of $\mathbb{Z}^n_{\geq 0}$ has a smallest element with respect to $<$.

Once a monomial ordering has been fixed, every non-zero polynomial acquires a distinguished piece of data: its leading monomial, leading coefficient, and leading term. These are the ingredients that drive the multivariable division process just as degree and leading coefficient drive ordinary polynomial division.

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(3) Prove that a monomial ordering $<$ on $\mathbb{Z}^n_{\geq 0}$ is equivalent to a relation $<$ on the set of monomials in $\mathbb{K}[x_1,\dots,x_n]$ satisfying analogous conditions to (i), (ii), and (iii) above. Explain why this justifies the moniker "monomial ordering".

(4) Prove or disprove that the following are monomial orderings:

(a) Lexicographic (Lex) Order: Given $\alpha=(\alpha_1,\dots,\alpha_n)$ and $\beta=(\beta_1,\dots,\beta_n)$ in $\mathbb{Z}^n_{\geq 0}$ we say $\beta <_{\text{lex}} \alpha$ if and only if the left-most non-zero entry of $\alpha-\beta$ is positive.

(b) Graded Lex Order: With α and β as in (a) we say that $\beta <_{\text{grlex}} \alpha$ if and only if

$$|\beta| := \sum_{i=1}^n \beta_i < |\alpha| = \sum_{i=1}^n \alpha_i \qquad \text{or} \qquad |\beta| = |\alpha| \text{ and } \beta <_{\text{lex}} \alpha.$$

(c) Reverse Lex (RevLex) Order: With α and β as in (a) we say that $\beta <_{\text{revlex}} \alpha$ if and only if the right-most non-zero entry of $\alpha-\beta$ is negative.

(d) Graded RevLex Order: With α and β as in (a) we say that $\beta <_{\text{grevlex}} \alpha$ if and only if

$$|\beta| := \sum_{i=1}^n \beta_i < |\alpha| = \sum_{i=1}^n \alpha_i \qquad \text{or} \qquad |\beta| = |\alpha| \text{ and } \beta <_{\text{revlex}} \alpha.$$

(e) Weight Order: Fix a vector $w = (w_1, \ldots, w_n) \in \mathbb{Z}^n$. With α and β as in (a) say $\beta <_w \alpha$ if and only if

$$\beta \cdot w = \sum_{i=1}^{n} \beta_i w_i < \alpha \cdot w = \sum_{i=1}^{n} \alpha_i w_i$$

(5) **Leading Terms.** Fix a monomial ordering $<$ on $\mathbb{Z}^n_{\geq 0}$, and use the corresponding ordering on monomials in $\mathbb{K}[x_1, \ldots, x_n]$. Let $f = \sum_\alpha c_\alpha x^\alpha \in \mathbb{K}[x_1, \ldots, x_n]$ be a non-zero polynomial. Define the *leading monomial* $\mathrm{LM}(f)$, *leading coefficient* $\mathrm{LC}(f)$, and *leading term* $\mathrm{LT}(f)$ with respect to the fixed monomial ordering.

(a) Compute $\mathrm{LM}(f)$, $\mathrm{LC}(f)$, and $\mathrm{LT}(f)$ for

$$f = x^3 + 3x^2y + 5xy^3 - 7z^2 \in \mathbb{K}[x, y, z]$$

with respect to Lex, Graded Lex, and Graded RevLex orderings (assume $x > y > z$).

(b) Prove that if $f, g \in \mathbb{K}[x_1, \ldots, x_n]$ are non-zero, then

$$\mathrm{LM}(fg) = \mathrm{LM}(f)\,\mathrm{LM}(g) \qquad \text{and} \qquad \mathrm{LT}(fg) = \mathrm{LT}(f)\,\mathrm{LT}(g).$$

(c) Explain which property of a monomial ordering is used in part (b).

(6) Let $G = (g_1, \ldots, g_s)$ be an ordered list of non-zero polynomials in $\mathbb{K}[x_1, \ldots, x_n]$.

(a) Write pseudocode for a division algorithm which takes as input f and G and returns polynomials $q_1, \ldots, q_s, r$ such that $f = q_1 g_1 + \cdots + q_s g_s + r$, where no monomial appearing in r is divisible by any of the monomials $\mathrm{LM}(g_1), \ldots, \mathrm{LM}(g_s)$.

(b) In the algorithm above, what should happen if the leading monomial of the current polynomial is not divisible by any $\mathrm{LM}(g_i)$?

(c) Prove that your algorithm terminates. What plays the role of "degree" in the proof of termination?

(7) Work in $\mathbb{K}[x, y]$ with Lex order and $x > y$. Let $= xy^2 - x$, $g_1 = xy - 1$, and $g_2 = y^2 - 1$.

(a) Divide f by the ordered list (g_1, g_2).

(b) Divide f by the ordered list (g_2, g_1).

(c) Show directly that $f \in \langle g_1, g_2 \rangle$.

(d) What do your computations show about division in several variables? In particular, is the remainder uniquely determined by the ideal $\langle g_1, g_2 \rangle$?

Question 7 shows the central obstacle in several variables: ordinary division is no longer canonical, so ideal membership cannot be read off from a unique remainder. In several variables, a zero remainder of f divided by $g_1, \ldots, g_s$ implies that $f \in \langle g_1, \ldots, g_s \rangle$, but a non-zero remainder does not rule out ideal membership. Fixing this will led to the theory of Gröbner bases, which we shall see later in the course.

WORKSHEET 2.1: THE ZARISKI TOPOLOGY PT. I

Throughout this course, "ring" means *commutative* ring with unity.

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(1) Recall that the spectrum and maximal spectrum of a ring R are the sets:

$$\mathrm{Spec}(R) := \{P \subset R \mid P \text{ is a prime ideal}\} \quad \text{and} \quad \mathrm{mSpec}(R) := \{P \subset R \mid P \text{ is a maximal ideal}\}$$

Given any subset $\mathscr{S} \subset R$ define

$$\mathbb{V}(\mathscr{S}) := \{P \in \mathrm{Spec}(R) \mid \mathscr{S} \subset P\}.$$

(a) Describe $\mathrm{Spec}(\mathbb{K})$ and $\mathrm{mSpec}(\mathbb{K})$ when $\mathbb{K}$ is a field.

(b) Describe both $\mathrm{Spec}(\mathbb{K}[x]/\langle x^3\rangle)$ and $\mathrm{mSpec}(\mathbb{K}[x]/\langle x^3\rangle)$ when $\mathbb{K}$ is a field.

(c) For an arbitrary ring R, compute $\mathbb{V}(\{0\})$ and $\mathbb{V}(\{1\})$.

(d) Explain the natural poset structure on $\mathrm{Spec}(R)$. Describe the poset for $\mathrm{Spec}(\mathbb{Z})$.

A *topological space* is a set X together with a collection of subsets τ, which satisfies the following conditions: i) $\varnothing \in \tau$ and $X \in \tau$, ii) τ is closed under arbitrary unions, and iii) τ is closed under finite intersections. We call τ a *topology* on X, the elements of τ the open subsets of X, and a subset $Z \subset X$ is closed if and only if $X \setminus Z$ is open (i.e., $X \setminus Z \in \tau$). If $S \subset X$ is a subset, the *subspace topology* on S is the topology where the open sets of S are all sets of the form $S \cap U \subset S$ where $U \subset X$ is open in X. A map $f : X \to Y$ of topological spaces is *continuous* if and only if $f^{-1}(U) \subset X$ is open for all open subsets $U \subset Y$, i.e., the preimage of an open set is open.

---

(2) **The Zariski Topology.** We will now place a topology on the set $\mathrm{Spec}(R)$, called the Zariski topology after Oscar Zariski.

(a) Prove that if $\mathscr{S}, \mathscr{T} \subset R$ are arbitrary sets then $\mathbb{V}(\mathscr{S}) \cap \mathbb{V}(\mathscr{T}) = \mathbb{V}(\mathscr{S} \cup \mathscr{T})$. Explain why your argument extends to an *arbitrary* intersection.

(b) Prove that if $\mathscr{S}, \mathscr{T} \subset R$ are arbitrary sets then $\mathbb{V}(\mathscr{S}) \cup \mathbb{V}(\mathscr{T}) = \mathbb{V}(\mathscr{S} \cdot \mathscr{T})$ where

$$\mathscr{S} \cdot \mathscr{T} := \{st \mid s \in \mathscr{S},\, t \in \mathscr{T}\}.$$

(c) Prove that the sets of the form $\mathbb{V}(\mathscr{S})$ are the closed sets of a topology on $\mathrm{Spec}(R)$. We call this topology the Zariski topology on $\mathrm{Spec}(R)$. (Hint: Rephrase the definition of a topology in terms of closed sets instead of open sets.)

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(d) Take a break and Google Oscar Zariski and read about his life.

(3) Describe the closed subsets of $\mathrm{mSpec}(R) \subset \mathrm{Spec}(R)$ in the subspace topology.

(4) Let $\mathbb{K}$ be an algebraically closed field. This exercise will describe the Zariski topology on $\mathrm{Spec}(\mathbb{K}[x])$.

(a) Describe the points of $\mathrm{Spec}(\mathbb{K}[x])$. (Hint: There are two types.)

(b) Prove that if $\mathscr{S} \subset \mathbb{K}[x]$ then $\mathbb{V}(\mathscr{S}) = \mathbb{V}(\{f\})$ where f is the greatest common divisor of all elements of $\mathscr{S}$.

(c) Show that the subspace topology on $\mathrm{mSpec}(\mathbb{K}[x])$ is the finite complement topology. (Recall the *finite complement topology* on a set X is defined by letting the open sets be the empty set and sets of the form $X \setminus T$ for $T \subset X$ a subset of finite cardinality.)

(5) **Dense Points.** If X is a topological space we say that a set $S \subset X$ is *dense* if $U \cap S \neq \varnothing$ for every non-empty open subset $U \subset X$.

(a) Let p be a point in a topological space X. Prove that $\{p\}$ is a dense subset of X if and only if the smallest closed subset which contains $\{p\}$ is X. We call such points *dense points* of X. (Note: no point in $\mathbb{R}^n$ or $\mathbb{C}^n$ with the Euclidean topology is dense in this sense.)

(b) Given a prime ideal $P \subset R$ we will frequently write $[P]$ to denote the corresponding point in $\mathrm{Spec}(R)$. Prove that a point $[P] \in \mathrm{Spec}(R)$ is dense if and only if $\mathbb{V}(P) = \mathrm{Spec}(R)$.

(c) Characterize all of the dense points in the following:

(i) $\mathrm{Spec}(\mathbb{K}[x])$

(ii) $\mathrm{Spec}(\mathbb{Z})$

(iii) $\mathrm{Spec}(\mathbb{Z}[t])$

(iv) $\mathrm{Spec}(\mathbb{K}[x]/\langle x^2 \rangle)$

(v) $\mathrm{Spec}(\mathbb{K}[x,y])$

(vi) $\mathrm{Spec}(\mathbb{Z}/\langle 6 \rangle)$.

(vii) $\mathrm{Spec}(\mathbb{Z}/\langle 4 \rangle)$.

(viii) $\mathrm{Spec}(\mathbb{Z}/\langle 12 \rangle)$.

(d) Prove that if R is a domain then $\mathrm{Spec}(R)$ has a unique dense point.

(e) Repeat part 5c replacing Spec with mSpec in the subspace topology. Can $\mathrm{mSpec}(R)$ ever have a dense point?

(f) Consider the following two properties (which together are implied by the Hausdorff condition) for a topological space X: i) for every $p \in X$ the set $\{p\}$ is closed and ii) given two distinct points $p, q \in X$ there exist open subsets $U, V \subset X$ such that $p \in U$ and $q \in V$ and $U \cap V = \varnothing$. Prove that if X is a topological space with a dense point, and X is not a single point, then X fails both parts of this definition.

(6) Let $\mathscr{S} \subset R$ be any subset. Prove that $\mathbb{V}(\mathscr{S}) = \mathbb{V}(\langle \mathscr{S} \rangle)$ where $\langle \mathscr{S} \rangle$ denotes the ideal generated by $\mathscr{S}$.

(7) In this exercise we wish to understand the Zariski topology on $\mathrm{Spec}(\mathbb{C}[x,y])$. Fix a point $p = (a,b) \in \mathbb{C}^2$.

(a) Use the *evaluation-at-p* map: $\mathrm{ev} : \mathbb{C}[x,y] \to \mathbb{C}$ to prove that the set of polynomial functions vanishing at p is a maximal ideal $\mathfrak{m}_p \subset \mathbb{C}[x,y]$.

(b) Find generators for the ideal $\mathfrak{m}_p$. Is this ideal principal?

(c) Prove that the following map is injective:

$$\begin{array}{ccc} \mathbb{C}^2 & \xrightarrow{\eta} & \mathrm{Spec}(\mathbb{C}[x,y]) \\ p & \longmapsto & \mathfrak{m}_p \end{array}.$$

(d) Show that the map η from part 7c is not surjective by finding at least two distinct points in $\mathrm{Spec}(\mathbb{C}[x,y])$ not in the image of η.

(e) Describe the closed subsets of $\mathrm{Spec}(\mathbb{C}[x,y])$. (Hint: They come in four types)

A function $f : X \to Y$ of topological spaces is *continuous* if and only if $f^{-1}(U) \subset X$ is open for all open subsets $U \subset Y$, i.e., the preimage of an open is open. A *homeomorphism* $f : X \to Y$ is a continuous map of topological spaces with a continuous inverse $f^{-1} : Y \to X$. A useful fact is that a continuous map $f : X \to Y$ is a homeomorphism if and only if it is bijective and $f(U) \subset Y$ is open for every open subsets $U \subset X$.

---

(8) **Ring Maps Induce Continuous Maps.** Let $\phi : R \to S$ be a ring homomorphism. The goal of this exercise is to show ϕ induces a map $\phi^{\#} : \mathrm{Spec}(S) \to \mathrm{Spec}(R)$, which is continuous in the Zariski topology.

(a) Prove that if $[P] \in \mathrm{Spec}(S)$, then $[\phi^{-1}(P)] \in \mathrm{Spec}(R)$.

(b) Prove that the map below is continuous in the Zariski topology:

$$\begin{array}{ccc} \mathrm{Spec}(S) & \xrightarrow{\phi^{\#}} & \mathrm{Spec}(R) \\ [P] & \longmapsto & \left[\phi^{-1}(P)\right] \end{array}$$

(c) For the ring map $\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ sending each integer to its residue class modulo 2, describe the induced map on spectra explicitly.

(d) For the ring map $\mathbb{Z} \hookrightarrow \mathbb{Q}$, describe the induced map on spectra explicitly.

(e) If $\phi : R \hookrightarrow S$ is an injective ring map, give a simple description of the induced map on spectra.

A *category* is a collection of objects together with, for each pair of objects, a set of morphisms between them, equipped with an associative composition law and an identity morphism for each object. A *functor* between categories **C** and **D** is a mapping that assigns to each object of **C** an object of **D** and to each morphism in **C**

a morphism in **D**, preserving identity morphisms and composition. A functor is *contravariant* if it reverses the order of composition and *covariant* if it preserves the order of composition.

---

(9) **Spec as a Functor.** In this problem we package together much of this worksheet to define the Spec functor. Consider the map:

$$\textbf{comRing} \longrightarrow \textbf{Top}$$

$$R \longmapsto \operatorname{Spec}(R)$$

from the category of commutative rings with unit with ring homomorphisms to the category of topological spaces with continuous maps.

(a) Explain using Question 8 how this mapping defines a contravariant functor from **comRing** to **Top**. To check functoriality, verify

$$(\mathrm{Id}_R)^{\#} = \mathrm{Id}_{\operatorname{Spec}(R)} \qquad \text{and} \qquad \left(\psi \circ \phi\right)^{\#} = \phi^{\#} \circ \psi^{\#}$$

for ring maps $\phi : R \to S$ and $\psi : S \to T$. We call this the Spec functor.

(b) Do isomorphic rings have Spectra that are homeomorphic topological spaces? Explain.

(c) If two rings have homeomorphic spectra, are they necessarily isomorphic rings? Explain.

(10) Find a homeomorphism between $\operatorname{Spec}(\mathbb{C}[x])$ and $\operatorname{Spec}(\mathbb{C}[x,y]/\langle x-y\rangle)$. Construct the corresponding ring map explicitly in *Macaulay2* and compute its kernel.

(11) **Closed Embeddings I.** Let $I \subset R$ be an ideal and $\pi : R \to R/I$ the natural quotient map. Consider the induced map on spectra

$$\operatorname{Spec}(R/I) \xrightarrow{\pi^{\#}} \operatorname{Spec}(R)$$

In this exercise we will show this map is a closed embedding. A *closed embedding* is a map of topological spaces whose image is closed and which is a homeomorphism onto its image.

(a) Using the correspondence between ideals in R and R/I give an explicit description of $\pi^{\#}$.

(b) Prove that the image of $\pi^{\#}$ is equal to $\mathbb{V}(I)$.

(c) Explain why the ideal correspondence implies that $\pi^{\#}$ is injective.

(d) Prove that $\pi^{\#}$ is a closed embedding onto $\mathbb{V}(I)$, in the subspace topology, by showing that if $J \subset R/I$ is an ideal then

$$\pi^{\#}(\mathbb{V}(J)) = \mathbb{V}\left(\pi^{-1}(J)\right).$$

WORKSHEET 2.2: ALGEBRAIC SETS

Throughout this course, "ring" means *commutative* ring with unity. In this worksheet $\mathbb{K}$ will denote a field. Our goal is to study subsets of $\mathbb{K}^n$ cut out by systems of polynomial equations. We begin by studying zero loci of polynomial systems and the maximal ideals arising from evaluation at points. This leads to the following foundational theorem:

Theorem 1 (Hilbert's Nullstellensatz I)**.** *If $\mathbb{K}$ is algebraically closed, then every maximal ideal of $\mathbb{K}[x_1,\dots,x_n]$ is of the form $\langle x_1-a_1,\dots,x_n-a_n\rangle$ for some point $(a_1,\dots,a_n)\in\mathbb{K}^n$.*

Note this is just the first of a series of theorems that will go by Hilbert's Nullstellensatz. Recall that a $\mathbb{K}$-algebra is a ring R together with a specified ring homomorphism $\mathbb{K}\to R$. A morphism of $\mathbb{K}$-algebras $\varphi: R\to S$ is a ring homomorphism that preserves scalars, i.e. $\varphi(c\cdot 1_R)=c\cdot 1_S$ for every $c\in\mathbb{K}$. In particular, $\mathbb{K}[x_1,\dots,x_n]$ is a $\mathbb{K}$-algebra, every evaluation map $\mathrm{ev}_p:\mathbb{K}[x_1,\dots,x_n]\to\mathbb{K}$ is a $\mathbb{K}$-algebra homomorphism, and if $\mathfrak{m}$ is a maximal ideal then $R/\mathfrak{m}$ is naturally a field equipped with a $\mathbb{K}$-algebra structure. When we say below that a residue field is "isomorphic to $\mathbb{K}$," we mean *as a $\mathbb{K}$-algebra*.

---

(1) **Evaluation Maps.** For a point $p=(a_1,\dots,a_n)\in\mathbb{K}^n$ the *evaluation-at-p* map is given by

$$\begin{array}{ccc}\mathbb{K}[x_1,\dots,x_n] & \xrightarrow{\mathrm{ev}_p} & \mathbb{K} \\ f & \longmapsto & f(p)\end{array}.$$

(a) Prove that ev_p is a ring homomorphism and find generators for $\mathfrak{m}_p := \ker(\mathrm{ev}_p)$.

(b) Prove that there is a bijection, induced by these evaluation maps, between $\mathbb{K}^n$ and the subset of $\mathrm{Spec}(\mathbb{K}[x_1,\dots,x_n])$ consisting of maximal ideals whose residue field is isomorphic to $\mathbb{K}$ as a $\mathbb{K}$-algebra.

(c) Find a maximal ideal of $\mathbb{R}[x,y]$ that is not of the form $\mathfrak{m}_p$ for any $p\in\mathbb{R}^2$.

(d) Restate Hilbert's Nullstellensatz I (above) in terms of $\mathfrak{m}_p$.

(2) **Algebraic Sets.** Let $\mathscr{S}\subset\mathbb{K}[x_1,\dots,x_n]$ be a subset. Define the corresponding *algebraic set* in $\mathbb{K}^n$ to be:

$$\mathbb{V}(\mathscr{S}) := \left\{p\in\mathbb{K}^n \;\middle|\; f(p)=0 \text{ for all } f\in\mathscr{S}\right\}.$$

(a) Prove that $\mathbb{V}$ is order-reverse, i.e., if $\mathscr{S}\subset\mathscr{T}$ then $\mathbb{V}(\mathscr{T})\subset\mathbb{V}(\mathscr{S})$.

(b) Prove that $\mathbb{V}(\mathscr{S})=\mathbb{V}(\langle\mathscr{S}\rangle)=\mathbb{V}(\sqrt{\langle\mathscr{S}\rangle})$.

(c) Describe each of the following algebraic sets, and when possible draw a picture.

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(i) $\mathscr{S} := \{x^2 - y\} \subset \mathbb{R}[x, y]$.

(ii) $\mathscr{S} := \{x^2 - xy\} \subset \mathbb{R}[x, y]$.

(iii) $\mathscr{S} := \{x^6 - 3x^5y + 3x^4y^2 - x^3y^3\} \subset \mathbb{R}[x, y]$.

(iv) $\mathscr{S} := \{x^4 - 2x^3y + x^2y^2\} \subset \mathbb{R}[x, y]$.

(v) $\mathscr{S} := \{x^2 + y^2 - 1\} \subset \mathbb{R}[x, y]$.

(vi) $\mathscr{S} := \{n(x^2 + y^2 - z^2) \mid n \in \mathbb{Z}_{\geq 0}\} \subset \mathbb{R}[x, y, z]$.

(vii) $\mathscr{S} := \{x^2 + 1\} \subset \mathbb{R}[x]$.

(viii) $\mathscr{S} := \{x^2 + 1\} \subset \mathbb{C}[x]$.

(ix) $\mathscr{S}$ is a collection of any linear polynomials in $\mathbb{K}[x_1, \ldots, x_n]$.

(d) Explain why $\mathrm{SL}_n(\mathbb{K})$ has the structure of an algebraic set.

(e) Explain why the set of $m \times n$ matrices of rank $< t$ is an algebraic set contained in $\mathbb{K}^{m \times n}$.

(3) **Defining Ideal of an Algebraic Set.** Let $V \subset \mathbb{K}^n$ be an algebraic set. The *defining ideal* of V is the ideal in $\mathbb{K}[x_1, \ldots, x_n]$ given by

$$\mathbb{I}(V) := \left\{ f \in \mathbb{K}[x_1, \ldots, x_n] \;\middle|\; f(p) = 0 \text{ for all } p \in V \right\}.$$

(a) Prove that $\mathbb{I}(V)$ is in fact an ideal.

(b) Prove that $\mathbb{I}(V)$ is radical.

(c) If $V = \mathbb{V}(\mathscr{S})$ for some subset $\mathscr{S} \subset \mathbb{K}[x_1, \ldots, x_n]$, show that $\langle \mathscr{S} \rangle \subset \sqrt{\langle \mathscr{S} \rangle} \subset \mathbb{I}(V)$.

(d) Let $\mathbb{K} = \mathbb{R}$. Give an example to show that the inclusion $\sqrt{\langle \mathscr{S} \rangle} \subset \mathbb{I}(V)$ can be proper.

(e) If $V = \mathbb{V}(\mathscr{S})$ for some subset $\mathscr{S} \subset \mathbb{K}[x_1, \ldots, x_n]$, show that $\mathbb{V}(\mathscr{S}) = \mathbb{V}(\mathbb{I}(V))$.

So far you have built two order-reversing operations:

$$\mathscr{S} \longmapsto \mathbb{V}(\mathscr{S}) \qquad \text{and} \qquad V \longmapsto \mathbb{I}(V).$$

You have shown that passing from equations to a zero set only remembers the radical of the ideal they generate, and Question 3d shows that over a non-algebraically closed field this can lose information. The next theorem says that over an algebraically closed field this is the only ambiguity.

Theorem 2 (Hilbert's Nullstellensatz II)**.** *If $\mathbb{K}$ is algebraically closed and $V = \mathbb{V}(\mathscr{S}) \subset \mathbb{K}^n$ is an algebraic set, for some subset $\mathscr{S} \subset \mathbb{K}[x_1, \ldots, x_n]$, then $\mathbb{I}(V) = \sqrt{\langle \mathscr{S} \rangle}$.*

At this point the basic affine dictionary is taking shape; Question (4) asks you to make the correspondence between algebraic sets and radical ideals completely explicit. After that we turn to coordinate rings, which package an algebraic set V into algebra and will let us recover from $\mathbb{K}[V]$ its points, its topology, and ultimately its morphisms.

---

(4) Let $\mathbb{K}$ be algebraically closed. Use Hilbert's Nullstellensatz II (above) to give an explicit inclusion-reversing bijection between algebraic sets in $\mathbb{K}^n$ and radical ideals.

(5) **Coordinate Rings.** Let $V \subset \mathbb{K}^n$ be an algebraic set and $\mathscr{R} = \mathrm{Hom}_{\mathbf{Set}}(V, \mathbb{K})$ be the ring of all (set) functions from V to $\mathbb{K}$. In this question we study the restriction map from $\mathbb{K}[x_1, \ldots, x_n]$ to $\mathscr{R}$ given by:

$$\begin{array}{ccc} \mathbb{K}[x_1, \ldots, x_n] & \xrightarrow{\quad\rho\quad} & \mathscr{R} \\ f & \longmapsto & f|_V \end{array}.$$

Here we are viewing $f \in \mathbb{K}[x_1, \ldots, x_n]$ as defining a function $\mathbb{K}^n \to \mathbb{K}$ given by $p \mapsto f(p)$. We call the image of ρ the *coordinate ring* of V and denote it $\mathbb{K}[V]$.

(a) Prove that ρ is in fact a ring homomorphism.

(b) Verify that each of the following functions lies in the coordinate ring of the given algebraic set.

(i) Let $V = \mathbb{V}(x^2 + y^2 - 1) \subset \mathbb{R}^2$ and consider the function $V \to \mathbb{R}$ given by $(x, y) \mapsto e^{x^2+y^2}$.

(ii) Let $V = \mathbb{V}(xy - 1) \subset \mathbb{K}^2$ and consider the function $V \to \mathbb{K}$ given by $(x, y) \mapsto \frac{1}{x}$.

(c) Identify the kernel of ρ and use it to give an explicit presentation for $\mathbb{K}[V]$ as a $\mathbb{K}$-algebra.

(d) The i-th coordinate of $\mathbb{K}^n$ is the linear functional $e_i^\vee : \mathbb{K}^n \to \mathbb{K}$ sending $(a_1, \ldots, a_n) \mapsto a_i$. Explain why $e_i^\vee \in \mathbb{K}[\mathbb{K}^n]$, and under the presentation given in part 5c, which polynomial corresponds to $e_i^\vee$.

(e) Explain why $\mathbb{K}[V]$ is called the coordinate ring of V. Hint: Think about the previous part.

(f) Describe the coordinate ring for the following algebraic sets:

(i) $\mathbb{V}(y - x^2) \subset \mathbb{K}^2$.

(ii) $\mathbb{V}(y^2, xy, x^{17}) \subset \mathbb{K}^2$.

(iii) $\mathbb{V}(x^6 - 3x^5y + 3x^4y^2 - x^3y^3) \subset \mathbb{K}^2$.

(iv) The set of 3×3 matrices over $\mathbb{F}_5$ of determinant 1 and trace 0. (Hint/Warning: There will likely be 9 more polynomials in the defining ideal than you expect, coming from the fact that finite fields have a special automorphism. This highlights another way we might get a counterexample for Question 3d.)

You should now think of $\mathbb{K}[V]$ as the algebraic avatar of V: its elements are precisely the polynomial functions on V, and the coordinate functions on $\mathbb{K}^n$ descend to generators of $\mathbb{K}[V]$. The next step is to recognize points of V purely algebraically, either as $\mathbb{K}$-algebra maps $\mathbb{K}[V] \to \mathbb{K}$ or as maximal ideals whose residue fields are isomorphic to $\mathbb{K}$ as $\mathbb{K}$-algebras.

---

(6) Let R be a finitely generated $\mathbb{K}$-algebra. Fix a presentation $R \cong \mathbb{K}[x_1, \ldots, x_n]/I$. Describe a natural bijection between the following three sets:

(i) points in $\mathbb{V}(I) \subset \mathbb{K}^n$,

(ii) maximal ideals in R with residue field isomorphic to $\mathbb{K}$ as a $\mathbb{K}$-algebra, and

(iii) elements of $\mathrm{Hom}_{\mathbb{K}-\mathbf{alg}}(R,\mathbb{K})$, the set of morphisms from R to $\mathbb{K}$ in the category of $\mathbb{K}$-algebras.

(7) **Zariski Topology on Algebraic Sets.** Let $V \subseteq \mathbb{K}^n$ be an algebraic set. Define the *Zariski topology* on V by declaring the closed sets to be the subsets of V of the form $\mathbb{V}(I)$ for some ideal $I \subset \mathbb{K}[x_1,\ldots,x_n]$. In other words, the closed sets are precisely the *algebraic subsets* of V.

(a) Verify that this definition indeed determines a topology on V.

(b) For $f \in \mathbb{K}[V]$, define

$$D(f) = \{p \in V \mid f(p) \neq 0\}.$$

Show that $\{D(f)\}$ form a basis for the Zariski topology on V.

(c) For each point $p \in V$, let

$$\mathfrak{m}_p = \{f \in \mathbb{K}[V] \mid f(p) = 0\}$$

be the corresponding maximal ideal. Show that the map

$$V \longrightarrow \mathrm{Spec}(\mathbb{K}[V])$$

$$p \longmapsto \mathfrak{m}_p$$

is a homeomorphism onto its image, which consists of those maximal ideals whose residue field is isomorphic to $\mathbb{K}$ as a $\mathbb{K}$-algebra.

(d) If $\mathbb{K}$ is algebraically closed how can the preceding part be rephrased in terms of $\mathrm{mSpec}(\mathbb{K}[V])$?

(e) Is the Zariski topology on V Hausdorff in general? If not, when is it?

By now you have recovered from $\mathbb{K}[V]$ not just the points of V, but also its Zariski topology. The next goal is to understand maps in the same way: a morphism of algebraic sets should be encoded contravariantly by a $\mathbb{K}$-algebra homomorphism of coordinate rings.

---

(8) **Morphisms of Algebraic Sets.** A *polynomial map* from $\mathbb{K}^n$ to $\mathbb{K}^m$ is a function of the form:

$$\Phi : \mathbb{K}^n \longrightarrow \mathbb{K}^m \quad p \mapsto (f_1(p),\ldots,f_m(p)).$$

where $f_1,\ldots,f_m \in \mathbb{K}[x_1,\ldots,x_n]$. A *morphism of algebraic sets* $V \to W$ is defined to be the restriction of such a polynomial map Φ with $\Phi(V) \subset W$. Fix a polynomial map Φ as above.

(a) Show that there is an induced $\mathbb{K}$-algebra homomorphism

$$\phi : \mathbb{K}[\mathbb{K}^m] \longrightarrow \mathbb{K}[\mathbb{K}^n] \quad \text{given by} \quad g \mapsto g \circ \Phi.$$

Describe ϕ explicitly in terms of the coordinate function generators of $\mathbb{K}[\mathbb{K}^m]$.

(b) Let $V \subset \mathbb{K}^n$ and $W \subset \mathbb{K}^m$ be algebraic sets. Show that $\Phi(V) \subset W$ if and only if $\phi(\mathbb{I}(W)) \subset \mathbb{I}(V)$.

(c) Deduce that Φ restricts to a morphism of algebraic sets $\Phi|_V : V \longrightarrow W$ if and only if the composition

$$\mathbb{K}[\mathbb{K}^m] \xrightarrow{\phi} \mathbb{K}[\mathbb{K}^n] \xrightarrow{\rho} \mathbb{K}[V]$$

factors through $\mathbb{K}[W]$. Here ρ is the restriction map.

(d) For each of the given maps of coordinate rings, describe the corresponding map of algebraic sets. When possible draw pictures. Compare to the corresponding maps on spectra.

(i) $\mathbb{C}[x,y] \to \mathbb{C}[t]$ given by $x \mapsto t$ and $y \mapsto t$.

(ii) $\mathbb{C}[t] \to \mathbb{C}[x,y]$ given by $t \mapsto x$.

(iii) $\mathbb{C}[t] \to \mathbb{C}[x,y,t]/\langle xy - t\rangle$ given by $t \mapsto t$.

(9) Prove that a morphism of algebraic sets is continuous with respect to the Zariski topology.

(10) Show that the image of $\mathbb{K} \to \mathbb{K}^3$ sending $t \to (t, t^2, t^3)$ is an algebraic set defined by two polynomials. Verify your answer by constructing the corresponding ring map in *Macauly2*.

(11) Prove or disprove: The image of every polynomial map $\mathbb{K}^n \to \mathbb{K}^m$ is an algebraic set. (Warning: this might be field dependent. As a hint, if $\mathbb{K}$ is finite, which subsets of $\mathbb{K}^n$ are algebraic sets?)

You now have the main algebra-geometry dictionary in hand. Over any field $\mathbb{K}$, every algebraic set $V \subset \mathbb{K}^n$ has a coordinate ring $\mathbb{K}[V]$, points of V are the same as $\mathbb{K}$-algebra maps $\mathbb{K}[V] \to \mathbb{K}$, and morphisms $V \to W$ correspond contravariantly to $\mathbb{K}$-algebra homomorphisms $\mathbb{K}[W] \to \mathbb{K}[V]$. Thus the assignment

$$V \longmapsto \mathbb{K}[V]$$

always gives a contravariant bridge from geometry to algebra. When $\mathbb{K}$ is algebraically closed, Hilbert's Nullstellensatz says that every finitely generated reduced $\mathbb{K}$-algebra comes from an algebraic set, so this bridge becomes an anti-equivalence of categories.

---

(12) **Putting It All Together.** Assume that $\mathbb{K}$ is algebraically closed. Use the previous parts together with Hilbert's Nullstellensatz II to show that every finitely generated reduced $\mathbb{K}$-algebra is isomorphic to $\mathbb{K}[V]$ for some algebraic set V. Conclude that the assignment $V \mapsto \mathbb{K}[V]$ defines a (anti-) equivalence from the category of algebraic sets over $\mathbb{K}$ to the category of finitely generated reduced $\mathbb{K}$-algebras.

WORKSHEET 2.3: VALUATION RINGS

Fix a field $\mathbb{K}$. In this worksheet we will explore a special, but important type of rings called valuation rings.

Definition 1. A *valuation ring* of a field $\mathbb{K}$ is a subring $R \subset \mathbb{K}$ with the property that for all $x \in \mathbb{K}^\times$ either $x \in R$ or $x^{-1} \in R$.

Note that, since a valuation ring R of a field $\mathbb{K}$ is a subring of $\mathbb{K}$ it is necessarily a domain. We first study valuation rings directly, then show how certain functions that broadly measure some notion of divisibility, called valuations, produce them. Lastly, we compare discrete and non-discrete examples.

---

(1) **Examples of Valuation Rings.** Recall $\mathbb{C}(x,y)$ is the field of formal rational functions in x and y, which concretely is the set of elements f/g where $f,g \in \mathbb{C}[x,y]$ with $g \neq 0$. For each of the following subsets $\mathscr{S} \subset \mathbb{C}(x,y)$ determine whether or not they are valuation rings by showing whether for every $h \in \mathbb{C}(x,y)^\times$ either $h \in \mathscr{S}$ or $h^{-1} \in \mathscr{S}$.

(a) Hint: Use that $\mathbb{C}[x,y]$ is a UFD to understand when g is divisible by y.

$$\mathscr{S}_1 := \left\{ \frac{f}{g} \;\middle|\; f,g \in \mathbb{C}[x,y] \text{ and } g \notin \langle y \rangle \right\}$$

(b) Hint: Consider the element $\frac{x}{y}$.

$$\mathscr{S}_2 := \left\{ \frac{f}{g} \;\middle|\; f,g \in \mathbb{C}[x,y] \text{ and } g(0,0) \neq 0 \right\}$$

(c) Define the *order* of a non-zero polynomial $f \in \mathbb{C}[x,y]$ to be $\operatorname{ord}(f)$, the lowest degree of a monomial in f and $\operatorname{ord}(0) = \infty$.

$$\mathscr{S}_3 := \left\{ \frac{f}{g} \;\middle|\; f,g \in \mathbb{C}[x,y],\ g \neq 0,\ \text{and } \operatorname{ord}(f) \geq \operatorname{ord}(g) \right\}$$

(2) **Ideals in Valuation Rings.** Let R be a valuation ring of $\mathbb{K}$.

(a) Show that for any two non-zero elements $x, y \in R$, either $x|y$ or $y|x$.

(b) Show that every finitely generated ideal of R is principal.

(c) Prove that the ideals of R form a totally ordered set. That is, for any two ideals I, J in R, either $I \subseteq J$ or $J \subseteq I$. (Hint: Given $x \in I$ with $x \notin J$ consider whether an element $y \in J$ can divide x.)

(d) What does the previous parts say about the poset structure on $\operatorname{Spec}(R)$?

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(3) **Local Rings & Valuation Rings.** An arbitrary ring R is said to be a *local ring* if it has exactly one maximal ideal. We often write a local ring R as a pair $(R, \mathfrak{m})$ where $\mathfrak{m} \subset R$ is the unique maximal ideal.

(a) Describe $\mathrm{mSpec}(R)$ if R is a local ring.

(b) Describe the closed points of $\mathrm{Spec}(R)$ if R is a local ring.

(c) Prove that a valuation ring is a local ring.

(4) **p-Adic Integers.** Fix a prime p. The *p-adic numbers* $\mathbb{Q}_p$ are the completion of $\mathbb{Q}$ with respect to the p-adic absolute value $|\cdot|_p$, defined as follows: for a nonzero rational number x, write $x = p^k \frac{a}{b}$ where $a, b \in \mathbb{Z}$ are not divisible by p. Then

$$|x|_p = p^{-k}, \qquad |0|_p = 0.$$

Note $|\cdot|_p$ extends uniquely to $\mathbb{Q}_p$. The *p-adic integers* are defined by

$$\mathbb{Z}_p = \left\{ x \in \mathbb{Q}_p \;\middle|\; |x|_p \leq 1 \right\}.$$

Similar to decimal expansion in $\mathbb{R}$, it is a fact that every element $x \in \mathbb{Q}_p$ can be written uniquely as a formal series

$$x = \sum_{i=k}^{\infty} a_i p^i \qquad \text{where} \qquad a_i \in \{0, 1, \ldots, p-1\}, \quad a_k \neq 0,$$

and that

$$\mathbb{Z}_p = \left\{ \sum_{i=0}^{\infty} a_i p^i \;\middle|\; a_i \in \{0, 1, \ldots, p-1\} \right\}.$$

(a) Prove that the p-adic absolute value is well-defined. (Hint: Use the fact that $\mathbb{Z}$ is a UFD to show that every element $x \in \mathbb{Q}$ can be written uniquely (up to sign and reordering) as $p_1^{a_1} \ldots p_t^{a_t}$ where the p_i are distinct primes and $a_i \in \mathbb{Z}$.)

(b) Prove the p-adic absolute value satisfies the *non-Archimedean (ultrametric) inequality*:

$$|x + y|_p \leq \max\left\{ |x|_p, |y|_p \right\}.$$

(c) Prove the p-adic absolute value is multiplicative: $|xy|_p = |x|_p |y|_p$.

(d) Use parts 4b and 4c to show that $\mathbb{Z}_p$ is closed under addition and multiplication.

(e) Conclude that $\mathbb{Z}_p$ is a ring (called the ring of p-adic integers).

(f) Prove that $\mathbb{Z}_p$ is a valuation ring.

The previous exercises suggest why valuation rings are called *valuation* rings: they arise from functions that measure divisibility, generalizing the p-adic absolute value. Such functions are called valuations. An *ordered abelian group* is an abelian group $(\Gamma, +)$ together with a total order $<$ on Γ that is compatible with addition, in the sense that if $\alpha < \beta$ then $\alpha + \gamma < \beta + \gamma$ for all $\alpha, \beta, \gamma \in \Gamma$. We will frequently extend the

total ordering on an ordered abelian group Γ to $\Gamma \cup \{\infty\}$ by making the convention that $\gamma < \infty$ for all $\gamma \in \Gamma$. Similarly, we will extend addition to $\Gamma \cup \{\infty\}$ by declaring $\infty + \gamma = \gamma + \infty = \infty$ for all $\gamma \in \Gamma$.

Definition 2. Let $(\Gamma, +, <)$ be an ordered abelian group. A Γ-*valued valuation*, or just a *valuation*, on $\mathbb{K}$ is a function: $v : \mathbb{K} \to \Gamma \cup \{\infty\}$, which satisfies the following axioms for all $x, y \in \mathbb{K}$:

(i) $v(xy) = v(x) + v(y)$

(ii) $v(x + y) \geq \min\{v(x), v(y)\}$, and

(iii) $v(x) = \infty$ if and only if $x = 0$.

Note that this means v defines a group homomorphism $v : \mathbb{K}^\times \to \Gamma$. Indeed, it is standard to define a valuation as a group homomorphism $v : \mathbb{K}^\times \to \Gamma$ satisfying axiom (ii), and then extend it to all of $\mathbb{K}$ by setting $v(0) = \infty$. This convention is implicit throughout the remainder of the worksheet.

The *value group* of a valuation $v : \mathbb{K}^\times \to \Gamma$ is the image of v. A valuation is a *discrete valuation* if its value group is isomorphic to $\mathbb{Z}$. If $\mathbb{F} \subset \mathbb{K}$ is a subfield, we say that v is a $\mathbb{F}$-valuation if $v(\lambda) = 0$ for all $\lambda \in \mathbb{F}^\times$.

---

(5) **Examples of Valuations.** Prove that the following are valuations.

(a) The p-adic valuation:

$$v_p : \mathbb{Q}^\times \to \mathbb{Z}, \qquad v_p\left(p^k \frac{a}{b}\right) = k$$

where $a, b \in \mathbb{Z}$ are not divisible by p.

(b) The order valuation on $\mathbb{C}(x)$:

$$v_0 : \mathbb{C}(x)^\times \to \mathbb{Z}, \qquad v_0(f/g) = \mathrm{ord}_{x=0}(f) - \mathrm{ord}_{x=0}(g).$$

(c) More generally, if $a \in \mathbb{C}$, define

$$v_a : \mathbb{C}(x)^\times \to \mathbb{Z}$$

by order of vanishing at $x = a$. Show this is again a valuation.

(d) On $\mathbb{C}(x, y)$, define

$$v(f/g) = \mathrm{ord}(f) - \mathrm{ord}(g),$$

where $\mathrm{ord}(h)$ is the degree of the lowest-degree nonzero term of $h \in \mathbb{C}[x, y]$. Show that this is a valuation with value group $\mathbb{Z}$.

(6) Prove that every ordered abelian group is torsion-free. (Hint: Reduce to the case when $x > 0$.)

(7) **Basic Properties of Valuations.** Let $v : \mathbb{K}^\times \to \Gamma$ be an arbitrary valuation. Fix $f, g \in \mathbb{K}^\times$.

(a) Show that $v(f^{-1}) = -v(f)$.

(b) Show that if $v(g) < v(f)$, then $v(f + g) = v(g)$. (Hint: Use that $(f + g) + (-f) = g$.)

(c) Show that $v(f) = v(-f)$. (Hint: Use part 7b.)

(8) Consider the field $\mathbb{C}(x,y)$. Throughout this question π refers to the number $\pi = 3.1415926...$.

(a) Show that there is a unique $\mathbb{C}$-valuation $v_\pi : \mathbb{C}(x,y)^\times \to (\mathbb{R},+)$ such that

$$v_\pi\left(x^a y^b\right) = a + \pi b.$$

What is the value group of v_π? (Hint: Use Question 3.)

(b) Consider the group $\Gamma = \mathbb{Z} \oplus \mathbb{Z}$ ordered by the lexicographical ordering. Show that there exists a unique $\mathbb{C}$-valuation $v : \mathbb{C}(x,y)^\times \to \Gamma$ such that $v(x^a y^b) = (a,b)$.

(c) Are either of these valuations constructed in parts 8a or 8b discrete?

(9) **Valuations Determine Valuation Rings.** Let v be an arbitrary valuation on $\mathbb{K}$.

(a) Prove that the set $R \subset \mathbb{K}$ of elements $f \in \mathbb{K}$ with $v(f) \geq 0$ is a valuation ring of $\mathbb{K}$. We call this the *valuation ring* of v. (Hint: Remember our convention about $v(0)$.)

(b) Show that the following is an ideal of R:

$$\mathfrak{m} := \{f \in \mathbb{K} \mid 0 < v(f)\}.$$

(c) Prove that if $f \in R$ and $f \notin \mathfrak{m}$ then f is a unit.

(d) Combine the previous parts to deduce $\mathfrak{m}$ is the unique maximal ideal of R.

(e) Fix an element $\gamma \in \Gamma$. Prove the sets below are ideals of R:

$$\{f \in R \mid v(f) > \gamma\} \quad \text{and} \quad \{f \in R \mid v(f) \geq \gamma\}.$$

(10) **Discrete Valuation Rings.** Let $v : \mathbb{K}^\times \to \mathbb{Z}$ be a discrete valuation on $\mathbb{K}$. Let $(R,\mathfrak{m})$ be the corresponding *discrete valuation ring*.

(a) Let $I \subset R$ be any non-zero ideal, explain why we can choose an element $f \in I$ such that $v(f)$ is minimal among elements in I.

(b) Prove that every ideal in R is principal. In particular, a discrete valuation ring is a PID.

(c) By part 10b there exists an element $t \in R$ such that $\langle t \rangle = \mathfrak{m}$. Show that every ideal of R is either 0 or of the form $\langle t^n \rangle$ for some $n \geq 0$.

(11) Consider the valuation v_π on $\mathbb{C}(x,y)$ constructed in Question 8a. Let $(R,\mathfrak{m})$ be the corresponding valuation ring for v_π.

(a) For a real number $x \in \mathbb{R}_{\geq 0}$ let $\{x\} := x - \lfloor x \rfloor$ denote its fractional part. Fix a positive integer N and consider the sequence consisting of the fractional parts of the first $N+1$ multiples of π: $\{0\},\{\pi\},\{2\pi\},\ldots,\{N\pi\} \in [0,1)$. Divide the interval $[0,1)$ into N evenly spaced subintervals:

$$[0,1) = \left[0,\frac{1}{N}\right) \cup \left[\frac{1}{N},\frac{2}{N}\right) \cup \cdots \cup \left[\frac{N-1}{N},1\right).$$

Prove there exist integers $0 \le k, j \le N$ such that $\{k\pi\}$ and $\{j\pi\}$ lie in the same subinterval. Deduce that for any positive integer N there exist integers $0 \le k, j \le N$ such that $|\{k\pi\} - \{j\pi\}| \le \frac{1}{N}$. (Hint: Think about pigeons.)

(b) Using that π is irrational show that $\{j\pi\} \neq \{k\pi\}$ whenever $j \neq k$.

(c) Prove for every real number $\epsilon > 0$, there exist integers $a, b \in \mathbb{Z}$ such that $0 < a + b\pi < \epsilon$.

(d) Prove that $\mathfrak{m}$ is not finitely generated. (Hint: Use part 11c.)

(e) Prove that $\mathfrak{m}$ is the only non-zero radical ideal of R. (Hint: If $I \subset R$ is a non-zero radical ideal, let $f \in I$ be non-zero. For an arbitrary element $h \in \mathfrak{m}$ show that $h^n \in \langle f \rangle$ for $n \gg 0$.)

(f) Describe $\mathrm{Spec}(V)$.

(12) Consider the valuation v on $\mathbb{C}(x, y)$ constructed in Question 8b. Let $(R, \mathfrak{m})$ be the corresponding valuation ring for v.

(a) Show that the maximal ideal of R is

$$\mathfrak{m} = \left\{ f \in \mathbb{C}(x, y) \;\middle|\; v(f) \ge (0, 1) \right\} = \langle y \rangle.$$

(b) Show that the ideal $\langle x \rangle \subset R$ is not prime. (Hint: $x = xy^{-1} \cdot y$.)

(c) Show that the set below is a prime, non-maximal ideal of R:

$$P := \left\{ f \in R \;\middle|\; v(f) = (a, b) \text{ where } a \ge 1 \right\} \cup \{0\}.$$

(d) Use the valuation to construct an infinite strictly ascending chain of ideals in R:

$$\left\langle xy^{-1} \right\rangle \subset \left\langle xy^{-2} \right\rangle \subset \left\langle xy^{-3} \right\rangle \subset \cdots.$$

Explain why all inclusions are strict.

(13) **Valuation Rings Determine Valuations.** Let R be a valuation ring of $\mathbb{K}$ with $R \neq \mathbb{K}$. This problem shows how to recover a valuation from the divisibility structure of R.

(a) Let Γ' be the set of nonzero principal ideals of R:

$$\Gamma' = \left\{ \langle a \rangle \subset R \;\middle|\; a \in R,\ a \neq 0 \right\}.$$

Define a multiplication on Γ' by ideal multiplication: $\langle a \rangle \cdot \langle b \rangle = \langle ab \rangle$.

(i) Show that this operation is well-defined and makes Γ' into an abelian monoid (associative, commutative, binary operation with an identity element).

(ii) Show that Γ' is totally ordered by inclusion: for distinct elements $\langle a \rangle, \langle b \rangle \in \Gamma'$, either $\langle a \rangle \subseteq \langle b \rangle$ or $\langle b \rangle \subseteq \langle a \rangle$.

(b) Define an equivalence relation $\sim$ on $\mathbb{K}^\times$ by $x \sim y$ if and only if $x = uy$ for some unit $u \in R^\times$. Let $\Gamma = \mathbb{K}^\times / \sim$ denote the set of equivalence classes. Given $x \in \mathbb{K}^\times$ write $[x]$ its equivalence class in Γ.

(i) Show that $\sim$ is a well-defined equivalence relation.

(ii) Show that multiplication in $\mathbb{K}^\times$ induces a well-defined operation on Γ, making it into an abelian group.

(iii) Define $[x] \leq [y]$ if and only if $yx^{-1} \in R$. Show this is a well-defined total order on Γ.

(iv) Deduce that Γ is an ordered abelian group.

(v) Show that the map $\Gamma' \to \Gamma$ given by $\langle a \rangle \mapsto [a]$ is an order-preserving embedding whose image consists exactly of the elements $\geq [1]$.

(c) Define $v : \mathbb{K}^\times \to \Gamma$ by $v(x) = [x]$. Prove that v is a valuation on $\mathbb{K}$ with value group Γ and

$$R = \left\{ x \in \mathbb{K} \;\middle|\; v(x) \geq [1] \right\}.$$

Moreover, $v(x) \in \Gamma'$ if and only if $x \in R$.

WORKSHEET 3.1: GRÖBNER BASES PT. II

Fix a field $\mathbb{K}$. Unless otherwise specified the ring throughout this worksheet will be $\mathbb{K}[x_1,\ldots,x_n]$. In the previous worksheet we introduced monomial orderings and leading terms. In this worksheet we return to the multivariable division algorithm, and then use it to study monomial ideals, initial ideals, and Gröbner bases. The main goal is to push the story all the way to a major finiteness theorem: every ideal of $\mathbb{K}[x_1,\ldots,x_n]$ is finitely generated.

Recall a *monomial ordering* on $\mathbb{K}[x_1,\ldots,x_n]$ is a total well-ordering $<$ on $\mathbb{Z}^n_{\geq 0}$ such that if $\alpha < \beta$ then $\alpha+\gamma < \beta+\gamma$ for all $\gamma \in \mathbb{Z}^n_{\geq 0}$. Using the bijection between monomials in $\mathbb{K}[x_1,\ldots,x_n]$ and elements of $\mathbb{Z}^n_{\geq 0}$ we think of a monomial ordering as giving us a way to compare monomials by saying $x^\alpha < x^\beta$ if and only if $\alpha < \beta$. In Worksheet 1.3 we saw several important monomial orders: lexicographic (lex) order, graded lexicographic (grlex) order, and graded reverse lexicographic (grevlex) order. Please review that worksheet for their definitions.

Returning to our goal of having a division algorithm in $\mathbb{K}[x_1,\ldots,x_n]$ we are faced with two challenges compared to the familiar setting of $\mathbb{K}[x]$:

(i) In several variables the notion of "leading term" is ambiguous. In one variable, the leading term is simply the term of highest degree. In several variables this no longer suffices: for instance, what should the leading term of x^2+xy+y^2 be? All three terms have total degree 2.

(ii) We want a division algorithm that can answer questions about ideals, i.e., membership, intersections, sums, products, radicals. In $\mathbb{K}[x]$ every ideal is principal, so dividing by a single generator is enough. For $n>1$, however, $\mathbb{K}[x_1,\ldots,x_n]$ is no longer a PID, so ideals generally require multiple generators. We therefore need a way to divide a polynomial by an entire *list* of polynomials.

As seen on Worksheet 1.3, once a monomial ordering has been fixed, we give a meaningful definition of leading term, leading monomial, and leading coefficient; solving issue (i). Note while we normally write $\mathrm{LT}(f)$, $\mathrm{LM}(f)$ and $\mathrm{LC}(f)$ for these quantities; they do depend on the chosen order $<$, and so would more properly be called $\mathrm{LT}_<(f)$, $\mathrm{LM}_<(f)$ and $\mathrm{LC}_<(f)$ to highlight this fact.

To resolve issue (ii), we generalize long division as follows. In single-variable polynomial long division, we repeatedly ask: "Does the leading term of the divisor divide the current leading term of the dividend?" If yes, we subtract off the appropriate multiple; if not, we are done. The multivariable algorithm works the same way, except that instead of checking a single divisor we scan through an *ordered* list of divisors $G=(g_1,\ldots,g_s)$ and use the first one whose leading term divides the current leading term. More precisely, the algorithm maintains a "working polynomial" p (initialized to f), quotients $q_1,\ldots,q_s$ (initialized to 0), and a remainder r (initialized to 0), then repeats:

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- **Try to Divide.** Walk through the list $g_1, g_2, \ldots, g_s$ in order. If you find some g_i whose leading term $\mathrm{LT}(g_i)$ divides $\mathrm{LT}(p)$, subtract the appropriate multiple: update $q_i \leftarrow q_i + \mathrm{LT}(p)/\mathrm{LT}(g_i)$ and $p \leftarrow p - (\mathrm{LT}(p)/\mathrm{LT}(g_i))g_i$. Then return to the start of the list and try again with the new p.
- **Move to Remainder.** If *no* g_i in the list has a leading term dividing $\mathrm{LT}(p)$, then the current leading term of p cannot be reduced further. Move it to the remainder: $r \leftarrow r + \mathrm{LT}(p)$ and $p \leftarrow p - \mathrm{LT}(p)$.

We continue until the working polynomial p is equal to 0. Proving this algorithm terminates as expected, which you will do later in this worksheet, amounts essentially to proving the following theorem

Theorem 1. *Fix a monomial ordering $<$ on $\mathbb{K}[x_1, \ldots, x_n]$ and let $G = (g_1, \ldots, g_s)$ be an ordered list of nonzero polynomials. If $f \in \mathbb{K}[x_1, \ldots, x_n]$ then there exist $q_1, \ldots, q_s, r \in \mathbb{K}[x_1, \ldots, x_n]$ such that*

$$f = q_1 g_1 + \cdots + q_s g_s + r,$$

where no monomial appearing in r is divisible by any of $\mathrm{LM}(g_1), \ldots, \mathrm{LM}(g_s)$.

Written in pseudocode the algorithm for multivariable polynomial division is shown below.

Algorithm 1 Division by an ordered list $G = (g_1, \ldots, g_s)$

Require: $f, g_1, \ldots, g_s$
Ensure: $q_1, \ldots, q_s, r$
  $q_1, \ldots, q_s \leftarrow 0$; $r \leftarrow 0$; $p \leftarrow f$
  while $p \neq 0$ **do**
    divides $\leftarrow 0$
    for $i = 1, \ldots, s$ **do**
      if $\mathrm{LT}(g_i)$ divides $\mathrm{LT}(p)$ **then**
        $q_i \leftarrow q_i + \mathrm{LT}(p)/\mathrm{LT}(g_i)$
        $p \leftarrow p - (\mathrm{LT}(p)/\mathrm{LT}(g_i))g_i$
        divides $\leftarrow 1$
        break
      end if
    end for
    if divides $= 0$ **then**
      $r \leftarrow r + \mathrm{LT}(p)$
      $p \leftarrow p - \mathrm{LT}(p)$
    end if
  end while

(1) **Review of Monomial Orders.** Work in $\mathbb{K}[x, y, z]$ and assume $x > y > z$.

(a) Order the monomials x^2y, xy^2, xz^2, y^3, yz^4 from largest to smallest with respect to lex, graded lex, and graded reverse lex.

(b) Compute LM(f), LC(f), and LT(f) for $f = 4x^2y - 3xy^2 + y^4 + 2z^5$ with respect to lex and grevlex.

(c) Prove that 1 is the smallest monomial for every monomial ordering.

(d) Deduce that every strictly decreasing sequence of monomials terminates.

(e) Explain why part (d) should make you optimistic that a division algorithm might terminate.

(2) **Examples of Multivariable Division.** For this question, we will work in $\mathbb{K}[x,y]$ considered with the lex order where $x > y$.

(a) Divide $f = x^2y + xy + y - x - 2$ by the ordered list $G = (xy-1, y-1)$. Record the successive values of p, the quotients, and the remainder after each update.

(b) Divide $f = xy^2 - x$ by the ordered list $G = (xy-1,\ y^2-1)$. Record the successive values of p, the quotients, and the remainder after each update.

(c) Repeat (b), but with the ordered list $(y^2-1,\ xy-1)$, i.e. where the order of G is swapped.

(d) Show directly that $xy^2 - x \in \langle xy-1, y^2-1\rangle$.

(e) What do your computations show about division in several variables? In particular, what is always true when the remainder is 0, and what can go wrong when the remainder is non-zero?

(3) **Termination of Division Algorithm** Let $G = (g_1,\ldots,g_s)$ be an ordered list of non-zero polynomials.

(a) Show that throughout the algorithm one always has $f = q_1g_1 + \cdots + q_sg_s + p + r$.

(b) Show that after each pass through the `while` loop, the leading monomial of p strictly decreases with respect to the given monomial order.

(c) Use Question 1(d) to prove that the algorithm terminates.

(d) Prove that on termination the output satisfies $f = q_1g_1 + \cdots + q_sg_s + r$, and no monomial appearing in r is divisible by any of $\mathrm{LM}(g_1),\ldots,\mathrm{LM}(g_s)$.

(e) Explain why a zero remainder certifies ideal membership: if dividing f by G produces remainder 0, then $f \in \langle g_1,\ldots,g_s\rangle$.

Question 2 reveals an important new feature of polynomial division in several variables. Unlike the one-variable case, the remainder obtained by dividing a polynomial f by polynomials $g_1,\ldots,g_t$ depends on two choices: i) the order of our list of divisors and ii) the monomial order we have fixed. Because of this, division does *not* immediately solve the ideal membership problem in several variables. In other words, even if dividing f by the list $(g_1,\ldots,g_t)$ produces a nonzero remainder, it may still happen that $f \in \langle g_1,\ldots,g_t\rangle$. So, in several variables, a nonzero remainder does *not* by itself show that $f \notin \langle g_1,\ldots,g_t\rangle$.

We will eventually develop a way to fix this problem. Before we do, however, it is useful to study a special class of ideals for which ideal membership *is* controlled purely by divisibility of monomials.

Definition 2. An ideal $I \subseteq \mathbb{K}[x_1,\dots,x_n]$ is a *monomial ideal* if it can be generated by monomials.

We shall see shortly that a monomial ideal is entirely determined by the set of monomials belonging to the ideal. Since monomials in $\mathbb{K}[x_1,\dots,x_n]$ are in bijection with $\mathbb{Z}^n_{\geq 0}$ this means we can – and often do think of monomial ideals as corresponding to subsets of $\mathbb{Z}^n_{\geq 0}$. With this viewpoint, the following classical result guarantees that monomial ideals are well-behaved.

Theorem 3 (Dickson's Lemma)**.** *Every subset of $\mathbb{Z}^n_{\geq 0}$ has only finitely many minimal elements with respect to the coordinatewise order $\leq$.*

Motivated by this theorem we say a *minimal monomial generating set* for a monomial ideal I is a finite set of monomials $\{m_1,\dots,m_t\}$ such that $I = \langle m_1,\dots,m_t\rangle$ and no m_i divides m_j for $i \neq j$. Equivalently, no generator is redundant: removing any m_i from the set yields a strictly smaller ideal.

---

(4) **Properties of Monomial Ideals.**

(a) Decide which of the following ideals are monomial ideals:

$$\langle x^2, xy^3, z\rangle, \qquad \langle x+y, y^2\rangle, \qquad \langle x^2-y, y^3\rangle, \qquad \langle x^2, xy, y^4+x\rangle.$$

(b) Let $m_1,\dots,m_t$ and m be monomials. Prove that $m \in \langle m_1,\dots,m_t\rangle$ if and only if $m_i \mid m$ for some i.

(c) Deduce that a polynomial $f \in \mathbb{K}[x_1,\dots,x_n]$ lies in I if and only if every monomial appearing in f is divisible by one of $m_1,\dots,m_t$.

(d) Find a minimal monomial generating set for each of the following ideals:

$$J = \langle x^3,\, x^2y,\, xy^2,\, y^4,\, x^4y,\, x^2y^3\rangle \qquad \text{and} \qquad L = \langle x^2z,\, xyz,\, y^2z,\, z^3,\, xz^4\rangle.$$

Draw a picture of the subset of $\mathbb{Z}^2_{\geq 0}$ corresponding to the monomials in J.

(5) **Understanding Dickson's Lemma.**

(a) Let $A \subseteq \mathbb{Z}^n_{\geq 0}$ and set $I_A := \langle x^\alpha \mid \alpha \in A\rangle$. If $A_{\min}$ denotes the set of minimal elements of A, show that every monomial in I_A is divisible by some x^α with $\alpha \in A_{\min}$.

(b) Explain why the statement "every subset of $\mathbb{Z}^n_{\geq 0}$ has finitely many minimal elements with respect to $\leq$" is equivalent to the statement "every monomial ideal in $\mathbb{K}[x_1,\dots,x_n]$ is finitely generated."

(c) Restate Dickson's Lemma entirely in terms of monomial ideals in $\mathbb{K}[x_1,\dots,x_n]$.

(d) Prove your monomial ideal version of Dickson's Lemma in the case when $n = 1$.

(6) **Proving Dickson's Lemma.** We now prove the monomial ideal version of Dickson's Lemma by induction on the number of variables. Your work in Question 5d serves as the base case. Assume Dickson's Lemma holds in $n-1$ variables. Let $I \subseteq \mathbb{K}[x_1,\dots,x_n]$ be a monomial ideal. For $k \geq 0$ let

$$J_k = \left\{ f \in \mathbb{K}[x_1,\dots,x_{n-1}] \;\middle|\; f x_n^k \in I \right\} \cup \{0\}.$$

(a) Prove that J_k is a monomial ideal in $\mathbb{K}[x_1,\dots,x_{n-1}]$ for all $k \geq 0$.

(b) Show that the J_k form an ascending chain.

$$J_0 \subseteq J_1 \subseteq J_2 \subseteq \cdots.$$

(Hint: What happens when you multiply something in J_k by x_n?)

(c) Let $L = \bigcup_{d \geq 0} J_d$. Show that L is a monomial ideal in $\mathbb{K}[x_1,\dots,x_{n-1}]$.

(d) Prove that there exists $N \in \mathbb{Z}_{\geq 0}$ such that $L = J_k$ for all $k \geq N$. (Hint: Use the inductive hypothesis to show that L is finitely generated.)

(e) For each $0 \leq k \leq N$, choose monomial generators $m_{k,1},\dots,m_{k,r_k}$ for J_k. Via the inclusion $\mathbb{K}[x_1,\dots,x_{n-1}] \subset \mathbb{K}[x_1,\dots,x_n]$ view these as elements in $\mathbb{K}[x_1,\dots,x_n]$. Prove that I is generated by the finite set

$$\left\{ m_{k,i} x_n^k \;\middle|\; 0 \leq k \leq N,\ 1 \leq i \leq r_k \right\}.$$

(f) Conclude Dickson's Lemma for all n.

(7) **Minimal monomial generators.** Let I be a monomial ideal, and let $M(I)$ be the set of monomials in I which are minimal under divisibility.

(a) Use Dickson's Lemma to show that $M(I)$ is finite.

(b) Prove that $I = \langle M(I) \rangle$.

(c) Prove that every monomial generating set of I contains $M(I)$. Conclude that $M(I)$ is the unique minimal monomial generating set of I.

Monomial ideals are easier to study because membership is determined entirely by divisibility of monomials, and Dickson's Lemma gives strong finiteness properties. For a general ideal, we would like to recover similar structures by extracting the leading term of each polynomial. This leads to the following.

Definition 4. Let $<$ be a monomial order on $\mathbb{K}[x_1,\dots,x_n]$. The *initial ideal* of a nonzero ideal $I \subset \mathbb{K}[x_1,\dots,x_n]$ with respect to $<$ is:

$$\mathrm{in}_<(I) := \langle \mathrm{LT}(f) \;\middle|\; f \in I \rangle.$$

A finite subset $\mathcal{G} = \{g_1,\dots,g_s\} \subset I$ is a *Gröbner basis* for I with respect to $<$ if $\mathrm{in}_<(I) = \langle \mathrm{LT}(g_1),\dots,\mathrm{LT}(g_s) \rangle$.

It is standard to set $\text{in}_<(\langle 0\rangle) = \langle 0\rangle$. Note a priori there is no reason for Gröbner bases to exist in general or be interesting. In the next few exercises you will show that Dickson's Lemma not only implies that Gröbner bases exist, but also they will give an answer to the ideal membership problem.

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(8) **Initial Ideals of Monomial Ideals.** Let $I \subset \mathbb{K}[x_1,\dots,x_n]$ be a monomial ideal. Prove that $\text{in}_<(I) = I$ for every monomial order $<$.

(9) For this problem consider $\mathbb{K}[x,y]$ with the lex order where $x > y$. Let $I = \langle xy-1, y^2-1\rangle$.

(a) Find explicit generators for the ideal generated by the leading terms of $xy-1$ and y^2-1.

(b) Prove that $\{xy-1, y^2-1\}$ is not a Gröbner basis for I by finding an element in I not contained in the ideal computed in 9a. (Hint: Consider the element $x-y$.)

(c) Use *Macaulay2* to find a Gröbner basis for I.

(10) **Existence of Gröbner Bases.** Let $I \subseteq \mathbb{K}[x_1,\dots,x_n]$ be an ideal and $<$ be a monomial order.

(a) Show that $\text{in}_<(I)$ is a monomial ideal.

(b) By Dickson's Lemma, $\text{in}_<(I) = \langle m_1,\dots,m_t\rangle$ for a minimal monomial generating set $m_1,\dots,m_t$. Show that there exist elements $g_1,\dots,g_t \in I$ such that $\text{LT}(g_i)$ divides m_i for all i.

(c) Show that the minimality of $m_1,\dots,m_t$ implies $\text{LT}(g_i) = c_i m_i$ for some scalar $c_i \in \mathbb{K}^\times$

(d) Conclude that $\mathcal{G} = \{g_1,\dots,g_t\}$ is a Gröbner basis of I.

(11) **Ideal Membership & Gröbner Bases.** Let $I \subseteq \mathbb{K}[x_1,\dots,x_n]$ be an ideal and $<$ be a monomial order. Fix a Gröbner basis $G = (g_1,\dots,g_t)$ for I, i.e. elements $g_i \in I$ such that $m_i := \text{LT}(g_i)$ generate $\text{in}_<(I)$.

(a) Given $f \in \mathbb{K}[x_1,\dots,x_n]$ consider the division of f by $\mathcal{G} = (g_1,\dots,g_t)$ obtaining

$$f = q_1 g_1 + \cdots + q_t g_t + r,$$

as in Theorem 1. Show that $f \in I$ if and only if $r \in I$.

(b) Assume $f \in I$ and $r \neq 0$. Show that $\text{LT}(r) \in \text{in}_<(I)$, and conclude $\text{LT}(r)$ is divisible by one of $m_1,\dots,m_t$. Why does this contradict the defining property of the remainder? Conclude that every $f \in I$ has remainder 0 on division by G.

(c) Membership Test. Prove that if $\mathcal{G}$ is a Gröbner basis for I, then for any polynomial $f \in \mathbb{K}[x_1,\dots,x_n]$

$$f \in I \quad \Longleftrightarrow \quad \text{the remainder on dividing } f \text{ by } \mathcal{G} \text{ is } 0.$$

(d) Prove that $I = \langle g_1,\dots,g_t\rangle$. (Hint: The inclusion $\langle g_1,\dots,g_t\rangle \subset I$ should be clear. For the other, let $f \in I$ then apply the division algorithm and the membership test.)

Notice that Questions 10 and 11 together show; given any ideal $I \subseteq \mathbb{K}[x_1,\ldots,x_n]$ and any monomial order $<$, a Gröbner basis for I exists (Question 10), and every Gröbner basis generates I (Question 11(d)). Since a Gröbner basis is finite by construction, every ideal of $\mathbb{K}[x_1,\ldots,x_n]$ is finitely generated. This naturally leads us to our first encounter with the following important finiteness condition.

Definition 5. A ring R is *Noetherian* if and only if every ascending chain of ideals:

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$$

stabilizes in the sense that there exists $N \in \mathbb{Z}_{\geq 1}$ such that $I_n = I_{n+1}$ for all $n \geq N$.

It is common to say a ring is Noetherian if it has the ACC property where ACC is short for ascending chain condition. You will show below a ring R being Noetherian is equivalent to every ideal of R being finitely generated. Thus, you have proven that $\mathbb{K}[x_1,\ldots,x_n]$ is Noetherian. Since a field $\mathbb{K}$ is Noetherian, you have actually proven a special, but important case of the following theorem.

Theorem 6 (Hilbert's Basis Theorem)**.** *Let R be a ring. If R is Noetherian, then $R[x]$ is Noetherian.*

The remainder of this worksheet is devoted to proving the equivalence between the ACC and finite generation of ideals, and proving Hilbert's basis theorem in general.

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(12) **Properties of Noetherian Rings.** For this question let R be any commutative ring, and consider an ascending chain of ideals in R:

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots.$$

(a) Show that $J = \bigcup_{k\geq 1} I_k$ is an ideal of R.

(b) If J is finitely generated, say $J = \langle a_1,\ldots,a_t\rangle$, explain why there exists an integer $N \in \mathbb{Z}_{\geq 1}$ such that $a_1,\ldots,a_t \in I_N$. Conclude that the chain of ideals stabilizes.

(c) Explain why you just showed that if every ideal of R is finitely generated then R is Noetherian.

(d) Let I be an ideal of R and suppose every ascending chain of ideals in R stabilizes. Pick any $a_1 \in I$. If $\langle a_1\rangle \neq I$, pick $a_2 \in I \setminus \langle a_1\rangle$, obtaining $\langle a_1\rangle \subsetneq \langle a_1, a_2\rangle$. Continue in this way. Use the ascending chain condition to show this process must terminate, and conclude that I is finitely generated.

(e) Show that if R is Noetherian and $I \subseteq R$ is an ideal, then R/I is Noetherian.

(13) **Hilbert's Basis Theorem.** Let R be a Noetherian ring, and let $I \subseteq R[x]$ be an ideal. For $k \in \mathbb{Z}_{\geq 0}$, define

$$J_k := \left\{ a \in R \;\middle|\; \begin{array}{c}\text{there exists } f \in I \\ \mathrm{LT}(f) = ax^k\end{array} \right\} \cup \{0\}.$$

Equivalently, $a \in J_k$ if and only if there exists $f \in I$ with $\mathrm{LT}(f) = ax^k$ (or $a = 0$). Thus J_k is the set of all leading coefficients of polynomials of degree k in I, together with 0. Note that we are viewing

$R[x]$ as a polynomial ring in the single variable x over R. We define the leading term of a polynomial $f = a_k x^k + a_{k-1}x^{k-1} + \cdots + a_1 x + a_0$ (with $a_k \neq 0$) to be $\mathrm{LT}(f) = a_k x^k$.

(a) Show that J_k is an ideal of R for all $k \geq 0$.

(b) Show that the J_k form an ascending chain $J_0 \subseteq J_1 \subseteq J_2 \subseteq \cdots$.

(c) Let $L = \bigcup_{d \geq 0} J_d$. Show that L is an ideal in R.

(d) Prove that there exists $N \in \mathbb{Z}_{\geq 0}$ such that $L = J_k$ for all $k \geq N$. (Hint: Use that R is Noetherian.)

For each $0 \leq k \leq N$, choose generators $a_{k,1}, \ldots, a_{k,r_k}$ for J_k and polynomials $f_{k,i} \in I$ such that $\mathrm{LT}(f_{k,i}) = a_{k,i}x^k$. We will now show that I is equal to

$$M := \left\langle f_{k,i} \;\middle|\; 0 \leq k \leq N,\ 1 \leq i \leq r_k \right\rangle.$$

The inclusion $M \subset I$ is clear. We prove the reverse inclusion $I \subseteq M$ by induction on degree. For the base case suppose $g \in I$ and $\deg(g) = 0$. Since $g \in I$ and $\mathrm{LT}(g) = cx^0 = c$, we have $c \in J_0$. Using that $J_0 = \langle a_{0,1}, \ldots, a_{0,r_0} \rangle$ we may write c as $c = b_i a_{0,} + \cdots b_{r_0} a_{0,r_0}$ with $b_i \in R$. For each i we have $\mathrm{LT}(f_{0,i}) = a_{0,i}x^0 = a_{0,i}$ so $f_{0,i} = a_{0,i}$. Hence $a_{0,i}$ is in M for all i implying $g \in M$ proving the base case.

Now suppose every element of I with degree less than d belongs to M. Let $g \in I$ with $\deg(g) = d$ and $\mathrm{LT}(g) = cx^d$. Our goal is to find $h \in M$ with $\mathrm{LT}(h) = cx^d$, so that $g - h \in I$ has degree less than d and the inductive hypothesis applies. Set $n = \min\{d, N\}$. Since the ascending chain $J_0 \subseteq J_1 \subseteq \cdots$ stabilizes at J_N, we have $J_d = J_n$, so $c \in J_n = \langle a_{n,1}, \ldots, a_{n,r_n} \rangle$. Write

$$c = b_1 a_{n,1} + \cdots + b_{r_n} a_{n,r_n}$$

for some $b_i \in R$, and define

$$h = \sum_{i=1}^{r_n} b_i x^{d-n} f_{n,i}.$$

(Note: if $d \leq N$ then $n = \min\{d, N\} = d$ and the $x^{d-n} = 1$.) Each $f_{n,i} \in M$, so $h \in M$.

(e) Show that $\mathrm{LT}(h) = cx^d$. From this deduce that $g - h \in I$ and either $\deg(g - h) < d$ or $g - h = 0$.

(f) Using the inductive hypothesis conclude that $g - h \in M$, and show this implies $g \in M$. Conclude that $I \subset M$. Explain why this concludes the proof of Hilbert's Basis Theorem.

(g) Prove via induction on n that if R is Noetherian then $R[x_1, \ldots, x_n]$ is Noetherian.

(h) Compare this proof to the proof of Dickson's Lemma we gave above.

This worksheet has introduced two important concepts that will recur throughout the course: the usefulness of Gröbner bases and the importance of finiteness conditions such as Noetherianity. A significant open thread raised by this worksheet is that while we know theoretically that Gröbner bases exist, we do not yet know how to actually compute them. We will return to this question later.

It is fitting that Gröbner bases and Noetherianity are so linked. Hilbert's original proof of his Basis Theorem was non-constructive, even over $\mathbb{K}[x_1,\ldots,x_n]$, which – as the story goes – came as a shock to many mathematicians of the time. Paul Gordan, a prominent algebraist, reportedly remarked, perhaps seriously, perhaps in jest, *"This is not mathematics. This is theology."* (Gordan later gave another proof of the theorem, one which in a sense also proved Dickson's Lemma some 10 years before Dickson.) On the other hand, Gröbner bases have come to serve as the foundational tool of computational commutative algebra, allowing us to compute and make explicit a great many things that Hilbert's methods could not.

WORKSHEET 3.1: THE NULLSTELLENSATZ

Throughout this course, "ring" means *commutative* ring with unity. In this worksheet $\mathbb{K}$ will denote a field. In the previous worksheet you built the order-reversing maps

$$I \longmapsto \mathbb{V}(I) \qquad \text{and} \qquad V \longmapsto \mathbb{I}(V)$$

connecting ideals in $\mathbb{K}[x_1,\dots,x_n]$ with algebraic sets in $\mathbb{K}^n$. Our goal now is to prove that over an algebraically closed field these constructions *almost* recover each other exactly. The only restriction being that since $\mathbb{I}(V)$ is always a radical ideal, these functions are only bijections when restricted to the set of radical ideals in $\mathbb{K}[x_1,\dots,x_n]$. More precisely, our goal is to prove the first two versions of Hilbert's Nullstellensatz under the condition that $\mathbb{K}$ is uncountable (i.e. $\mathbb{C}$). Although our work will also show the second version follows from the first for any algebraically closed field.

Theorem 1 (Hilbert's Nullstellensatz I)**.** *If $\mathbb{K}$ is algebraically closed, then every maximal ideal of $\mathbb{K}[x_1,\dots,x_n]$ is of the form $\langle x_1-a_1,\dots,x_n-a_n\rangle$ for some point $(a_1,\dots,a_n)\in\mathbb{K}^n$.*

Theorem 2 (Hilbert's Nullstellensatz II)**.** *If $\mathbb{K}$ is algebraically closed and $V=\mathbb{V}(\mathscr{S})\subset\mathbb{K}^n$ is an algebraic set, for some subset $\mathscr{S}\subset\mathbb{K}[x_1,\dots,x_n]$, then $\mathbb{I}(V)=\sqrt{\langle\mathscr{S}\rangle}$.*

For a point $p=(a_1,\dots,a_n)\in\mathbb{K}^n$, we will often write $\mathfrak{m}_p$ for the ideal $\langle x_1-a_1,\dots,x_n-a_n\rangle\subset\mathbb{K}[x_1,\dots,x_n]$, which by the previous worksheet is the kernel of the evaluation-at-p map $\mathrm{ev}_p:\mathbb{K}[x_1,\dots,x_n]\to\mathbb{K}$. An alternative statement of Hilbert's Nullstellensatz I is to say there is a bijection between $\mathbb{K}^n$ and maximal ideals in $\mathbb{K}[x_1,\dots,x_n]$ given by $p\mapsto\mathfrak{m}_p$. We say that this statement can fail when $\mathbb{K}$ is not algebraically closed, e.g., $\langle x^2+1\rangle\subset\mathbb{R}[x]$.

Since points are themselves algebraic sets, the second version of Hilbert's Nullstellensatz can be thought of as a generalization of the first. In particular Nullstellensatz II can be thought of as saying that $\mathbb{V}$ and $\mathbb{I}$ are mutual inverses giving bijections

$$\left\{\begin{array}{c}\text{algebraic sets}\\ \text{in } \mathbb{K}^n\end{array}\right\} \underset{\mathbb{V}}{\overset{\mathbb{I}}{\rightleftarrows}} \left\{\begin{array}{c}\text{radical ideals}\\ \text{in } \mathbb{K}[x_1,\dots,x_n]\end{array}\right\}$$

Somewhat amazingly, the entire difficulty of this theorem comes down to only one of the four inclusions needed. In particular, last week we showed that $\mathbb{V}(\mathbb{I}(V))=V$ for any algebraic subset $V\subset\mathbb{K}^n$, regardless of the field. Similarly, we also proved that the inclusion $\sqrt{I}\subseteq\mathbb{I}(\mathbb{V}(I))$ holds over an arbitrary field. Thus, the hard part of the proof of Hilbert's Nullstellensatz II is to show that

$$\mathbb{I}(\mathbb{V}(I))\subseteq\sqrt{I}$$

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for every ideal $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$. This can fail over non-algebraically closed fields, as the example $I = \langle x^2 + 1 \rangle \subset \mathbb{R}[x]$ shows.

The strategy of this worksheet is to isolate that difficulty: first review the geometry–algebra dictionary, and then prove the point Hilbert's Nullstellensatz I for uncountable fields using field theory and a clever cardinality argument. We will then deduce what some call the weak Nullstellensatz:

Theorem 3 (Hilbert's Weak Nullstellensatz)**.** *If $\mathbb{K}$ is algebraically closed and $I \subset \mathbb{K}[x_1, \ldots, x_n]$ is an ideal then $\mathbb{V}(I) = \varnothing$ if and only if $I = \langle 1 \rangle$.*

Note the difficult direction of the weak Nullstellensatz is the forward direction, the reverse is immediate. Intuitively this is saying that a system of polynomial equations has no solution over $\mathbb{K}$ if and only if the "obvious" obstruction – namely you can write 1 in terms of your polynomials – occurs. Finally we will use Rabinowitsch's trick to obtain the full statement. Along the way we will also pause to recall some field theory language, since the key obstruction is the difference between algebraic and transcendental field extensions.

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(1) **Warm-Up: Recalling the Correspondence.** Recall the two order-reversing maps

$$I \longmapsto \mathbb{V}(I) \qquad \text{and} \qquad V \longmapsto \mathbb{I}(V).$$

Before we move to proving the main theorems let us warm up with a few examples.

(a) Give the definitions of $\mathbb{V}(I)$ and $\mathbb{I}(V)$.

(b) If $V = \{p\} \subset \mathbb{K}^n$ for $p = (a_1, \ldots, a_n)$, describe $\mathbb{I}(V)$ both intrinsically and by giving generators.

(c) The ideals $\langle 0 \rangle$ and $\langle 1 \rangle$ are the extreme cases. Describe $\mathbb{V}(\langle 0 \rangle)$ and $\mathbb{V}(\langle 1 \rangle)$ as subsets of $\mathbb{K}^n$.

(d) Over $\mathbb{R}$, describe $\mathbb{V}(\langle x, y \rangle)$ and $\mathbb{V}(\langle x^2 + y^2 \rangle) \subseteq \mathbb{R}^2$.

(e) Over $\mathbb{F}_p$, describe $\mathbb{V}(\langle 0 \rangle)$ and $\mathbb{V}(\langle x^p - x, xy^p - x^p y \rangle) \subseteq \mathbb{F}_p^2$.

(2) **A Parameterized Example.** Consider the map

$$\phi : \overline{\mathbb{F}}_{17}^2 \longrightarrow \overline{\mathbb{F}}_{17}^3, \qquad (u, v) \longmapsto (u^2, uv, v^2).$$

(a) Show that every point in the image of ϕ satisfies the equation $xz - y^2 = 0$.

(b) Let $(a, b, c) \in \overline{\mathbb{F}}_{17}^3$ satisfy $ac = b^2$. Show that if $a \neq 0$, then (a, b, c) lies in the image of ϕ. Handle the case $a = 0$ separately.

(c) Deduce that the image of ϕ is equal to $\mathbb{V}(xz - y^2) \subseteq \overline{\mathbb{F}}_{17}^3$.

(d) Show that $xz - y^2$ is irreducible in $\overline{\mathbb{F}}_{17}[x, y, z]$ and conclude that the corresponding radical ideal is $\langle xz - y^2 \rangle$.

(3) **Reduction to the Core Difficulty.** Recall what you proved last time about the interaction of $\mathbb{V}$ and $\mathbb{I}$ over an arbitrary field.

(a) Show that if $V \subseteq \mathbb{K}^n$ is algebraic, then $\mathbb{V}(\mathbb{I}(V)) = V$.

(b) Show that for every ideal $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$ one has $I \subseteq \sqrt{I} \subseteq \mathbb{I}(\mathbb{V}(I))$.

(c) Show via an example – other than $\langle x^2 + 1 \rangle \subset \mathbb{R}[x]$ – that the inclusion in 3b can be proper when $\mathbb{K}$ is not algebraically closed

(d) Explain why proving Hilbert's Nullstellensatz II reduces to showing that when $\mathbb{K}$ is algebraically closed, $\mathbb{I}(\mathbb{V}(I)) \subset \sqrt{I}$ for every ideal $I \subset \mathbb{K}[x_1, \ldots, x_n]$.

Before the next question, recall some field theory language. If $\mathbb{K}$ is a field and $\mathbb{E}$ is a field containing $\mathbb{K}$, we call $\mathbb{E}/\mathbb{K}$ a *field extension*. In this situation $\mathbb{E}$ is naturally a vector space over $\mathbb{K}$, and we write $[\mathbb{E} : \mathbb{K}] := \dim_{\mathbb{K}} \mathbb{E}$ for its dimension, called the *degree* of the extension. If $[\mathbb{E} : \mathbb{K}]$ is finite we say the extension is *finite*.

Fixing a field extension $\mathbb{E}/\mathbb{K}$, we say an element $\zeta \in \mathbb{E}$ is *algebraic* over $\mathbb{K}$ if there exists a nonzero polynomial $f \in \mathbb{K}[x]$ with $f(\zeta) = 0$; otherwise ζ is *transcendental* over $\mathbb{K}$. The extension $\mathbb{E}/\mathbb{K}$ is called *algebraic* if every element of $\mathbb{E}$ is algebraic over $\mathbb{K}$. Every finite extension is algebraic: if $[\mathbb{E} : \mathbb{K}] = d$, then for any $\zeta \in \mathbb{E}$ the $d + 1$ elements $\zeta^0, \zeta, \ldots, \zeta^d$ are $\mathbb{K}$-linearly dependent, giving a polynomial relation for ζ over $\mathbb{K}$.

More generally, elements $\zeta_1, \ldots, \zeta_r \in \mathbb{E}$ are *algebraically independent* over $\mathbb{K}$ if no nonzero polynomial $f \in \mathbb{K}[x_1, \ldots, x_r]$ satisfies $f(\zeta_1, \ldots, \zeta_r) = 0$. In particular, a single element ζ is transcendental over $\mathbb{K}$ exactly when $\{\zeta\}$ is algebraically independent over $\mathbb{K}$. One helpful characterization of transcendental elements can be phrased in terms of our trusty friend evaluation maps.

Lemma 4. *Let $\mathbb{E}/\mathbb{K}$ be a field extension. An element $\zeta \in \mathbb{E}$ is transcendental over $\mathbb{K}$ if and only if the evaluation-at-ζ homomorphism below is injective:*

$$\begin{array}{ccc} \mathbb{K}[x] & \xrightarrow{\text{ev}_\zeta} & \mathbb{E} \\ f & \longmapsto & f(\zeta) \end{array}.$$

A useful consequence of this lemma is that if $\zeta \in \mathbb{E}$ is transcendental over $\mathbb{K}$, then the map ev_ζ identifies $\mathbb{K}[x]$ with the subring $\mathbb{K}[\zeta] \subseteq \mathbb{E}$. Passing to fields of fractions then identifies the rational function field $\mathbb{K}(x)$ with the subfield $\mathbb{K}(\zeta) \subseteq \mathbb{E}$. You will prove the converse in the exercises as well.

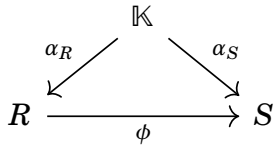
A field $\mathbb{K}$ is *algebraically closed* if every nonconstant polynomial in $\mathbb{K}[x]$ has a root in $\mathbb{K}$. An equivalent formulation is that $\mathbb{K}$ has no nontrivial algebraic extensions. That is, if $\mathbb{E}/\mathbb{K}$ is algebraic, then $\mathbb{E} = \mathbb{K}$. The field $\mathbb{C}$ of complex numbers is the prototypical example.

A $\mathbb{K}$-*algebra* is a ring R equipped with a ring homomorphism $\alpha : \mathbb{K} \to R$. Given such a map, R becomes a $\mathbb{K}$-vector space by defining $\lambda \cdot a := \alpha(\lambda)a$ for $\lambda \in \mathbb{K}$ and $a \in R$. This scalar multiplication is compatible with

the ring multiplication in the sense that

$$\lambda \cdot (ab) = (\lambda \cdot a)b = a(\lambda \cdot b)$$

for all $a, b \in R$ and $\lambda \in \mathbb{K}$. Conversely, any ring R that is simultaneously a $\mathbb{K}$-vector space satisfying the above compatibility condition is a $\mathbb{K}$-algebra; the structure map is $\alpha(\lambda) = \lambda \cdot 1_R$. A *homomorphism of $\mathbb{K}$-algebras* $\phi : R \to S$ is a ring homomorphism that also respects the $\mathbb{K}$-action, i.e. $\phi(\lambda \cdot_R a) = \lambda \cdot_S \phi(a)$ for all $\lambda \in \mathbb{K}$ and $a \in R$. Equivalently, the diagram



commutes, where α_S and α_R are the respective structure maps. In other words, ϕ is a ring homomorphism that is also $\mathbb{K}$-linear. For example, $R = \mathbb{K}[x_1, \ldots, x_n]$ is a $\mathbb{K}$-algebra via the inclusion of constants, and any quotient R/I inherits a $\mathbb{K}$-algebra structure from the surjection $\pi : R \twoheadrightarrow R/I$. If $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$ is an ideal, the quotient $\mathbb{K}[x_1, \ldots, x_n]/I$ is a $\mathbb{K}$-algebra. It is a field if and only if I is maximal. In that case, the quotient map π gives a field extension:

$$\mathbb{K} \longrightarrow \mathbb{K}[x_1, \ldots, x_n] \xrightarrow{\pi} \mathbb{K}[x_1, \ldots, x_n]/I$$

gives a field extension of $\mathbb{K}$. Every element of this quotient is a $\mathbb{K}$-linear combination of the images of monomials $x_1^{a_1} \cdots x_n^{a_n}$, so properties of this extension (such as its dimension over $\mathbb{K}$) are controlled by the combinatorics of monomials together with the relations imposed by I.

The problem below uses all of these ideas to prove a beautiful dictionary between algebra and geometry: when $\mathbb{K}$ is both uncountable and algebraically closed (e.g. $\mathbb{K} = \mathbb{C}$), the maximal ideals of $\mathbb{K}[x_1, \ldots, x_n]$ are in natural bijection with the points of $\mathbb{K}^n$.

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(4) **Hilbert's Nullstellensatz I ($\mathbb{K}$ uncountable).** Let $R := \mathbb{K}[x_1, \ldots, x_n]$ and $\mathfrak{m} \subset R$ be a maximal ideal. Write $L := R/\mathfrak{m}$ for the residue field, which contains $\mathbb{K}$ via the natural quotient map. The goal is to show that if $\mathbb{K}$ is uncountable and algebraically closed, then in fact $L \cong \mathbb{K}$.

(a) Show that the monomials in R form a countable $\mathbb{K}$-basis of R.

(b) Deduce that L has at most countable dimension as a vector space over $\mathbb{K}$.

(c) Let $\mathbb{E}/\mathbb{K}$ be a field extension and let $\zeta \in \mathbb{E}$. Explain why the following are equivalent:

(i) ζ is transcendental over $\mathbb{K}$;

(ii) $\{\zeta\}$ is algebraically independent over $\mathbb{K}$;

(iii) the evaluation map $\mathbb{K}[x] \to \mathbb{E}$ sending $f(x)$ to $f(\zeta)$ is injective.

In particular, if a field extension of $\mathbb{K}$ is not algebraic, then it contains a transcendental element.

(d) Let $\mathbb{K}$ be any infinite field, and let $\zeta \in \mathbb{E}$ be transcendental over $\mathbb{K}$ for some field extension $\mathbb{E}/\mathbb{K}$. Prove that the set of elements in $\mathbb{E}$:

$$\left\{ \frac{1}{\zeta - \lambda} \;\middle|\; \lambda \in \mathbb{K} \right\} \subseteq \mathbb{E}$$

is linearly independent over $\mathbb{K}$. (Hint: Suppose $\sum_{i=1}^{m} \frac{c_i}{\zeta - \lambda_i} = 0$ for distinct $\lambda_1, \ldots, \lambda_m \in \mathbb{K}$ and coefficients $c_1, \ldots, c_m \in \mathbb{K}$. Multiply both sides by $\prod_{j=1}^{m}(\zeta - \lambda_j)$ to obtain $\sum_{i=1}^{m} c_i \prod_{j \neq i}(\zeta - \lambda_j) = 0$. Now define $P(x) = \sum_{i=1}^{m} c_i \prod_{j \neq i}(x - \lambda_j) \in \mathbb{K}[x]$. Then $P(\zeta) = 0$. Since ζ is transcendental over $\mathbb{K}$, conclude that $P(x)$ must be the zero polynomial. Evaluate $P(x)$ at $x = \lambda_k$ for each k to show that $c_k = 0$.)

(e) Now assume that $\mathbb{K}$ is uncountable. Using part 4d, show that if a field extension $\mathbb{E}/\mathbb{K}$ contains a transcendental element, then $[\mathbb{E} : \mathbb{K}]$ is uncountable.

(f) Assume $\mathbb{K}$ is uncountable. Show if $\mathfrak{m} \subseteq R$ is a maximal ideal, then the extension $\mathbb{K} \hookrightarrow R/\mathfrak{m} := L$ is algebraic.

(g) Now assume that $\mathbb{K}$ is both uncountable and algebraically closed. Show that every algebraic extension of $\mathbb{K}$ is trivial, so $L \cong \mathbb{K}$ as $\mathbb{K}$-algebras.

(h) With the notation from part 4g let $\psi : R \to \mathbb{K}$ be the ring map given by the composition

$$R \xrightarrow{\pi} R/\mathfrak{m} = L \xrightarrow{\sim} \mathbb{K}$$

where π is the quotient map and the second map is the identification of L with $\mathbb{K}$ from part 4g. Note $\ker(\psi) = \mathfrak{m}$. Define $a_i := \psi(x_i) \in \mathbb{K}$ for $i = 1, \ldots, n$. Prove that $\mathfrak{m} = \langle x_1 - a_1, \ldots, x_n - a_n \rangle$.

(i) Deduce that when $\mathbb{K}$ is uncountable and algebraically closed, the assignment $p \mapsto \mathfrak{m}_p$ gives a bijection between points of $\mathbb{K}^n$ and maximal ideals of $\mathbb{K}[x_1, \ldots, x_n]$.

We have now proven Hilbert's Nullstellensatz I under the additional assumption that $\mathbb{K}$ is uncountable, using virtually only field theory! Note this applies to $\mathbb{C}$ and $\mathbb{C}_p$ as these are both algebraically closed and uncountable, but does not apply to $\overline{\mathbb{Q}}$ and $\overline{\mathbb{F}_p}$ which while algebraically closed are countable.

In what follows we will *only* assume $\mathbb{K}$ is algebraically closed, and will use Hilbert's Nullstellensatz I to deduce the weak Nullstellensatz. After that we will use what has come to be known as *Rabinowitsch's trick* to upgrade the weak Nullstellensatz to the full statement of Hilbert's Nullstellensatz II.

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(5) **Weak Nullstellensatz.** Let $\mathbb{K}$ be algebraically closed and let $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$ be an ideal.

(a) Show that if I is proper, then I is contained in some maximal ideal $\mathfrak{m}$. (Hint: Use Zorn's Lemma.)

(b) Assuming Hilbert's Nullstellensatz I, show that if I is proper then $\mathbb{V}(I) \neq \varnothing$. (Hint: $\mathbb{V}(\mathfrak{m}) \subset \mathbb{V}(I)$.)

(c) Conclude that $\mathbb{V}(I) = \varnothing$ if and only if $I = \langle 1 \rangle$.

(6) **Hilbert's Nullstellensatz II. (Rabinowitsch's Trick)** Let $\mathbb{K}$ be algebraically closed and $R := \mathbb{K}[x_1,\dots,x_n]$. Let $I = \langle f_1,\dots,f_t\rangle \subset R$ be an ideal. Choose any element $g \in \mathbb{I}(\mathbb{V}(I))$, we will show that some power of g lies in I. Note if $g = 0$ there is nothing to prove as $0 \in I$. Thus, we assume g is non-zero.

(a) In the larger polynomial ring $R[z]$, define $J := \langle f_1,\dots,f_t, 1-gz\rangle$. Show that $\mathbb{V}(J) = \varnothing$ without appealing to the weak Nullstellensatz. (Hint: If $(a_1,\dots,a_{n+1}) \in \mathbb{V}(J)$ then what do we know about $g(a_1,\dots,a_n)$ since $g \in \mathbb{I}(\mathbb{V}(I))$.)

(b) Use Question 5 to conclude that $J = \langle 1\rangle \subset R[z]$. Deduce there exist polynomials $h_1,\dots,h_t,h_{t+1} \in R[z]$ such that the following relation holds.

$$1 = h_1f_1 + \cdots + h_tf_t + h_{t+1}(1-gz) \in R[z].$$

(c) Consider the ring $R[1/g]$ consisting of expressions of the form p/g^k where $p \in R$ and $k \geq 0$, in other words, we enlarge R by formally allowing division by powers of g. Consider the map of $\mathbb{K}$-algebras:

$$\begin{array}{ccc} R[z] & \xrightarrow{\psi} & R\left[\frac{1}{g}\right] \\ z & \longmapsto & \frac{1}{g} \end{array}$$

Explain why ψ is a well-defined $\mathbb{K}$-algebra homomorphism. (Hint: What does it take to define a $\mathbb{K}$-algebra map out of a polynomial ring $R[z]$?)

(d) Prove that $\psi(1-gz) = 0$ and deduce that $1 = \psi(h_1)f_1 + \cdots + \psi(h_t)f_t$ in $R[1/g]$.

(e) Explain why there exists $k \in \mathbb{N}$ such that $g^k\psi(h_i) \in R$ for every $i = 1,\dots,t$. Multiply the equation in part 6d by g^k to conclude that $g^k \in I$. (Hint: Each $\psi(h_i)$ is an element of $R[1/g]$, so it can be written as p_i/g^{k_i} for some $p_i \in R$ and $k_i \geq 0$.)

(f) Conclude that $g \in \sqrt{I}$, and hence $\mathbb{I}(\mathbb{V}(I)) \subseteq \sqrt{I}$.

(g) Combine this with Question 3 to finish the proof of Hilbert's Nullstellensatz II assuming Hilbert's Nullstellensatz I. (The latter we have proven over every uncountable algebraically closed field, in particular over $\mathbb{C}$.)

You have now proved Hilbert's Nullstellensatz for every *uncountable* algebraically closed field, including $\mathbb{C}$. The only remaining gap in the proof over an arbitrary algebraically closed field is the point case: one still needs to show that every maximal ideal is of the form $\mathfrak{m}_p$ without assuming uncountability. We will return to that later, after developing more tools.

For the remainder of the worksheet, assume Hilbert's Nullstellensatz I and II over all algebraically closed fields and use them freely. The point now is to unpack what the theorem says about coordinate rings, closed subsets, irreducibility, and functoriality. In particular, we would like a version of the Nullstellensatz that describes algebraic subsets not of just $\mathbb{K}^n$, but of any algebraic set $V \subset \mathbb{K}^n$. Towards this we will define

and study relative versions of the $\mathbb{I}$ and $\mathbb{V}$ operators. In particular, for an algebraic subset $W \subset V$ and an ideal $J \subset \mathbb{K}[V]$ define

$$\mathbb{I}_V(W) := \{\phi \in \mathbb{K}[V] \mid \phi(p) = 0 \text{ for all } p \in W\}. \qquad \text{and} \qquad \mathbb{V}_V(J) := \{p \in V \mid \phi(p) = 0 \text{ for all } \phi \in J\}.$$

Using these we how to prove the following relative version of the Nullstellensatz connecting algebraic subsets of V and radical ideals in $\mathbb{K}[V]$.

Theorem 5 (Hilbert's Nullstellensatz III)**.** *Let $\mathbb{K}$ be an algebraically closed field. If $V \subset \mathbb{K}^n$ is an algebraic subset then the maps $\mathbb{I}_V$ and $\mathbb{V}_V$ induce inverse, inclusion-reversing bijections:*

$$\left\{\begin{array}{c}\textit{algebraic subsets}\\ \textit{in } V\end{array}\right\} \underset{\mathbb{V}_V}{\overset{\mathbb{I}_V}{\rightleftarrows}} \left\{\begin{array}{c}\textit{radical ideals}\\ \textit{in } \mathbb{K}[V]\end{array}\right\}$$

Note that since the coordinate ring of $\mathbb{K}^n$ is canonically isomorphic to $\mathbb{K}[x_1, \ldots, x_n]$ we could view Hilbert's Nullstellensatz II as a special case of this third version. However, the real difficulty in proving the third version boils down to applying the second. As you will see many of the following exercises boil down to the same ideas and arguments we used for $\mathbb{I}$ and $\mathbb{V}$, at times almost verbatim.

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(7) **Properties of $\mathbb{I}_V$.** Assume $\mathbb{K}$ is algebraically closed, let $V \subseteq \mathbb{K}^n$ be an algebraic set with coordinate ring $\mathbb{K}[V]$. Let $W \subseteq V$ be an algebraic subset and $\rho_W : \mathbb{K}[V] \to \mathbb{K}[W]$ be the restriction homomorphism taking a function $\phi : V \to \mathbb{K}$ and restricting it to W to get a function $\phi|_W : W \to \mathbb{K}$. (If we wanted to feel fancy we could say $\phi|_W$ is the composition of $\phi : V \to \mathbb{K}$ with the inclusion $\iota : W \to V$, but in this setting that is needlessly complex.)

(a) Show that $\mathbb{I}_V(W)$ is an ideal of $\mathbb{K}[V]$, and if $Z \subset W \subset V$ then $\mathbb{I}_V(W) \subset \mathbb{I}_V(W)$.

(b) Since V is an algebraic subset of itself, as a reality check what is $\mathbb{I}_V(V)$?

(c) Prove that ρ_W is a surjective $\mathbb{K}$-algebra homomorphism with kernel $\mathbb{I}_V(W)$.

(d) Prove that ρ_W induces an $\mathbb{K}$-algebra isomorphism $\mathbb{K}[V]/\mathbb{I}_V(W) \cong \mathbb{K}[W]$.

(e) Show that if $W \subset V$ is an algebraic subset then the following diagram commutes:

$$\begin{array}{ccc} \mathbb{K}[x_1, \ldots, x_n] & \xrightarrow{\rho} & \mathbb{K}[V] \\ & \searrow^{\rho'} & \downarrow{\scriptstyle \rho_W} \\ & & \mathbb{K}[W] \end{array}$$

where $\rho' : \mathbb{K}[x_1, \ldots, x_n] \to \mathbb{K}[W]$ is the restriction of functions map defining $\mathbb{K}[W]$. Deduce that $\mathbb{I}_V(W) = \mathbb{I}(W)/\mathbb{I}(V)$.

(f) Conclude that if $W \subset V$ is an algebraic subset then $\mathbb{I}_V(W)$ is a radical ideal of $\mathbb{K}[V]$.

(8) **Properties of $\mathbb{V}_V$.** Let $\mathbb{K}$ be an algebraically closed field, and let $V \subseteq \mathbb{K}^n$ be an algebraic set with coordinate ring $\mathbb{K}[V]$.

(a) Let $J \subset \mathbb{K}[V]$ be an ideal. Show that $\mathbb{V}_V(J)$ is an algebraic subset of V by showing that $\mathbb{V}_V(J) = \mathbb{V}(\rho^{-1}(J)) \subseteq \mathbb{K}^n$. (Hint: Since $\ker(\rho) = \mathbb{I}(V) \subseteq \rho^{-1}(J)$, every point of $\mathbb{V}(\rho^{-1}(J))$ already lies in V.)

(b) Again exploring the extremes what algebraic subsets of V are $\mathbb{V}_V(\langle 1 \rangle)$ and $\mathbb{V}_V(\langle 0 \rangle)$?

(c) Show that if $I \subset J$ are ideals in $\mathbb{K}[V]$ then $\mathbb{V}_V(J) \subset \mathbb{V}_V(I)$.

(9) **Proving Hilbert's Nullstellensatz III.** Assume $\mathbb{K}$ is algebraically closed, let $V \subseteq \mathbb{K}^n$ be an algebraic set with coordinate ring $\mathbb{K}[V]$.

(a) Use Hilbert's Nullstellensatz in $\mathbb{K}[x_1, \ldots, x_n]$ to prove that $\mathbb{I}_V(\mathbb{V}_V(J)) = \sqrt{J}$ (Hint: First identify $\mathbb{I}_V(\mathbb{V}_V(J))$ with $\mathbb{I}(\mathbb{V}(\rho^{-1}(J)))/\mathbb{I}(V)$, then use $\mathbb{I}(\mathbb{V}(-)) = \sqrt{-}$.)

(b) Show that if $W \subseteq V$ is an algebraic subset, then $\mathbb{V}_V(\mathbb{I}_V(W)) = W$.

(c) Conclude that the assignments $W \longmapsto \mathbb{I}_V(W)$ and $J \mapsto \mathbb{V}_V(J)$ induce inverse, inclusion-reversing bijections between algebraic subsets of V and radical ideals in $\mathbb{K}[V]$.

(d) Prove that under the bijections in the previous parts the maximal ideals of $\mathbb{K}[V]$ correspond to the points of V, thought of as algebraic subsets. Explain why we should think of this as a relative version of Hilbert's Nullstellensatz I.

(10) **Irreducibility and Prime Ideals.** An algebraic set $V \subset \mathbb{K}^n$ is *irreducible* if there do not exist two proper algebraic subsets $W_1, W_2 \subsetneq V$ such that $V = W_1 \cup W_2$. Equivalently, V is not the union of two proper algebraic subsets. An algebraic set that is not irreducible is said to be *reducible*.

(a) Which of the following algebraic sets in $\mathbb{K}^n$ (with $\mathbb{K}$ algebraically closed) do you think are irreducible. Draw pictures when possible.

(i) $V = \mathbb{V}(x^2 - y^2) \subset \mathbb{C}^2$.

(ii) $V = \mathbb{V}(y - x^3) \subset \mathbb{C}^2$.

(iii) $V = \mathbb{V}(xz, yz) \subset \mathbb{C}^3$.

(iv) $V = \mathbb{V}(x^2 + y^2 - 1) \subset \mathbb{C}^2$.

(v) $V = \mathbb{V}(xy - z^2) \subset \mathbb{C}^3$.

(vi) $V = \mathbb{V}(3x^2 - 2xy) \subset \mathbb{C}^2$.

(b) Assume $\mathbb{I}(V)$ is a prime ideal. Show that V is irreducible. (Hint: Suppose for contradiction that $V = W_1 \cup W_2$ with $W_1, W_2 \subsetneq V$. Since $W_i \subsetneq V$, there exist $f_i \in \mathbb{I}(W_i) \setminus \mathbb{I}(V)$. What can you say about the product $f_1 f_2$?)

(c) Show that if $fg \in \mathbb{I}(V)$, then $V \subseteq \mathbb{V}(g) \cup \mathbb{V}(f)$. (Hint: Use that $\mathbb{V}$ is inclusion reversing, together with some facts you proved when checking the Zariski topology is a topology.)

(d) Prove that if V is irreducible then $\mathbb{I}(V)$ is prime. (Hint: Given $fg \in \mathbb{I}(V)$ we must show that $f \in \mathbb{I}(V)$ or $g \in \mathbb{I}(V)$. By 10c if $fg \in \mathbb{I}(V)$ then $V \subset \mathbb{V}(f) \cup \mathbb{V}(g)$ what does V being irreducible imply here?)

(e) Conclude that an algebraic set V is irreducible if and only if $\mathbb{I}(V)$ is prime. State a bijection between irreducible algebraic sets and prime ideals. Why do we not need to say radical?

(11) **Algebraic Sets and Spec.** Continue to assume that $\mathbb{K}$ is an algebraically closed field and that $V \subset \mathbb{K}^n$ is an algebraic subset with coordinate ring $\mathbb{K}[V]$. In this question we will prove an inclusion-reversing bijection between $\mathrm{Spec}(\mathbb{K}[V])$ and the irreducible algebraic subsets of V. Keep the notation from the previous problem, and use the Hilbert's Nullstellensatz III from Questions 7-9.

(12) Show that the correspondence in the previous part induces a homeomorphism between $\mathrm{mSpec}(\mathbb{K}[V])$ (with the subspace Zariski topology inherited from $\mathrm{Spec}(\mathbb{K}[V])$) and V (with its Zariski topology).

(a) If $W_1, W_2 \subseteq V$ are algebraic, show that $\mathbb{I}_V(W_1 \cup W_2) = \mathbb{I}_V(W_1) \cap \mathbb{I}_V(W_2)$. (Note: You proved the same holds of $\mathbb{I}$ when you checked the Zariski topology on algebraic sets is well-defined.)

(b) Let $W \subseteq V$ be an algebraic subset. Show that if $\mathbb{I}_V(W)$ is prime, then W is irreducible.

(c) Let $W \subseteq V$ be an algebraic subset. Prove that given $f, g \in \mathbb{K}[V]$, if $fg \in \mathbb{I}_V(W)$, then $W \subseteq \mathbb{V}_V(f) \cup \mathbb{V}_V(g)$. (This should feel very familiar to Question 10, except that now we are working with $\mathbb{V}_V$ instead of $\mathbb{V}$.)

(d) Prove that if W is irreducible, then $\mathbb{I}_V(W)$ is a prime ideal of $\mathbb{K}[V]$.

(e) Deduce that the assignment $[P] \mapsto \mathbb{V}_V(P)$ induces an inclusion reversing bijection between $\mathrm{Spec}(\mathbb{K}[V])$ and the set of irreducible algebraic subsets of V.

(f) Show that the correspondence in the previous part induces a homeomorphism between $\mathrm{mSpec}(\mathbb{K}[V])$, with the subspace Zariski topology inherited from $\mathrm{Spec}(\mathbb{K}[V])$, and V with its Zariski topology.

(13) **Functor-of-Points Viewpoint.** Let $\mathbb{K}$ be an arbitrary field, and let $I \subset \mathbb{K}[x_1, \ldots, x_n]$ be an ideal. For a field extension $\mathbb{E}/\mathbb{K}$ define:

$$\mathcal{V}_I(\mathbb{E}) := \left\{(a_1, \ldots, a_n) \in \mathbb{E}^n \;\middle|\; f(a_1, \ldots, a_n) = 0 \text{ for all } f \in I\right\} \subset \mathbb{E}^n$$

where we are viewing the elements of I as elements of $\mathbb{E}[x_1, \ldots, x_n]$ via the natural inclusion, $\mathbb{K}[x_1, \ldots, x_n] \subset \mathbb{E}[x_1, \ldots, x_n]$. Extend $\mathcal{V}_I$ to a function of sets by defining

$$\left\{\begin{matrix}\text{field extensions} \\ \text{of } \mathbb{K}\end{matrix}\right\} \xrightarrow{\quad \mathcal{V}_I \quad} \textbf{Set}$$

$$\mathbb{E} \longmapsto \mathcal{V}_I(\mathbb{E}).$$

(a) For the ideal $I = \langle x^2 + y^2 \rangle \subseteq \mathbb{Q}[x, y]$, describe the sets $\mathcal{V}_I(\mathbb{Q})$, $\mathcal{V}_I(\mathbb{R})$, and $\mathcal{V}_I(\mathbb{C})$.

(b) For the ideal $J = \langle x^2 + 1 \rangle \subseteq \mathbb{Q}[x]$, describe the sets $\mathcal{V}_J(\mathbb{Q})$, $\mathcal{V}_J(\mathbb{R})$, and $\mathcal{V}_J(\mathbb{C})$.

(c) Let $\iota : \mathbb{E} \hookrightarrow \mathbb{F}$ be a $\mathbb{K}$-embedding of field extensions, i.e., a field homomorphism that fixes $\mathbb{K}$ pointwise. Show that ι induces a well-defined map of sets

$$\iota_* : \mathcal{V}_I(\mathbb{E}) \longrightarrow \mathcal{V}_I(\mathbb{F}), \qquad (a_1, \ldots, a_n) \longmapsto (\iota(a_1), \ldots, \iota(a_n)).$$

(Hint: If $f \in \mathbb{K}[x_1, \ldots, x_n]$ and $f(a_1, \ldots, a_n) = 0$, why is $f(\iota(a_1), \ldots, \iota(a_n)) = 0$? Use the fact that ι fixes the coefficients of f.)

(d) Show that $\mathcal{V}_I$ is a *functor* from the category of field extensions of $\mathbb{K}$ (with $\mathbb{K}$-embeddings as morphisms) to the category of sets.

(e) Revisit part 13a; the inclusion $\mathbb{R} \hookrightarrow \mathbb{C}$ gives a map $\mathcal{V}_I(\mathbb{R}) \to \mathcal{V}_I(\mathbb{C})$. Describe this map explicitly. Is it injective? Is it surjective?

(f) Generalize the map $\mathcal{V}_I$ by enlarging the source from the category of field extensions to the category of $\mathbb{K}$-algebra, $\mathcal{V}_I : \mathbb{K} - \mathbf{alg} \to \mathbf{Set}$ Prove that your generalization remains a functor.

The ideas presenedt in Question 13 is one standard description of the affine scheme of finite type cut out by I. If these words do not mean anything to you, imagine a geometric space similar to algebraic set $\mathbb{V}(I)$, but somehow remembering the ideal I itself not just $\sqrt{I}$. This viewpoint is often called the *functor of points* because we think of I as remembering not only its $\mathbb{K}$-valued solutions, but also its solutions in every test $\mathbb{K}$-algebra. That functorial viewpoint is what survives even when $\mathbb{K}$ is not algebraically closed, and it is one of the main bridges from classical algebraic sets to affine schemes. However, much of this jump from algebraic sets to schemes is beyond the scope of this course.

WORKSHEET 4.1: MODULES PT. I

Throughout this course, "ring" means *commutative* ring with unity. In this worksheet $\mathbb{K}$ will denote a field. A *R-module* is an abelian group $(M,+)$ together with a scalar multiplication $R\times M\to M$, $(r,m)\mapsto rm$, satisfying the following:

(i) Associativity: $(rs)m = r(sm)$ for all $r,s\in R$ and $m\in M$,

(ii) Distribution over R: $(r+s)m = rm+sm$ for all $r,s\in R$ and $m\in M$,

(iii) Distribution over M: $r(m+n) = rm+rn$ for all $r\in R$ and $m,n\in M$, and

(iv) Identity: $1_R m = m$ for all $m\in M$.

Since M is an abelian group, it has a distinguished additive identity, which we normally denote by 0 or by 0_M if we need to distinguish it from 0_R. When the base ring is clear from context, we might simply say that M is a module, or we might say that M is a module over R. As we will see, modules are the analogue of vector spaces in which the scalars come from a ring instead of a field. They include ideals, quotient rings, abelian groups, vector spaces, and many other naturally occurring algebraic objects.

If M is an R-module, a subset $N\subseteq M$ is an *R-submodule* of M if and only if $rm+sn\in N$ for all $r,s\in R$ and $m,n\in N$. That is, a submodule is a subgroup of M such that the scalar multiplication of R on M restricts to a scalar multiplication on N.

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(1) **Basic Identities.** Let R be a ring and M an R-module. Prove the following identities.

(a) $r0_M = 0_M$ for all $r\in R$.

(b) $0_R m = 0_M$ for all $m\in M$.

(c) $(-r)m = r(-m)$ for all $r\in R$ and all $m\in M$.

(2) **Basic Examples of Modules.**

(a) If $\mathbb{K}$ is a field, explain why a $\mathbb{K}$-module is the same as a $\mathbb{K}$-vector space.

(b) Explain why R is itself an R-module in a natural way.

(c) Show that if $I\subseteq R$ is an ideal, then the multiplication on R naturally makes I into an R-module.

(d) Prove that if $I\subseteq R$ is an ideal, then R/I is an R-module.

(e) Prove that a $\mathbb{Z}$-module is the same thing as an abelian group.

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(f) If $I \subseteq R$ is an ideal and M is an R-module, show that

$$IM := \left\{ \sum_{i=1}^{n} r_i m_i \;\middle|\; n \geq 0,\ r_i \in I,\ m_i \in M \right\}$$

is a submodule of M.

(3) **Submodules Generated by Elements.** Let M be an R-module. Given elements $m_1, \ldots, m_t \in M$, the submodule of M generated by $m_1, \ldots, m_t$ is the set

$$Rm_1 + Rm_2 + \cdots + Rm_t := \{r_1 m_1 + \cdots + r_t m_t \mid r_1, \ldots, r_t \in R\} \subseteq M.$$

We say that $m_1, \ldots, m_t$ are *generators* for M if $Rm_1 + Rm_2 + \cdots + Rm_t = M$. A submodule $N \subseteq M$ is a *cyclic submodule* if $N = Rm$ for some $m \in M$.

(a) Prove that $Rm_1 + Rm_2 + \cdots + Rm_t$ is an R-submodule of M.

(b) Prove that $Rm_1 + Rm_2 + \cdots + Rm_t$ is the smallest R-submodule of M containing $m_1, \ldots, m_t$.

(c) Prove that a ring R is itself a cyclic R-module by finding a generator.

(d) In the $\mathbb{Z}$-module $\mathbb{Z}/12\mathbb{Z}$, compute the submodules generated by $\overline{8}$ and $\overline{6}$.

(e) Let $R = \mathbb{K}[x]$ and $M = R/\langle x^3 \rangle$. Describe the submodules generated by $\overline{1}$, $\overline{x}$, and $\overline{x}^2$.

A *homomorphism* of R-modules is a group homomorphism $\phi : M \to N$ such that $\phi(rm) = r\phi(m)$ for all $r \in R$ and $m \in M$. Two R-modules M and N are *isomorphic* if and only if there exist R-module homomorphisms $\phi : M \to N$ and $\psi : N \to M$ such that $\phi \circ \psi = \mathrm{Id}_N$ and $\psi \circ \phi = \mathrm{Id}_M$. Equivalently, there is a bijective homomorphism of R-modules $\phi : M \to N$. We call such maps *isomorphisms* of R-modules. If $\phi : M \to N$ is a homomorphism of R-modules, there are two natural submodules associated to it: its *kernel* and its *image*:

$$\ker(\phi) := \{m \in M \mid \phi(m) = 0_N\} \subseteq M \qquad \text{and} \qquad \mathrm{img}(\phi) := \{\phi(m) \mid m \in M\} \subseteq N.$$

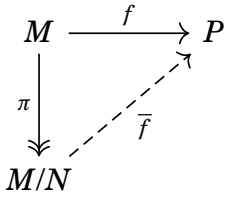
One should readily check that both the kernel and image of a module homomorphism are submodules. As with linear maps, group homomorphisms, and ring homomorphisms, the kernel measures the failure of injectivity and the image measures the failure of surjectivity.

Given a submodule $N \subseteq M$, the quotient abelian group M/N inherits an R-module structure via

$$r(m + N) := rm + N.$$

Well-definedness uses that N is a submodule, not merely a subgroup. The projection $\pi : M \twoheadrightarrow M/N$ is a homomorphism with $\ker(\pi) = N$. The quotient M/N is the module obtained by forcing $N = 0$. Precisely: if

$f\colon M \to P$ is a homomorphism with $N \subseteq \ker(f)$, there is a unique $\overline{f}\colon M/N \to P$ with $f = \overline{f} \circ \pi$:



Phrased differently, there is a natural isomorphism

$$\mathrm{Hom}_R(M/N, P) \cong \{f \in \mathrm{Hom}_R(M,P) \mid N \subseteq \ker(f)\};$$

That is, homomorphisms out of M/N are exactly homomorphisms out of M that kill N. This is usually summarized as the First Isomorphism Theorem.

Theorem 1 (First Isomorphism Theorem)**.** *Every homomorphism $\phi\colon M \to N$ of R-modules induces an isomorphism*

$$M/\ker(\phi) \xrightarrow{\sim} \mathrm{img}(\phi), \quad m + \ker(\phi) \longmapsto \phi(m).$$

In other words, quotienting by the kernel accounts for all the redundancy: every surjection is a quotient map.

Theorem 2 (Third Isomorphism Theorem)**.** *Let $N \subseteq M$ be a submodule and $\pi\colon M \twoheadrightarrow M/N$ the projection. There is an inclusion-preserving bijection*

$$\{\textit{submodules of } M \textit{ containing } N\} \xleftrightarrow{1:1} \{\textit{submodules of } M/N\}$$

given by $L \mapsto L/N = \pi(L)$ with inverse $\overline{L} \mapsto \pi^{-1}(\overline{L})$. Moreover, for $N \subseteq L \subseteq M$:

$$M/L \;\cong\; (M/N)\big/(L/N).$$

The correspondence says that the submodule lattice of M/N is the interval $[N, M]$ in the submodule lattice of M. The supplementary isomorphism $(M/N)/(L/N) \cong M/L$ is sometimes called the *Third Isomorphism Theorem*: quotienting in two stages is the same as quotienting all at once.

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(4) **Basic Examples of Module Homomorphisms.**

(a) If $\mathbb{K}$ is a field, explain why a homomorphism of $\mathbb{K}$-modules is the same as a $\mathbb{K}$-linear map.

(b) Explain why a homomorphism of $\mathbb{Z}$-modules is the same as a homomorphism of abelian groups.

(5) **Cyclic Modules & Annihilators.** Let M be an R-module. The *annihilator* of an element $m \in M$ is $\mathrm{Ann}_R(m) \coloneqq \{r \in R \mid rm = 0\}$.

(a) Prove that $\mathrm{Ann}_R(m)$ is an ideal of R.

(b) What is $\mathrm{Ann}_R(0_M)$?

(c) Consider the multiplication by m function:

$$\phi_m : R \to M \qquad \text{given by} \qquad r \mapsto rm.$$

Prove that ϕ_m is a homomorphism of R-modules.

(d) Show that $\ker(\phi_m) = \mathrm{Ann}_R(m)$ and $\mathrm{img}(\phi_m) = Rm$.

(e) Prove that a submodule $N \subseteq M$ is cyclic if and only if $N \cong R/I$ for some ideal $I \subseteq R$.

Let $\{M_i\}_{i\in I}$ be a family of R-modules indexed by a set I. The *direct product* is

$$\prod_{i\in I} M_i = \{(m_i)_{i\in I} \mid m_i \in M_i\},$$

with componentwise addition and scalar multiplication. The *direct sum* is the submodule

$$\bigoplus_{i\in I} M_i = \left\{ (m_i)_{i\in I} \in \prod_{i\in I} M_i \mid m_i = 0 \text{ for all but finitely many } i \right\}.$$

For each $j \in I$ there is a projection map π_j and an inclusion map ι_j:

$$\pi_j : \prod_{i\in I} M_i \to M_j, \quad (m_i)_{i\in I} \longmapsto m_j, \qquad \text{and} \qquad \iota_j : M_j \to \bigoplus_{i\in I} M_i, \quad m \longmapsto (0,\ldots,0,m,0,\ldots).$$

The product is characterized by maps to the factors, while the direct sum is characterized by maps from the summands. These are the universal properties that make products and sums so useful.

---

(6) **Direct Sums and Products of Modules.** Let $\{M_i\}_{i\in I}$ be a family of R-modules indexed by a set I.

(a) Prove that both $\prod_{i\in I} M_i$ and $\bigoplus_{i\in I} M_i$ are R-modules.

(b) Explain why, if I is finite, $\prod_{i\in I} M_i = \bigoplus_{i\in I} M_i$.

(c) Consider the $\mathbb{Z}$-modules $M_n = \mathbb{Z}/n\mathbb{Z}$ for $n \geq 1$. Describe explicitly an element of $\prod_{n=1}^{\infty} M_n$ that is not an element of $\bigoplus_{n=1}^{\infty} M_n$.

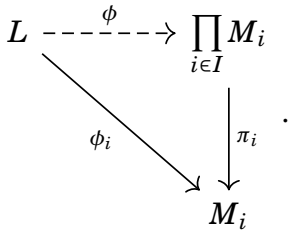
(d) Let $M_i = R$ for all $i \in \mathbb{N}$. Show that the submodule of $\prod_{i\in\mathbb{N}} R$ generated by the standard basis vectors $e_1, e_2, e_3, \ldots$ is $\bigoplus_{i\in\mathbb{N}} R$, not the whole product, provided $R \neq 0$.

(e) For two modules M and N, write down the inclusion maps ι_M, ι_N and projection maps π_M, π_N for $M \oplus N$. Verify the following identities

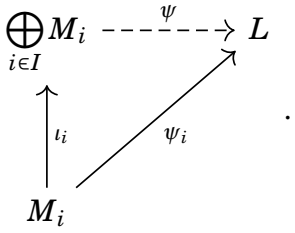
$$\pi_M \circ \iota_M = \mathrm{Id}_M, \quad \pi_N \circ \iota_N = \mathrm{Id}_N, \quad \pi_M \circ \iota_N = 0, \quad \pi_N \circ \iota_M = 0, \quad \text{and} \quad \iota_M \circ \pi_M + \iota_N \circ \pi_N = \mathrm{Id}_{M\oplus N}.$$

(7) **Universal Properties of Products and Sums.** Let $\{M_i\}_{i\in I}$ be a family of R-modules indexed by I.

(a) Let L be an R-module and suppose that for every $i \in I$ we are given a homomorphism $\phi_i : L \to M_i$. Prove that there exists a unique homomorphism $\phi : L \to \prod_{i \in I} M_i$ such that the following diagram commutes for all $i \in I$:



(b) Let L be an R-module and suppose that for every $i \in I$ we are given a homomorphism $\psi_i : M_i \to L$. Prove that there exists a unique homomorphism $\psi : \bigoplus_{i \in I} M_i \to L$ such that the following diagram commutes for all $i \in I$:



(c) Deduce natural isomorphisms of R-modules

$$\mathrm{Hom}_R\left(L, \prod_{i \in I} M_i\right) \cong \prod_{i \in I} \mathrm{Hom}_R(L, M_i) \qquad \text{and} \qquad \mathrm{Hom}_R\left(\bigoplus_{i \in I} M_i, L\right) \cong \prod_{i \in I} \mathrm{Hom}_R(M_i, L).$$

(d) Explain why a family of maps $\psi_i : M_i \to L$ does not generally define a map $\prod_{i \in I} M_i \to L$. Use the case $M_i = L = \mathbb{Z}$ and $\psi_i = \mathrm{Id}_{\mathbb{Z}}$ to illustrate the problem.

(e) When I is finite, explain why the two universal properties describe the same module. This is why finite direct sums are also finite direct products.

Over a field, every module (i.e. vector space) has a basis, and a linear map is determined by its values on a basis. Over a general ring, modules need not have bases at all—but those that do are of fundamental importance. The goal of this section is to set up the theory of free modules and to understand precisely how they differ from vector spaces.

Let M be an R-module. A subset $B \subseteq M$ *generates* (or *spans*) M if every element of M can be written as a finite R-linear combination of elements of B, i.e. if

$$M = \left\{ \sum_{i=1}^{n} r_i m_i \;\middle|\; n \geq 0,\, r_i \in R,\, m_i \in B \right\}.$$

We say B is *linearly independent* if, whenever $m_1, \ldots, m_n \in B$ are distinct and $\sum_{i=1}^{n} r_i m_i = 0$, we have $r_i = 0$ for all i. A *basis* for M is a subset that is both generating and linearly independent, or equivalently,

a subset $\{b_s\}_{s\in S}$ such that every element of M can be written *uniquely* as a finite R-linear combination $\sum_s r_s b_s$.

Over a field, these notions interact exactly as in undergraduate linear algebra: every spanning set contains a basis, every linearly independent set extends to one, and all bases have the same cardinality. Over a general ring, the first two statements can fail; in particular, many modules do not admit bases at all. A module that does admit a basis is called *free*. Given a set S, the *free R-module on S* is

$$R^{(S)} := \bigoplus_{s\in S} R.$$

Concretely, $R^{(S)}$ consists of all functions $S \to R$ with finite support. It has a distinguished element e_s for each $s \in S$, where e_s has a 1 in the s-coordinate and 0 elsewhere. Every element of $R^{(S)}$ can then be written uniquely as a finite sum

$$\sum_{s\in S} r_s e_s,$$

so $\{e_s \mid s \in S\}$ is a basis, called the *standard basis*. When $S = \{1,\dots,n\}$ we write R^n instead of $R^{(S)}$.

The key feature of a free module is that a homomorphism out of it is completely determined by where the basis goes. Precisely: for every function of sets $\alpha\colon S \to M$ (where M is any R-module), there is a unique R-module homomorphism $\widetilde{\alpha}\colon R^{(S)} \to M$ such that $\widetilde{\alpha}(e_s) = \alpha(s)$ for all $s \in S$. In categorical language, the functor $S \mapsto R^{(S)}$ is left adjoint to the forgetful functor from R-modules to sets. This universal property characterizes $R^{(S)}$ up to unique isomorphism and is the reason free modules play a role analogous to that of vector spaces—they are the modules we can map out of "for free."

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(8) **Free Modules and Their Universal Property.**

(a) Verify directly that every element of $R^{(S)}$ can be written uniquely as a finite R-linear combination of the standard basis elements e_s.

(b) Prove the universal property of $R^{(S)}$ stated above.

(c) Deduce that there is a natural isomorphism

$$\mathrm{Hom}_R(R^{(S)}, M) \cong \prod_{s\in S} M.$$

Why is the right side a product rather than a direct sum?

(d) Show that an R-module M is generated by $m_1,\dots,m_n$ if and only if there is a surjective homomorphism $R^n \twoheadrightarrow M$ sending e_i to m_i.

(e) Conclude that every finitely generated R-module is a quotient of a finite free module.

(f) More generally, show that every R-module is a quotient of a free R-module.

(g) For $M = R/I$, write M as a quotient of a free module and identify the kernel.

(h) For $R = \mathbb{K}[x]$ and $M = R/\langle x^3\rangle$, describe M as a module with one generator and one relation.

(9) **Free Modules Are Not Quite Vector Spaces.**

(a) Show that $\mathbb{Z}/n\mathbb{Z}$ is not a free $\mathbb{Z}$-module for any $n > 1$.

(b) More generally, if R is an integral domain and $0 \neq I \subsetneq R$ is an ideal, show that R/I is not a free R-module.

(c) In the $\mathbb{Z}$-module $\mathbb{Z}$, show that $\{2,3\}$ is a generating set but not a basis.

(d) In the $\mathbb{Z}$-module $\mathbb{Z}$, show that $\{2\}$ is linearly independent but does not generate $\mathbb{Z}$.

(e) Let $R = \mathbb{Z}/6\mathbb{Z}$. Show that the element $\overline{2} \in R$ is nonzero but not part of a basis for the free R-module R.

(f) Explain why the previous examples show that bases over rings behave differently from bases over fields.

(10) **Matrices and Maps Between Finite Free Modules.**

(a) Let $A = (a_{ij})$ be an $m \times n$ matrix with entries in R. Define a homomorphism

$$\phi_A : R^n \to R^m, \quad e_j \longmapsto \sum_{i=1}^{m} a_{ij} e_i.$$

Prove that every homomorphism $R^n \to R^m$ arises uniquely in this way.

(b) Under this correspondence, prove that composition of homomorphisms corresponds to multiplication of matrices.

(c) When $R = \mathbb{K}$ is a field, explain how this recovers the usual matrix description of linear maps between finite-dimensional vector spaces.

(d) Let $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ and view A as a homomorphism $\mathbb{Z}^2 \to \mathbb{Z}^2$. Describe $\operatorname{coker}(A) = \mathbb{Z}^2/\operatorname{img}(A)$.

(e) Let $R = \mathbb{K}[x]$ and consider multiplication by x as a map $R^2 \to R^2$. Write its matrix with respect to the standard basis, and compute its cokernel.

For two R-modules M and N, we write $\operatorname{Hom}_R(M,N)$ for the set of R-module homomorphisms $M \to N$. Since R is commutative, $\operatorname{Hom}_R(M,N)$ is itself an R-module: addition and scalar multiplication are defined pointwise by

$$(\phi + \psi)(m) = \phi(m) + \psi(m), \qquad (r\phi)(m) = r\phi(m).$$

This construction is one of the most important ways of building new modules from old ones. There is already a great deal to understand: Hom out of a free module is easy, Hom out of a quotient records annihilators, and Hom interacts cleanly with direct sums and products through their universal properties.

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(11) $\mathrm{Hom}_R(M,N)$ **as a Module.**

(a) Prove that $\mathrm{Hom}_R(M,N)$ is an abelian group under pointwise addition. What is the zero element?

(b) Prove that the formula $(r\phi)(m) = r\phi(m)$ makes $\mathrm{Hom}_R(M,N)$ into an R-module.

(c) Show that evaluation at $1 \in R$ defines an isomorphism of R-modules

$$\mathrm{Hom}_R(R,M) \cong M.$$

(d) More generally, prove that

$$\mathrm{Hom}_R(R/I,M) \cong \{m \in M \mid Im = 0\}.$$

The isomorphism should again be given by evaluation at $\overline{1} \in R/I$.

(e) Deduce that $\mathrm{Hom}_R(R^n,M) \cong M^n$.

(f) Prove that $\mathrm{End}_R(R) \cong R$ as rings, where the ring structure on $\mathrm{End}_R(R)$ is given by addition and composition.

(g) If $I \subseteq R$ is an ideal, identify $\mathrm{Hom}_R(R/I,R)$ in terms of an annihilator ideal of R.

(12) **Hom Computations.**

(a) Describe $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z},\mathbb{Z}/n\mathbb{Z})$ explicitly for positive integers m,n. Show that it is isomorphic to $\mathbb{Z}/d\mathbb{Z}$, where $d = \gcd(m,n)$.

(b) Prove that $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Q},\mathbb{Z}) = 0$.

(c) Prove that $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z},M) \cong M$ for every abelian group M.

(d) Let $R = \mathbb{K}[x]$ and let $M = R/\langle f\rangle$, $N = R/\langle g\rangle$. Use the preceding computation of $\mathrm{Hom}_R(R/I,-)$ to describe $\mathrm{Hom}_R(M,N)$ as a submodule of N.

(e) In the special case $f = x^a$ and $g = x^b$, compute $\mathrm{Hom}_{\mathbb{K}[x]}(\mathbb{K}[x]/\langle x^a\rangle, \mathbb{K}[x]/\langle x^b\rangle)$ up to isomorphism.

(13) **Functoriality of Hom.** Let M,M',N,N' be R-modules.

(a) If $\alpha : M' \to M$ is a homomorphism, prove that precomposition defines an R-module homomorphism

$$\alpha^* : \mathrm{Hom}_R(M,N) \to \mathrm{Hom}_R(M',N), \quad \phi \longmapsto \phi \circ \alpha.$$

(b) If $\beta : N \to N'$ is a homomorphism, prove that postcomposition defines an R-module homomorphism

$$\beta_* : \mathrm{Hom}_R(M,N) \to \mathrm{Hom}_R(M,N'), \quad \phi \longmapsto \beta \circ \phi.$$

(c) Check that $(\mathrm{Id}_M)^* = \mathrm{Id}_{\mathrm{Hom}_R(M,N)}$ and $(\mathrm{Id}_N)_* = \mathrm{Id}_{\mathrm{Hom}_R(M,N)}$.

(d) If $M'' \xrightarrow{\gamma} M' \xrightarrow{\alpha} M$, prove that $(\alpha \circ \gamma)^* = \gamma^* \circ \alpha^*$.

(e) If $N \xrightarrow{\beta} N' \xrightarrow{\delta} N''$, prove that $(\delta \circ \beta)_* = \delta_* \circ \beta_*$.

(f) Let $M^* := \mathrm{Hom}_R(M,R)$ be the *dual module*. If $M = R^n$, describe the dual basis of M^* and prove that $M^* \cong R^n$.

Many natural constructions in algebra are *bilinear*: matrix multiplication, the dot product, and the evaluation pairing between a module and its dual. In each case, the map is additive in each variable separately, but not additive as a function of both variables together—so it is not a homomorphism from the product module. The tensor product is the device that resolves this tension. It provides a module $M \otimes_R N$ that converts bilinear maps *out of* $M \times N$ into honest linear maps out of $M \otimes_R N$.

Let R be a commutative ring with identity, and let M, N, and P be R-modules. An *R-bilinear map* from $M \times N$ to P is a function

$$B \colon M \times N \to P$$

satisfying the following three conditions for all $r \in R$, $m, m' \in M$, and $n, n' \in N$:

(i) $B(m + m', n) = B(m,n) + B(m',n)$,

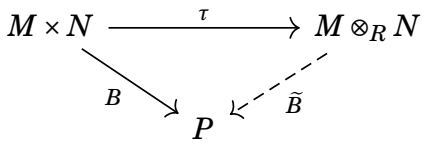
(ii) $B(m, n + n') = B(m,n) + B(m,n')$,

(iii) $B(rm, n) = rB(m,n) = B(m, rn)$.

In other words, B is R-linear in each argument when the other is held fixed. A *tensor product* of M and N is a pair $(M \otimes_R N, \tau)$ consisting of an R-module $M \otimes_R N$ and a bilinear map

$$\tau \colon M \times N \to M \otimes_R N, \qquad (m,n) \longmapsto m \otimes n,$$

that is *universal* in the following sense: for every R-module P and every R-bilinear map $B \colon M \times N \to P$, there exists a *unique* R-module homomorphism $\widetilde{B} \colon M \otimes_R N \to P$ such that $B = \widetilde{B} \circ \tau$.



The content of this property is that τ is the "most general" bilinear map out of $M \times N$: every other one factors through it, and does so in exactly one way. One important consequence is that to define a linear map *out of* a tensor product, it suffices to specify its values on the generators $m \otimes n$ and verify that the assignment is bilinear—the universal property then guarantees a unique, well-defined homomorphism. This strategy is used repeatedly in the exercises below.

The universal property tells us what a tensor product must *do*, but not that one exists. Here is a direct construction. Begin with the free R-module F on the set $M \times N$; its basis consists of formal symbols $[m,n]$,

one for each pair $(m,n) \in M \times N$. Let $K \subseteq F$ be the submodule generated by all elements of the forms

$$\begin{aligned}
[m+m', n] - [m,n] - [m',n],\\
[m, n+n'] - [m,n] - [m,n'],\\
[rm, n] - r[m,n],\\
[m, rn] - r[m,n],
\end{aligned}$$

for all $r \in R$, $m, m' \in M$, and $n, n' \in N$. Set $M \otimes_R N = F/K$, and write $m \otimes n$ for the image of $[m,n]$ in the quotient. By construction, the map $(m,n) \mapsto m \otimes n$ is bilinear, and every element of $M \otimes_R N$ is a finite R-linear combination of such "simple tensors."

The identities that $m \otimes n$ satisfies are therefore precisely the bilinearity relations:

$$\begin{aligned}
(m+m') \otimes n &= m \otimes n + m' \otimes n,\\
m \otimes (n+n') &= m \otimes n + m \otimes n',\\
(rm) \otimes n &= r(m \otimes n) = m \otimes (rn).
\end{aligned}$$

Not every element of $M \otimes_R N$ is a simple tensor $m \otimes n$; a general element is a finite sum $\sum_i m_i \otimes n_i$. A common source of errors is to assume that every tensor can be written as a single $m \otimes n$.

---

(14) **Constructing Tensor Products.**

(a) Using the generators-and-relations construction above, prove the identities

$$(m+m') \otimes n = m \otimes n + m' \otimes n, \quad m \otimes (n+n') = m \otimes n + m \otimes n',$$

and

$$(rm) \otimes n = r(m \otimes n) = m \otimes (rn).$$

(b) Prove that $0 \otimes n = 0$, $m \otimes 0 = 0$, and $(-m) \otimes n = -(m \otimes n) = m \otimes (-n)$.

(c) Prove that the construction described above satisfies the universal property of the tensor product.

(d) Prove that the tensor product is unique up to unique isomorphism: if T and T' both satisfy the universal property for M and N, then there is a unique isomorphism $T \cong T'$ compatible with the two bilinear maps from $M \times N$.

(e) Use the universal property to construct a natural isomorphism

$$M \otimes_R N \cong N \otimes_R M.$$

(f) If $f : M \to M'$ and $g : N \to N'$ are homomorphisms, construct a homomorphism

$$f \otimes g : M \otimes_R N \to M' \otimes_R N'$$

satisfying $(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$.

(15) **Basic Tensor Product Computations.**

(a) Prove that the map

$$R \otimes_R M \to M, \quad r \otimes m \longmapsto rm$$

is an isomorphism of R-modules.

(b) Prove that $R^n \otimes_R M \cong M^n$.

(c) More generally, prove that

$$\left(\bigoplus_{i \in I} M_i\right) \otimes_R N \cong \bigoplus_{i \in I} (M_i \otimes_R N).$$

(d) Let $I \subseteq R$ be an ideal. Prove that

$$(R/I) \otimes_R M \cong M/IM,$$

where IM is the submodule defined earlier. What is the image of $\overline{r} \otimes m$ under this isomorphism?

(e) Deduce that for ideals $I, J \subseteq R$ one has $(R/I) \otimes_R (R/J) \cong R/(I+J)$.

(f) Compute $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$ for positive integers m, n.

(g) Compute $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$.

(h) If V and W are finite-dimensional vector spaces over a field $\mathbb{K}$, prove that

$$\dim_{\mathbb{K}}(V \otimes_{\mathbb{K}} W) = \dim_{\mathbb{K}}(V) \dim_{\mathbb{K}}(W).$$

(16) **Tensor Products as Linearized Bilinear Maps.**

(a) Let $e_1, \ldots, e_m$ be the standard basis of R^m and $f_1, \ldots, f_n$ the standard basis of R^n. Show that $R^m \otimes_R R^n$ is free with basis $\{e_i \otimes f_j \mid 1 \le i \le m,\ 1 \le j \le n\}$.

(b) Let $B : R^m \times R^n \to P$ be an R-bilinear map. Prove that B is uniquely determined by the mn elements $B(e_i, f_j) \in P$.

(c) If M is generated by $m_1, \ldots, m_a$ and N is generated by $n_1, \ldots, n_b$, prove that $M \otimes_R N$ is generated by the elements $m_i \otimes n_j$.

(d) Let $R = \mathbb{K}[x]$. Compute $R/\langle x^a \rangle \otimes_R R/\langle x^b \rangle$ up to isomorphism.

(e) Let $I, J \subseteq R$ be ideals. The multiplication map $I \times J \to IJ$, $(a, b) \mapsto ab$, is R-bilinear. Use the universal property to construct a natural surjective homomorphism

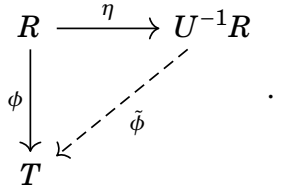
$$I \otimes_R J \twoheadrightarrow IJ.$$

(f) Show that if $I = R$, then the map in the previous part is an isomorphism. Then let $R = \mathbb{K}[\varepsilon]/\langle \varepsilon^2 \rangle$ and let $I = J = \langle \overline{\varepsilon} \rangle$. Compute $I \otimes_R I$ and IJ, and conclude that the multiplication map $I \otimes_R J \to IJ$ need not be injective.

WORKSHEET 4.2: LOCALIZATION

Throughout this course, "ring" means *commutative* ring with unity. In the previous worksheet we built the basic language of modules, quotient modules, free modules, Hom, and tensor products. The goal of this worksheet is to introduce one of the central operations in commutative algebra: *localization*. Informally, localization is the process of forcing a chosen collection of elements of a ring to become invertible. This construction simultaneously generalizes the passage from $\mathbb{Z}$ to $\mathbb{Q}$, the localization $\mathbb{Z}_{\langle p \rangle}$ at a prime number p, and the algebraic operation of restricting attention to a principal open subset $D(f)$ of $\mathrm{Spec}(R)$.

Let R be a ring. A subset $U \subset R$ is called *multiplicatively closed* if $1 \in U$ and $u, t \in U$ implies $ut \in U$. Given a multiplicative set $U \subset R$, the localization of R at U is a ring $U^{-1}R$ together with a ring homomorphism $\eta : R \to U^{-1}R$ satisfying the following universal property: For all ring homomorphisms $\phi : R \to T$ such that every element of $\phi(U)$ is a unit in T there exists a unique homomorphism $\tilde{\phi} : U^{-1}R \to T$ such that the following diagram commutes:



Roughly speaking we should think of $U^{-1}R$ as being the "smallest" ring where every element of U is a unit. As always the universal property guarantees that $U^{-1}R$ and the map η are unique up to unique isomorphism (assuming that they exist, which we will now discuss).

Informally, we think of elements of $U^{-1}R$ as obtained by formally inverting every element of U and imposing the necessary equivalence conditions to make it into a well-behaved ring. Elements of $U^{-1}R$ are formal fractions r/u where $r \in R$ and $u \in U$ subject to some equivalence relation. If R is an integral domain and $0 \notin U$, equality of fractions is the familiar cross-multiplication relation, i.e. $r_1/u_1 = r_2/u_2$ as elements in $U^{-1}R$ if and only if $r_1u_2 = r_2u_1$ as elements in R. In general, because zero divisors may be present, one must allow extra clearing denominators:

$$\frac{r_1}{u_1} = \frac{r_2}{u_2} \quad \text{in } U^{-1}R \quad \iff \quad t(r_1u_2 - r_2u_1) = 0 \quad \text{in } R \text{ for some } t \in U$$

Multiplication and addition in $U^{-1}R$ are then defined in the usual way for fractions:

$$\left(\frac{r_1}{u_1}\right)\left(\frac{r_2}{u_2}\right) = \frac{r_1r_2}{u_1u_2} \qquad \text{and} \qquad \frac{r_1}{u_1} + \frac{r_2}{u_2} = \frac{r_1u_2 + r_2u_1}{u_1u_2}$$

The additive identity is then the equivalence class of 0/1 and the multiplicative identity is 1/1. The function $\eta : R \to U^{-1}R$ is defined by $r \mapsto \frac{r}{1}$, which formally makes sense because $1 \in U$. Although this construction is relatively easy to define, there is work involved in checking that the operations and η are well-defined,

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that $U^{-1}R$ is a ring, and that it satisfies the correct universal property. You will carefully go through this construction in the exercises.

A few warnings: when R is not a domain localization can behave somewhat strangely. For example, from our experience with fractions since childhood and its unassuming definition one might expect η to be injective; this is not the case. The element $\eta(r) = r/1$ can be equal to zero in $U^{-1}R$. By definition this occurs when $r/1 = 0/1$, which translates to $tr = 0$ in R for some element $t \in U$. Even worse, if $0 \in U$ then the ring $U^{-1}R$ becomes the zero ring.

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(1) **Multiplicative Sets.**

(a) Verifying, using the definition directly, which of the following subsets are multiplicatively closed?

(i) $R = \mathbb{Z}$ and $U = \{1, 3, 9, 27, \ldots\}$.

(ii) $R = \mathbb{Z}$ and $U = \{1, 2, 3\}$.

(iii) $R = \mathbb{Z}$ and $U = \mathbb{Z} \setminus \langle 2 \rangle$.

(iv) $R = \mathbb{Z}$ and $U = \{n \geq 1\}$.

(v) $R = \mathbb{R}[x, y]$ and $U = R \setminus \langle x, y \rangle$.

(vi) $R = \mathbb{R}[x, y]$ and $U = \{f \in R \mid f(1, 0) \neq 0\}$.

(vii) $R = \mathbb{R}[x, y]/\langle y^2 - x^3 + x \rangle$ and $U = R \setminus \langle \overline{x}, \overline{y} \rangle$.

(viii) $R = \mathbb{R}[x, y]$ and $U = \{f \in R \mid f(0, 0) \neq 0$ and $f(1, 1) \neq 0\}$.

(b) Let $I \subset R$ be an ideal. Prove that $U = R \setminus I$ is multiplicatively closed if and only if I is prime.

(c) Let $U \subset R$ be a multiplicatively closed set. If $\phi : R \to S$ is a ring homomorphism, is $\phi(U) \subset S$ a multiplicatively closed subset?

(2) **First Examples.** Despite not having formally verified that our construction is well-defined, discuss ways to explicitly understand $U^{-1}R$ in each of the following cases.

(a) $R = \mathbb{Z}$ and $U = \mathbb{Z} \setminus \{0\}$.

(b) $R = \mathbb{Z}$ and $U = \{2^n \mid n \in \mathbb{Z}_{\geq 0}\}$.

(c) $R = \mathbb{Z}$ and $U = \mathbb{Z} \setminus \langle 2 \rangle$.

(d) $R = \mathbb{K}[x]$ and $U = \mathbb{K}[x] \setminus \{0\}$.

(e) $R = \mathbb{K}[x]$ and $U = \{1, x, x^2, x^3, \ldots\}$.

(f) $R = \mathbb{Z}/12\mathbb{Z}$ and $U = \{\overline{1}, \overline{2}, \overline{4}, \overline{8}\}$.

(g) $R = \mathbb{Z}/12\mathbb{Z}$ and $U = \{\overline{1}, \overline{3}, \overline{9}\}$.

(h) $R = \mathbb{Z}[x]/\langle nx - 1 \rangle$ and $U = \{n^k \mid k \in \mathbb{Z}_{\geq 0}\}$ for a fixed non-zero integer n.

(i) R an arbitrary ring and U a multiplicative closed set containing a nilpotent.

(3) **Constructing $U^{-1}R$.** Let R be a ring and $U \subset R$ a multiplicatively closed set. Define a relation on $R \times U$ by saying $(r_1, u_1) \sim (r_2, u_2)$ if and only if $t(r_1 u_2 - r_2 u_1) = 0$ in R for some $t \in U$.

(a) Prove that $\sim$ is an equivalence relation on $R \times U$.

(b) Let $\frac{r}{u}$ denote the equivalence class of a pair (r,u) under the equivalence relation $\sim$ defined in the previous part. Prove that the operations:

$$\left(\frac{r_1}{u_1}\right)\left(\frac{r_2}{u_2}\right) = \frac{r_1 r_2}{u_1 u_2} \qquad \text{and} \qquad \frac{r_1}{u_1} + \frac{r_2}{u_2} = \frac{r_1 u_2 + r_2 u_1}{u_1 u_2}$$

are well-defined. Hint: You must show that changing the representative of either input fraction does not change the resulting equivalence class.

(c) Define $U^{-1}R$ to be the set of equivalence classes of $\sim$. Show that the operations above make $U^{-1}R$ into a ring where $\frac{0}{1}$ is the zero element and $\frac{1}{1}$ is the multiplicative identity.

(d) Show that the function $\eta : R \to U^{-1}R$ given by $r \mapsto \frac{r}{1}$ is a well-defined ring homomorphism.

(e) Prove that if $u \in U$ then $\eta(u)$ is a unit in $U^{-1}R$. Explicitly what is its inverse?

(f) Where in this construction did you use that U was multiplicatively closed and contains 1?

(4) **First Properties of Localization.** Let R be a ring and $U \subset R$ a multiplicatively closed set.

(a) Prove that if $0 \in U$ then $U^{-1}R$ is the zero ring.

(b) Prove that if R is an integral domain and $0 \notin U$ then $\frac{r_1}{u_1} = \frac{r_2}{u_2}$ in $U^{-1}R$ if and only if $r_1 u_2 = r_2 u_1$.

(c) Let R be an integral domain and $U = R \setminus \{0\}$. Prove that $U^{-1}R$ is a field.

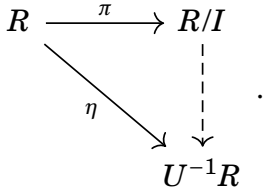
(d) Find an example of a non-zero ring R for which $R \setminus \{0\}$ is not multiplicatively closed. What goes wrong with trying to form $(R \setminus \{0\})^{-1}R$?

(5) **The Kernel of the Localization Map.** Let R be a ring, $U \subset R$ be a multiplicatively closed set, and $\eta : R \to U^{-1}R$ the localization map as described in Question 3.

(a) Prove that η is injective if and only if every $u \in U$ is a nonzerodivisor on R, i.e. $ur = 0$ implies $r = 0$ for all $r \in R$.

(b) Let $I = \{r \in R \mid ur = 0 \text{ for some } u \in U\}$. Prove that I is an ideal.

(c) Show that there exists an injective ring homomorphism $R/I \to U^{-1}R$ such that the following diagram commutes:



(d) Prove that $\ker(\eta) = I$.

(e) Revisit the examples in Question 2. For which of them is the localization map $R \to U^{-1}R$ injective?

(6) **Field of Fractions.** Let R be an integral domain. The *field of fractions* of R is the ring $U^{-1}R$ where $U = R \setminus \{0\}$. We denote the field of fractions of R by $\mathrm{Frac}(R)$.

(a) Show that $\mathrm{Frac}(R)$ is a field containing R.

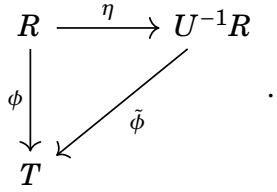
(b) Describe $\mathrm{Frac}(R)$ when $R = \mathbb{Z}$ and $R = \mathbb{K}[x]$.

(c) If R is an integral domain and $U \subset R$ is any multiplicatively closed set not containing zero, explain how we can view $U^{-1}R$ as a subring of $\mathrm{Frac}(R)$.

(7) **Universal Property of Localization.** In this exercise we shall show that $U^{-1}R$ and η as constructed in Question 3 satisfy the universal property of localization. Let R be a ring and $U \subset R$ a multiplicatively closed subset. Let $\phi : R \to T$ be a ring homomorphism such that $\phi(U) \subset T^\times$.

(a) Define a function $\tilde{\phi} : U^{-1}R \to T$ by $\frac{r}{u} \mapsto \phi(r)\phi(u)^{-1}$. Explain why $\tilde{\phi}$ is well-defined and prove it is a ring homomorphism.

(b) Show that $\tilde{\phi}$ makes the following diagram commute:



(c) Prove that $\tilde{\phi}$ is the unique ring homomorphism $U^{-1}R \to T$ making the diagram commute. Hint: If $\psi : U^{-1}R \to T$ is any such map, first determine $\psi\left(\frac{1}{u}\right)$ for $u \in U$, and then use the equality $\frac{r}{u} = \frac{r}{1}\frac{1}{u}$.

(8) **Applications of the Universal Property of Localization.** Let R be a ring and $U \subset R$ a multiplicatively closed subset.

(a) Use the universal property of localization to show that $U^{-1}R$ and η are unique up to unique isomorphism.

(b) Suppose every element of U is already a unit in R. Prove that the localization map $\eta : R \to U^{-1}R$ is an isomorphism.

(c) Suppose $V \subset R$ is another multiplicatively closed set and $U \subset V$. Use the universal property to construct a natural homomorphism $U^{-1}R \to V^{-1}R$.

(d) Let $\phi : R \to S$ be a ring homomorphism such that $\phi(U) \subset S^\times$. Let $\tilde{\phi} : U^{-1}R \to S$ be the induced map. Prove that if ϕ is surjective, then $\tilde{\phi}$ is surjective. Prove that if ϕ is injective, then $\tilde{\phi}$ is injective.

The two most important examples of localization are: i) when U is equal to the complement of a prime ideal and ii) when $U = \{1, f, f^2, f^3, \ldots\}$ for some non-zero element $f \in R$. Given a prime ideal $\mathfrak{p} \subset R$ you showed in Question 1b that $U = R \setminus \mathfrak{p}$ is a multiplicatively closed set. The ring $(R \setminus \mathfrak{p})^{-1}R$ occurs so frequently that we give it its own notation and name. The *localization of R at $\mathfrak{p}$* is the ring:

$$R_{\mathfrak{p}} := (R \setminus \mathfrak{p})^{-1}R.$$

The guiding feature of $R_{\mathfrak{p}}$ is that all elements outside of $\mathfrak{p}$ become invertible, while the elements in $\mathfrak{p}$ remain non-units. As we shall see shortly, this implies that $R_{\mathfrak{p}}$ has a unique maximal ideal, namely,

$$\mathfrak{p}R_{\mathfrak{p}} \coloneqq \left\{ \frac{r}{u} \mid r \in \mathfrak{p},\, u \in R \setminus \mathfrak{p} \right\}.$$

A ring with a unique maximal ideal is called a *local ring*. We often denote a local ring by $(R, \mathfrak{m})$, where $\mathfrak{m}$ is its unique maximal ideal. Local rings are algebraic analogues of rings of germs near a point.

Turning towards the second important example, fix a non-zero element $f \in R$ and consider the set $U = \{1, f, f^2, f^3, \ldots\}$. Note we should assume f is not a nilpotent element, since otherwise as we saw in Question 2i, the localization $U^{-1}R$ will be the zero ring. The localization $U^{-1}R$ consists of fractions of the form $\frac{r}{f^n}$ with $r \in R$ and $n \geq 0$. Every such fraction can be rewritten as $r \cdot \frac{1}{f^n} = r \cdot \left(\frac{1}{f}\right)^n$, so every element of $U^{-1}R$ is a polynomial expression in $\frac{1}{f}$ with coefficients in R. In other words, $U^{-1}R$ is generated, as a ring, by the image of R and the single new inverse $\frac{1}{f}$. We therefore write:

$$R\left[\tfrac{1}{f}\right] \coloneqq \{1, f, f^2, \ldots\}^{-1} R = \left\{ \frac{r}{f^n} \mid r \in R,\, n \geq 0 \right\}.$$

We will see shortly that we can make the idea that $R[\frac{1}{f}]$ is "polynomials in $\frac{1}{f}$ with coefficients in R" precise by showing $R[\frac{1}{f}]$ is isomorphic to $R[x]/\langle xf - 1\rangle$. Intuitively this makes sense since quotienting by $\langle xf - 1\rangle$ means setting $xf = 1$, i.e., let the formal variable x equal the inverse of f. (Note we saw something similar to this when we used the Rabinowitsch trick on the Nullstellensatz worksheet.)

Warning: Many texts use the notation R_f as shorthand for $R[\frac{1}{f}]$. This has the unfortunate effect of causing confusion when $f \in R$ is prime as it is easy to confuse $R_f \coloneqq R[\frac{1}{f}]$ with the localization $R_{\langle f\rangle}$ at the prime ideal $\langle f\rangle$. Despite the similar subscript notation, these constructions go in opposite directions: in $R_{\langle f\rangle}$, the elements outside $\langle f\rangle$ are inverted, whereas in R_f it is the powers of f itself that are inverted. Due to this confusion we will try our best to avoid using the R_f notation and instead just write $R[\frac{1}{f}]$.

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(9) **Localization at a Prime Ideal.** Let R be a ring and $\mathfrak{p} \subset R$ be a prime ideal.

(a) Prove that an element $\frac{r}{u} \in R_{\mathfrak{p}}$ is a unit if and only if $r \notin \mathfrak{p}$.

(b) Deduce that the non-units of $R_{\mathfrak{p}}$ are precisely the elements

$$\mathfrak{p}R_{\mathfrak{p}} \coloneqq \left\{ \frac{r}{u} \mid r \in \mathfrak{p},\, u \in R \setminus \mathfrak{p} \right\}.$$

(c) Prove that $\mathfrak{p}R_{\mathfrak{p}}$ is an ideal in $R_{\mathfrak{p}}$.

(d) Show that $\mathfrak{p}R_{\mathfrak{p}}$ is a maximal ideal in $R_{\mathfrak{p}}$. Hint: Show that any ideal strictly containing $\mathfrak{p}R_{\mathfrak{p}}$ must contain a unit.

(e) Show that $R_{\mathfrak{p}}$ does not have any other maximal ideal, i.e. $(R_{\mathfrak{p}}, \mathfrak{p}R_{\mathfrak{p}})$ is a local ring.

(f) More generally, show that a ring R is a local ring if and only if the set of non-units in R is an ideal.

(g) Let $p \in \mathbb{Z}$ be a prime number. Describe the units and unique maximal ideal of $\mathbb{Z}_{\langle p \rangle}$.

(h) Let $R = \mathbb{C}[x]$ and $\mathfrak{m} = \langle x - a \rangle$ for $a \in \mathbb{C}$. Which rational functions in $\mathbb{C}(x)$ lie in $R_{\mathfrak{m}}$? Which of them are units?

(10) **Inverting an Element.** Let R be a ring and let $f \in R$ be a nonzero, non-nilpotent element. Recall the ring $R[\frac{1}{f}]$ is defined to be the localization of R at $U = \{1, f, f^2, f^3, \ldots\}$.

(a) Show that the R-algebra homomorphism $\phi : R[x] \to R[\frac{1}{f}]$ determined by $\phi(r) = \frac{r}{1}$ for $r \in R$ and $\phi(x) = \frac{1}{f}$ is well-defined.

(b) Prove that ϕ is surjective by explicitly finding an element that maps to $\frac{r}{f^k}$ for $r \in R$ and $k \in \mathbb{Z}_{\geq 0}$.

(c) Show that $\ker(\phi) = \langle fx - 1 \rangle$.

(d) Conclude that $R[\frac{1}{f}]$ is isomorphic to $R[x]/\langle fx - 1 \rangle$.

(e) Give a different argument proving $R[\frac{1}{f}]$ and $R[x]/\langle fx - 1 \rangle$ are isomorphic using the universal property of localizations.

(f) What changes if f is nilpotent? Show that both $R[\frac{1}{f}]$ and $R[x]/\langle fx - 1 \rangle$ are the zero ring.

(g) Let $f, g \in R$. Use the universal property to prove that

$$\left(R\left[\frac{1}{f}\right]\right)\left[\frac{1}{g}\right] \cong R\left[\frac{1}{fg}\right].$$

Given any ring homomorphism $\phi : R \to S$, there is a natural interplay between ideals of R and ideals of S. If $I \subset R$ is an ideal, its *extension* is the ideal of S generated by the image of I under ϕ:

$$\phi(I) \cdot S := \big\langle \phi(I) \big\rangle = \left\{ \sum_{i=1}^{n} s_i \phi(r_i) \;\middle|\; r_i \in I,\, s_i \in S,\, n \geq 0 \right\}.$$

Note we must take the ideal generated by $\phi(I)$, not just the image of I itself, since as we previously saw $\phi(I)$ may not be an ideal of S.

Conversely, if $J \subseteq S$ is an ideal, its *contraction* is the preimage

$$\phi^{-1}(J) = \big\{ r \in R \;\big|\; \phi(r) \in J \big\},$$

which is always an ideal of R. Note there are numerous different notations for extension and contraction of ideals: some (namely Atiyah–MacDonald) write I^e and J^c for the extension and contraction respectively, others write IS and $J \cap R$ (as a nod to what occurs when ϕ is an injection). We generally avoid these notations to emphasize that contraction and extension are dependent on the homomorphism ϕ.

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(11) **Extension and Contraction of Ideals.** Let $\phi : R \to S$ be a homomorphism of rings. Let $I \subset R$ and $J \subset S$ be ideals.

(a) Prove that $\phi^{-1}(J)$ is an ideal of R.

(b) Give an explicit example showing that $\phi(I) \subset S$ need not be an ideal.

(c) Prove that $I \subset \phi^{-1}(\phi(I)S)$.

(d) Let $\phi : \mathbb{Z} \to \mathbb{Z}/\langle 6 \rangle$ be the natural quotient map and let $I = \langle 0 \rangle \subset \mathbb{Z}$. Explicitly describe $J = \phi(I)(\mathbb{Z}/\langle 6 \rangle)$ and $\phi^{-1}(J)$. Show that $I \subsetneq \phi^{-1}(J)$. Thus, the inclusion in the previous part can be strict.

(e) Prove that $\phi(\phi^{-1}(J))S \subset J$.

(f) Let $\phi : \mathbb{Z} \to \mathbb{Z}[x]$ be the inclusion map and let $J = \langle x \rangle \subset \mathbb{Z}[x]$. Explicitly describe $I = \phi^{-1}(J)$ and $\phi(I)\mathbb{Z}[x]$. Show that $\phi(I)\mathbb{Z}[x] \subsetneq J$. Thus, the inclusion in the previous part can be strict.

(g) Prove that if $J \subset S$ is prime then the contraction $\phi^{-1}(J) \subset R$ is prime.

(h) Give an example showing that if $J \subset S$ is maximal the contraction $\phi^{-1}(J) \subset R$ need not be maximal.

(i) Give an example showing that if $I \subset R$ is prime then the extension $\phi(I)S \subset S$ need not be prime.

(j) Let $\phi : R \hookrightarrow S$ be an injective ring homomorphism. Prove that

$$\phi^{-1}(J) = J \cap R \qquad \text{and} \qquad \phi(I)S = IS.$$

As we saw in the previous exercise, in general, extension and contraction are far from being inverse operations: $\phi^{-1}(\phi(I)S)$ may strictly contain I, and $\phi(\phi^{-1}(J))S$ may be strictly smaller than J. In the case of localization, extension and contraction along the canonical localization map $\eta : R \to U^{-1}R$ sending $r \mapsto r/1$ have very nice forms. For an ideal $I \subset R$ the extension along η is

$$\eta(I)\left(U^{-1}R\right) = U^{-1}I := \left\{ \frac{r}{u} \;\middle|\; r \in I,\, u \in U \right\} \subset U^{-1}R.$$

For an ideal $J \subset U^{-1}R$ the contraction along η is:

$$\eta^{-1}(J) = \left\{ r \in R \;\middle|\; \frac{r}{1} \in J \right\}.$$

Although the extension–contraction correspondence remains lossy for arbitrary ideals, it becomes perfectly behaved for prime ideals. If $\mathfrak{q} \subset U^{-1}R$ is a proper ideal, then no element of U can lie in its contraction. (This is because $\frac{u}{1}$ is a unit in $U^{-1}R$ for every $u \in U$.) Thus every prime ideal of $U^{-1}R$ contracts to a prime ideal of R which is disjoint from U. Conversely, if $\mathfrak{p} \subset R$ is a prime ideal disjoint from U, then its extension

$$U^{-1}\mathfrak{p} = \left\{ \frac{r}{u} \;\middle|\; r \in \mathfrak{p},\, u \in U \right\}$$

is a prime ideal of $U^{-1}R$. In fact, extension and contraction give an order-preserving bijection

$$\{\text{prime ideals of } U^{-1}R\} \longleftrightarrow \{\mathfrak{p} \in \operatorname{Spec}(R) \mid \mathfrak{p} \cap U = \emptyset\}.$$

Under this correspondence, a prime ideal $\mathfrak{q} \subset U^{-1}R$ is sent to its contraction $\eta^{-1}(\mathfrak{q})$, and a prime ideal $\mathfrak{p} \subset R$ disjoint from U is sent to its extension $U^{-1}\mathfrak{p}$.

This is especially important in the two basic examples above. If $U = R \setminus \mathfrak{p}$, then the prime ideals of $R_{\mathfrak{p}}$ correspond exactly to the prime ideals of R contained in $\mathfrak{p}$. If $U = \{1, f, f^2, \ldots\}$, then the prime ideals of $R[\frac{1}{f}]$ correspond exactly to the prime ideals of R which do not contain f.

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(12) **Ideals and Localization.** Let R be a ring, let $U \subset R$ be a multiplicatively closed set, and let $\eta : R \to U^{-1}R$ be the localization map.

(a) Let $I \subset R$ be an ideal. Prove that $U^{-1}I \coloneqq \{\frac{r}{u} \mid r \in I,\, u \in U\}$ is an ideal of $U^{-1}R$.

(b) Prove that $U^{-1}I$ is the extension of I along the localization map $\eta : R \to U^{-1}R$.

(c) Prove that $U^{-1}I = U^{-1}R$ if and only if $I \cap U \neq \varnothing$.

(d) Prove that $\frac{r}{u} \in U^{-1}I$ if and only if $tr \in I$ for some $t \in U$.

(e) Deduce that the contraction of $U^{-1}I$ back to R is

$$\eta^{-1}(U^{-1}I) = \left\{r \in R \;\middle|\; tr \in I \text{ for some } t \in U\right\}.$$

(f) Let $R = \mathbb{Z}$ and $U = \{1, 2, 4, 8, \ldots\}$. If $I = \langle 6 \rangle$, compute $U^{-1}I \subset U^{-1}\mathbb{Z}$ and its contraction back to $\mathbb{Z}$. Show that $I \subsetneq \eta^{-1}(U^{-1}I)$.

(g) Let $R = \mathbb{Z}$ and $U = \mathbb{Z} \setminus \langle 2 \rangle$. If $I = \langle 6 \rangle$, compute $U^{-1}I \subset \mathbb{Z}_{\langle 2 \rangle}$ and its contraction back to $\mathbb{Z}$.

(h) Let $R = \mathbb{K}[x, y]$ and $U = \{1, x, x^2, \ldots\}$. For each of the ideals $\langle x \rangle$, $\langle y \rangle$, and $\langle x, y \rangle$, describe its extension to $R[\frac{1}{x}]$. Which of these ideals become the whole ring after localization?

(13) **Prime Ideal Correspondence for Localization.** Let R be a ring, let $U \subset R$ be a multiplicatively closed set, and let $\eta : R \to U^{-1}R$ be the localization map.

(a) Let $\mathfrak{q} \subset U^{-1}R$ be a prime ideal. Prove that $\eta^{-1}(\mathfrak{q})$ is a prime ideal of R and that $\eta^{-1}(\mathfrak{q}) \cap U = \varnothing$.

(b) Let $\mathfrak{p} \subset R$ be a prime ideal such that $\mathfrak{p} \cap U = \varnothing$. Prove that $U^{-1}\mathfrak{p}$ is a proper prime ideal of $U^{-1}R$. Hint: Use the criterion from the previous exercise for when a fraction belongs to $U^{-1}\mathfrak{p}$.

(c) Let $\mathfrak{p} \subset R$ be a prime ideal such that $\mathfrak{p} \cap U = \varnothing$. Prove that the contraction of $U^{-1}\mathfrak{p}$ is $\mathfrak{p}$.

(d) Let $\mathfrak{q} \subset U^{-1}R$ be a prime ideal. Prove that the extension of its contraction is again $\mathfrak{q}$, i.e.

$$U^{-1}\eta^{-1}(\mathfrak{q}) = \mathfrak{q}.$$

Hint: If $\frac{r}{u} \in U^{-1}R$, then $\frac{r}{u} \in \mathfrak{q}$ if and only if $\frac{r}{1} \in \mathfrak{q}$, since $\frac{u}{1}$ is a unit.

(e) Conclude that extension and contraction give an order-preserving bijection

$$\operatorname{Spec}(U^{-1}R) \longleftrightarrow \left\{\mathfrak{p} \in \operatorname{Spec}(R) \;\middle|\; \mathfrak{p} \cap U = \varnothing\right\}.$$

(f) Specialize to the case $U = R \setminus \mathfrak{p}$. Prove that the prime ideals of $R_{\mathfrak{p}}$ correspond to the prime ideals of R contained in $\mathfrak{p}$.

(g) Specialize to the case $U = \{1, f, f^2, \ldots\}$. Prove that the prime ideals of $R[\frac{1}{f}]$ correspond to the prime ideals of R which do not contain f.

(h) Describe the prime ideals of $\mathbb{Z}_{\langle p \rangle}$ for a prime number p.

Localization is not limited to rings: we can also localize modules in essentially the same way. Let R be a ring and let M be an R-module. If $U \subset R$ is a multiplicatively closed subset, we want to build a new module $U^{-1}M$ by formally dividing elements of M by the elements of U. The elements of $U^{-1}M$ are fractions $\frac{m}{u}$, where $m \in M$ and $u \in U$. So the numerators come from the module M, while the denominators come from the set U which lives in the ring. Of course, we must introduce a criterion of when two fractions $\frac{m_1}{u_1}$ and $\frac{m_2}{u_2}$ are equal. Just as when localizing a ring we say

$$\frac{m_1}{u_1} = \frac{m_2}{u_2} \quad \text{in } U^{-1}M \quad \Longleftrightarrow \quad t(u_2 m_1 - u_1 m_2) = 0 \quad \text{in } M \text{ for some } t \in U$$

The factor of $t \in U$ is needed in order to ensure this is a transitive relation. Define addition on $U^{-1}M$ by

$$\frac{m_1}{u_1} + \frac{m_2}{u_2} = \frac{m_1 u_2 + m_2 u_1}{u_1 u_2}$$

which makes $U^{-1}M$ into an abelian group where the identity is 0/1. The abelian group $U^{-1}M$ has the structure of both an R-module and a $U^{-1}R$-module. The R-module structure and the $U^{-1}R$-module structure are given by

$$r\left(\frac{m}{u}\right) = \frac{rm}{u} \qquad \text{and} \qquad \left(\frac{r}{u_1}\right)\left(\frac{m}{u_2}\right) = \frac{rm}{u_1 u_2}$$

respectively. These module structures are compatible: the R-module structure on $U^{-1}M$ is the one obtained from the $U^{-1}R$-module structure by restriction of scalars along the localization map $\eta : R \to U^{-1}R$.

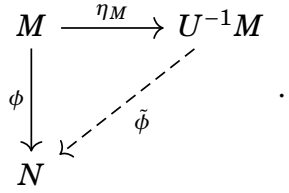
In general, if $\phi : R \to S$ is any ring homomorphism and N is an S-module, then N becomes an R-module by defining $rn := \phi(r)n$ for $r \in R$ and $n \in N$. This construction is called *restriction of scalars*. One checks immediately that this gives N a well-defined R-module structure, since ϕ preserves addition, multiplication, and the identity. In our situation, the localization map $\eta : R \to U^{-1}R$ is defined by $\eta(r) = r/1$. Since $U^{-1}M$ is a $U^{-1}R$-module, restriction of scalars along η gives $U^{-1}M$ an R-module structure via

$$r \cdot \frac{m}{u} = \eta(r) \cdot \frac{m}{u} = \frac{r}{1} \cdot \frac{m}{u} = \frac{rm}{u},$$

which is exactly the R-module structure we defined above. So there is really only one module structure at play: the $U^{-1}R$-module structure is the fundamental one, as the R-module structure is inherited from it.

Just as the localization of a ring is characterized by a universal property, so is the localization of a module. There is a natural R-module homomorphism $\eta_M : M \to U^{-1}M$ given by $m \mapsto \frac{m}{1}$, which we call the *localization map*. Note that the target $U^{-1}M$ carries a $U^{-1}R$-module structure, and in particular every element of U acts invertibly on $U^{-1}M$: for any $u \in U$, multiplication by $u/1 \in U^{-1}R$ is an isomorphism with inverse given by multiplication by $1/u$. The pair $U^{-1}M$ and η_M satisfies the following universal property: Let N be any $U^{-1}R$-module, viewed as an R-module by restriction of scalars along $\eta : R \to U^{-1}R$. Then for every

R-module homomorphism $\phi : M \to N$, there exists a unique $U^{-1}R$-module homomorphism $\tilde{\phi} : U^{-1}M \to N$ such that the following diagram commutes:



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(14) **Constructing $U^{-1}M$.** Let R be a ring, $U \subset R$ a multiplicatively closed set, and M be an R-module. Define a relation on $M \times U$ by $(m_1, u_1) \sim (m_2, u_2)$ if and only if $t(m_1 u_2 - m_2 u_1) = 0$ in M for some $t \in U$.

(a) Prove that $\sim$ is an equivalence relation on $M \times U$.

(b) Let $\frac{m}{u}$ denote the equivalence class of a pair (m, u) under the equivalence relation $\sim$ defined in the previous part. Prove that the operation:

$$\frac{m_1}{u_1} + \frac{m_2}{u_2} = \frac{m_1 u_2 + m_2 u_1}{u_1 u_2}$$

is well-defined. Show that this makes $U^{-1}M$ an abelian group.

(c) Show that the operations

$$r\left(\frac{m}{u}\right) = \frac{rm}{u} \qquad \text{and} \qquad \left(\frac{r}{u_1}\right)\left(\frac{m}{u_2}\right) = \frac{rm}{u_1 u_2}$$

are well-defined. Deduce that $U^{-1}M$ has the structure of both an R-module and a $U^{-1}R$-module.

(d) Let $\eta_M : M \to U^{-1}M$ be the function $m \mapsto \frac{m}{1}$. Show that η_M is an R-module homomorphism.

(e) Prove that $\eta_M(m) = 0$ if and only if $um = 0$ for some $u \in U$. Deduce that η_M is injective if and only if no element of U acts as a zero-divisor on M.

(15) **Examples of Localizing Modules.** Discuss ways to explicitly understand $U^{-1}M$ in each of the following cases. Describe the natural map $M \to U^{-1}M$ and determine whether it is injective.

(a) $R = \mathbb{Z}$, $U = \mathbb{Z} \setminus \{0\}$, and $M = \mathbb{Z}$.

(b) $R = \mathbb{Z}$, $U = \{1, 2, 4, 8, \ldots\}$, and $M = \mathbb{Z}$.

(c) $R = \mathbb{Z}$, $U = \{1, 2, 4, 8, \ldots\}$, and $M = \mathbb{Z}/\langle 6 \rangle$.

(d) $R = \mathbb{Z}$, $U = \mathbb{Z} \setminus \langle 2 \rangle$, and $M = \mathbb{Z}/\langle 6 \rangle$.

(e) $R = \mathbb{K}[x]$, $U = \{1, x, x^2, \ldots\}$, and $M = R/\langle x^3 \rangle$.

(f) $R = \mathbb{K}[x]$, $U = R \setminus \langle x \rangle$, and $M = R/\langle x^3 \rangle$.

(g) $R = \mathbb{K}[x, y]$, $U = \{1, y, y^2, \ldots\}$, and $M = R/\langle xy \rangle$.

(h) R is an arbitrary ring, M is an R-module, and U contains an element u such that $uM = 0$.

(16) **Universal Property of $U^{-1}M$.** Let R be a ring, let $U \subset R$ be a multiplicatively closed set, and let M be an R-module. Let N be a $U^{-1}R$-module, which we view as an R-module via the localization map $\eta : R \to U^{-1}R$. Let $\psi : M \to N$ be an R-module homomorphism.

(a) Define a function $\tilde{\psi} : U^{-1}M \to N$ by $\tilde{\psi}(\frac{m}{u}) = \frac{1}{u}\psi(m)$. Prove that $\tilde{\psi}$ is well-defined.

(b) Prove that $\tilde{\psi}$ is a homomorphism of $U^{-1}R$-modules.

(c) Show that $\tilde{\psi}$ makes the following diagram commute:

$$\begin{array}{ccc} M & \xrightarrow{\eta_M} & U^{-1}M \\ {\scriptstyle\psi}\downarrow & \swarrow_{\tilde{\psi}} & \\ N & & \end{array}$$

(d) Prove that $\tilde{\psi}$ is the unique $U^{-1}R$-module homomorphism making the diagram commute.

(e) Deduce that $U^{-1}M$ is unique up to unique isomorphism as a $U^{-1}R$-module satisfying this universal property.

(17) $U^{-1}M$ **as a Tensor Product.** Let R be a ring, $U \subset R$ be a multiplicatively closed set, and let M be an R-module. We view $U^{-1}R$ as an R-module via the localization map $\eta : R \to U^{-1}R$.

(a) Show that the function given below is a well-defined bilinear map of R-modules:

$$\begin{array}{ccc} U^{-1}R \times M & \xrightarrow{\overline{\alpha}} & U^{-1}M \\ \left(\frac{r}{u}, m\right) & \longmapsto & \frac{rm}{u} \end{array}$$

(b) Use the universal property of tensor products to construct an R-module homomorphism $\alpha : U^{-1}R \otimes_R M \to U^{-1}M$. Find a formula for $\alpha\left(\frac{r}{u} \otimes m\right)$.

(c) Show that the function β given below is a well-defined R-module homomorphism:

$$\begin{array}{ccc} U^{-1}M & \xrightarrow{\beta} & U^{-1}R \otimes_R M \\ \frac{m}{u} & \longmapsto & \frac{1}{u} \otimes m \end{array}.$$

(d) Prove that α and β are mutual inverses. Conclude that $U^{-1}M \cong U^{-1}R \otimes_R M$ as R-modules.

(e) Deduce that $U^{-1}(R^n) \cong (U^{-1}R)^n$ as $U^{-1}R$-modules.

(f) If $f : M \to N$ is an R-module homomorphism, define a natural $U^{-1}R$-module homomorphism $U^{-1}f : U^{-1}M \to U^{-1}N$. What is its formula on fractions?

WORKSHEET 4.3: ZARISKI TOPOLOGY PT. II

Throughout this course, "ring" means a *commutative* ring with unity. Recall that, given an ideal $I \subset R$, we define the vanishing set of I to be

$$\mathbb{V}(I) := \{[P] \in \mathrm{Spec}(R) \mid I \subset P\}.$$

The *Zariski topology* on $\mathrm{Spec}(R)$ is then defined by declaring the closed sets to be the subsets of the form $\mathbb{V}(I)$ as I ranges over all ideals of R. In Worksheet 2.1 we verified that this is indeed a topology and studied it in detail. For example, we showed that any ring map $\phi : R \to S$ induces a continuous map of topological spaces $\phi^{\#} : \mathrm{Spec}(S) \to \mathrm{Spec}(R)$ given by $[P] \mapsto [\phi^{-1}(P)]$. (More precisely, Spec is a contravariant functor from the category of commutative rings to the category of topological spaces.) Further, we showed that the natural quotient map $\pi : R \to R/I$ induces a closed embedding $\pi^{\#} : \mathrm{Spec}(R/I) \to \mathrm{Spec}(R)$ whose image is $\mathbb{V}(I)$.

Now that we have localization in our toolbox, we return to studying the Zariski topology on $\mathrm{Spec}(R)$. The main slogan is that localizing a ring lets us study the part of $\mathrm{Spec}(R)$ where a given "function" is nonzero. We have not yet defined functions on $\mathrm{Spec}(R)$, but in many familiar examples elements of R really are functions—for example, when R is the coordinate ring of an algebraic set. Recall the following theorem from the last worksheet.

Theorem 1. *Let R be a ring and let $U \subset R$ be a multiplicatively closed set. The localization map $\eta : R \to U^{-1}R$ induces an inclusion-preserving bijection:*

$$\left\{\begin{matrix}\textit{prime ideals in} \\ U^{-1}R\end{matrix}\right\} \xrightarrow{\sim} \left\{\begin{matrix}\textit{prime ideals in } R \\ \textit{disjoint from } U\end{matrix}\right\}.$$

$$P \longmapsto \eta^{-1}(P)$$

We begin by using this theorem in the case that $U = \{1, f, f^2, f^3, \ldots\}$ for an element $f \in R$ to study what are called the basic open sets of $\mathrm{Spec}(R)$. A *basic open subset* – also called a *distinguished open subset* – of $\mathrm{Spec}(R)$ is a set of the form

$$D(f) := \left\{[P] \in \mathrm{Spec}(R) \;\middle|\; f \notin P\right\} \subset \mathrm{Spec}(R)$$

for some element $f \in R$. We will show that these subsets form a nice basis for the Zariski topology on $\mathrm{Spec}(R)$. A *basis* for a topological space X is a collection $\mathscr{B}$ of open subsets of X such that for every open set $U \subset X$ and every point $x \in U$, there exists some $B \in \mathscr{B}$ with $x \in B \subset U$. Equivalently, every open subset of X can be written as a union of elements of $\mathscr{B}$.

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(1) **Basic Open Sets.** Let R be a ring and $f, g \in R$.

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(a) Prove directly from the definition of the Zariski topology that $D(f)$ is an open set.

(b) Compute $D(0)$ and $D(1)$.

(c) Prove that $D(f) = D(f^n)$ for all $n \geq 1$.

(d) Prove that $D(f) \cap D(g) = D(fg)$. Use induction to find a formula for the intersection of finitely many basic opens.

(e) Prove that $D(f) \subset D(g)$ if and only if $f \in \sqrt{\langle g \rangle}$. Hint: Take complements to rewrite $D(f) \subset D(g)$ as $\mathbb{V}(g) \subset \mathbb{V}(f)$, and use the fact that $\mathbb{V}(I) \subset \mathbb{V}(J)$ if and only if $\sqrt{J} \subset \sqrt{I}$.

(f) Prove that $D(f) = D(g)$ if and only if $\sqrt{\langle f \rangle} = \sqrt{\langle g \rangle}$.

(g) Let $U \subset \operatorname{Spec}(R)$ be an open subset and let $[\mathfrak{p}] \in U$. Show that there exists $f \in R$ such that $[\mathfrak{p}] \in D(f) \subset U$. Hint: Since U is open, write $\operatorname{Spec}(R) \setminus U = \mathbb{V}(I)$ for some ideal I. Then use the fact that $[\mathfrak{p}] \notin \mathbb{V}(I)$ to choose $f \in I \setminus \mathfrak{p}$.

(h) Deduce that the basic open sets form a basis for the Zariski topology on $\operatorname{Spec}(R)$.

(2) **Examples of Basic Opens.** Explicitly describe the basic open set $D(f) \subset \operatorname{Spec}(R)$ in each of the following cases.

(a) $R = \mathbb{K}[x]$ and $f = x$.

(b) $R = \mathbb{K}[x]$ and $f = x - a$ for $a \in \mathbb{K}$.

(c) $R = \mathbb{Z}$ and $f = p$ for a prime number p.

(d) $R = \mathbb{K}[x,y]/\langle xy \rangle$ and $f = \overline{x}$.

(e) $R = \mathbb{K}[x,y]/\langle xy \rangle$ and $f = \overline{y}$.

(f) $R = \mathbb{Z}/\langle 12 \rangle$ and $f = \overline{2}$.

(g) $R = \mathbb{Z}/\langle 12 \rangle$ and $f = \overline{3}$.

(h) $R = \mathbb{Z}/\langle 12 \rangle$ and $f = \overline{6}$.

(3) **Induced Map by Localization.** Let $U \subset R$ be a multiplicatively closed set and let $\eta : R \to U^{-1}R$ be the localization map. In this exercise we will study the induced map $\eta^{\#} : \operatorname{Spec}(U^{-1}R) \to \operatorname{Spec}(R)$.

(a) Use the correspondence between prime ideals in $U^{-1}R$ and R given in Theorem 1 to prove that $\eta^{\#}$ is injective and the image of $\eta^{\#}$ is

$$X_U := \left\{ [\mathfrak{p}] \in \operatorname{Spec}(R) \;\middle|\; \mathfrak{p} \cap U = \varnothing \right\} \subset \operatorname{Spec}(R).$$

(b) Let $R = \mathbb{Z}$ and $U = \mathbb{Z} \setminus \langle p \rangle$ for a prime $p \in \mathbb{Z}$. Show that $X_U = \{\langle 0 \rangle, \langle p \rangle\}$. Prove that this is not open in $\operatorname{Spec}(\mathbb{Z})$.

(c) Let $R = \mathbb{Z}$ and $U = \{2^i 3^j \mid i,j \in \mathbb{Z}_{\geq 0}\}$. Show that $X_U = \{\langle 0 \rangle\} \cup \{\langle p \rangle \mid p \text{ prime}, p \neq 2,3\}$. Show that this is an open subset of $\operatorname{Spec}(\mathbb{Z})$.

(d) Describe X_U when $R = \mathbb{C}[x,y]$ and $U = \{(xy)^n \mid n \in \mathbb{Z}_{\geq 0}\}$.

(e) Prove that $X_U = \bigcap_{f \in U} D(f)$.

(f) Deduce that if U is a multiplicative set generated by finitely many elements then X_U is open.

(g) Fix an element $\frac{r}{u} \in U^{-1}R$. Prove that $\eta^{\#}(D(\frac{r}{u})) = D(r) \cap X_U$. Hint: In $U^{-1}R$, the element u is a unit, so $D(\frac{r}{u}) = D(\frac{r}{1})$. Then use part (a) to identify the image of $\eta^{\#}$.

(h) Prove that $\eta^{\#} : \mathrm{Spec}(U^{-1}R) \to \mathrm{Spec}(R)$ is a homeomorphism onto its image X_U (considered with the subspace topology).

(i) Specializing to the case that $U = \{1, f, f^2, f^3, \ldots\}$, prove that $\eta^{\#} : \mathrm{Spec}(R[\frac{1}{f}]) \to \mathrm{Spec}(R)$ is a homeomorphism onto the basic open subset $D(f)$.

Let X be a topological space. An *open cover* of X is a collection $\{U_i\}_{i \in \Lambda}$ of open subsets of X such that $X = \bigcup_{i \in \Lambda} U_i$. A *finite subcover* is a finite subcollection $\{U_{i_1}, \ldots, U_{i_n}\}$ that still covers X, i.e. $X = U_{i_1} \cup \cdots \cup U_{i_n}$.

With these terms in hand, recall that X is *quasi-compact* if every open cover admits a finite subcover. More generally, a subset $Y \subset X$ is *quasi-compact* if every cover of Y by open subsets of X admits a finite subcover; equivalently, Y is quasi-compact in the subspace topology. In the Hausdorff setting this is simply compactness; the prefix "quasi-" is standard in algebraic geometry because Zariski topologies are virtually never Hausdorff, and one wants to keep the two notions distinct.

Every affine spectrum $\mathrm{Spec}R$ is quasi-compact. The proof is a clean illustration of how the basic opens $D(f)$ serve as a bridge between topology and commutative algebra. If $\{D(f_i)\}_{i \in \Lambda}$ covers $\mathrm{Spec}(R)$, then no prime ideal contains every f_i, so the f_i generate the unit ideal. But any expression $1 = \sum a_i f_i$ involves only finitely many terms, so a finite subfamily already generates the unit ideal and the corresponding $D(f_i)$'s already cover. The argument generalizes immediately: any open subset that can be written as a finite union of basic opens is itself quasi-compact.

This kind of translation—reformulating a covering condition as an ideal-membership statement, then exploiting the finiteness built into ring equations—is one of the workhorses of scheme theory.

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(4) **Quasi-compactness of the Zariski Topology.** Let R be a ring.

(a) Let $\Lambda \subset R$ be a collection of elements of R. Prove that $\{D(f)\}_{f \in \Lambda}$ is an open cover of $\mathrm{Spec}(R)$ if and only if $\langle f \mid f \in \Lambda \rangle$ is the unit ideal.

(b) Let $\Lambda \subset R$ be a collection of elements of R. Explain why if $\langle f \mid f \in \Lambda \rangle$ is the unit ideal then there exist finitely many elements $f_1, \ldots, f_n \in \Lambda$ such that $\langle f_1, \ldots, f_n \rangle$ is also the unit ideal.

(c) Deduce that every cover of $\mathrm{Spec}(R)$ by basic open subsets has a finite subcover.

(d) Use the fact that the $D(f)$ form a basis to prove that $\mathrm{Spec}(R)$ is quasi-compact. Hint: Refine each open set in the cover by basic open neighborhoods using the basis property from Problem 1.

(e) Prove that $\mathbb{V}(I)$ is quasi-compact for every ideal $I \subseteq R$. Hint: Use the homeomorphism $\mathrm{Spec}(R/I) \cong \mathbb{V}(I)$ from Worksheet 2.1.

(f) Let $I \subset R$ be an ideal and set $U := \mathrm{Spec}(R) \setminus \mathbb{V}(I)$.

(i) Show that $U = \bigcup_{f \in I} D(f)$.

(ii) Show that if U is quasi-compact, then $U = D(f_1) \cup \cdots \cup D(f_n)$ for some $f_1, \ldots, f_n \in I$.

(iii) Deduce that U is quasi-compact if and only if $\sqrt{I} = \sqrt{\langle f_1, \ldots, f_n \rangle}$ for some $f_1, \ldots, f_n \in I$.

Equivalently, U is quasi-compact if and only if $\sqrt{I}$ is the radical of a finitely generated ideal.

(g) Deduce that if R is Noetherian, every open subset of $\mathrm{Spec}(R)$ is quasi-compact. Hint: First show that $\mathrm{Spec}(R) \setminus \mathbb{V}(I) = \bigcup_{f \in I} D(f)$.

(5) **Counterexamples to Quasi-compactness of Subsets.** In the previous problem we saw that $\mathrm{Spec}(R)$ and every closed subset $\mathbb{V}(I) \subset \mathrm{Spec}(R)$ are quasi-compact in the Zariski topology. We also saw that when R is Noetherian every open subset of $\mathrm{Spec}(R)$ is quasi-compact. In this exercise we explore some pathologies for open subsets when R is not Noetherian.

(a) Let $R = \mathbb{K}[x_1, x_2, \ldots]$ and $I = \langle x_1, x_2, \ldots \rangle$. Consider the open subset $U := \mathrm{Spec}(R) \setminus \mathbb{V}(I)$. Prove that $\{D(x_i)\}_{i=1}^{\infty}$ is an open cover of U, which has no finite subcover. Hence an open subset of $\mathrm{Spec}(R)$ need not be quasi-compact. Hint: If you try to cover U by only $D(x_1), \ldots, D(x_n)$, consider the prime ideal $\langle x_1, \ldots, x_n \rangle$.

(b) Explain why each basic open set $D(x_i)$ is quasi-compact. Deduce that an arbitrary union of quasi-compact open subsets of $\mathrm{Spec}(R)$ need not be quasi-compact.

(6) **Checking Properties on a Finite Cover.** Let R be a ring. Suppose that $f_1, \ldots, f_n \in R$ are such that $\{D(f_i)\}_{i=1}^{n}$ is a finite open cover of $\mathrm{Spec}(R)$, i.e., $\mathrm{Spec}(R) = D(f_1) \cup D(f_2) \cup \cdots \cup D(f_n)$. We will see that certain ring-theoretic properties of R can be checked locally by looking at the rings $R[\frac{1}{f_i}]$.

(a) Let $r \in R$. Prove that if $r/1 = 0$ in $R[\frac{1}{f_i}]$ for all i then $r = 0$ in R. Hint: In $R[\frac{1}{f_i}]$, the equality $r/1 = 0$ means that $f_i^{n_i} r = 0$ for some $n_i \geq 0$.

(b) Deduce that if $R[\frac{1}{f_i}] = 0$ for all i then $R = 0$.

(c) Prove that if $R[\frac{1}{f_i}]$ is reduced for all i, then R is reduced.

An element $r \in R$ is *nilpotent* if $r^n = 0$ for some integer $n \geq 1$. If $I \subset R$ is an ideal, the *radical* of I is

$$\sqrt{I} := \left\{ r \in R \;\middle|\; r^n \in I \text{ for some } n \geq 1 \right\},$$

which we previously checked is an ideal of R containing I and $\sqrt{\sqrt{I}} = \sqrt{I}$. An ideal $I \subset R$ is *radical* if and only if $I = \sqrt{I}$. We saw that radical ideals play a crucial role in the study of algebraic sets as $\mathbb{I}(V)$ is always radical and Hilbert's Nullstellensatz gave an inclusion-reversing bijection between radical ideals in $\mathbb{K}[x_1, \ldots, x_n]$ and algebraic subsets of $\mathbb{K}^n$ when $\mathbb{K}$ is an algebraically closed field. These facts highlight the following motto: *(classical) geometry does not see nilpotents.* (Here classical means anything one encounters pre-scheme theory; topology, classical algebraic geometry, complex geometry, etc.)

We shall see that the invisibility of nilpotents to geometry also applies to spectra of rings. The *nilradical* of R is the set of all nilpotent elements of R, equivalently the radical of the zero ideal:

$$\mathrm{Nil}(R) := \sqrt{\langle 0 \rangle} = \left\{ r \in R \;\middle|\; r^n = 0 \text{ for some } n \geq 1 \right\}.$$

If $f \in \mathrm{Nil}(R)$ and $f^n = 0$, then $f^n \in I$ for every ideal $I \subset R$. In particular, if $P \subset R$ is a prime ideal, then $f^n = 0 \in P$, which implies that $f \in P$. Thus the nilradical of R is contained in every prime ideal of R. In the exercises below you will prove the reverse inclusion, giving the fundamental characterization

$$\mathrm{Nil}(R) = \bigcap_{\mathfrak{p} \in \mathrm{Spec}(R)} \mathfrak{p}.$$

Since prime ideals of $R/\mathrm{Nil}(R)$ are in bijection with prime ideals of R containing $\mathrm{Nil}(R)$, the equality above implies that prime ideals of R are in bijection with prime ideals of $R/\mathrm{Nil}(R)$. Geometrically, this means that the quotient map $\pi : R \to R/\mathrm{Nil}(R)$ induces a map of topological spaces

$$\mathrm{Spec}(R/\mathrm{Nil}(R)) \xrightarrow{\pi^\#} \mathrm{Spec}(R)$$

that is bijective and continuous. We will see shortly that this map is actually a homeomorphism; hence, topologically $\mathrm{Spec}(R)$ and $\mathrm{Spec}(R/\mathrm{Nil}(R))$ are indistinguishable. This is another instance of the motto: geometry does not see nilpotents. A ring R is called *reduced* if $\mathrm{Nil}(R) = \langle 0 \rangle$, i.e., if the only nilpotent element is 0. We often call $R/\mathrm{Nil}(R)$ the reduced ring associated to R and denote it $R_{\mathrm{red}} := R/\mathrm{Nil}(R)$.

More generally, the nilradical characterization extends to arbitrary ideals. One checks that $\mathrm{Nil}(R/I) = \sqrt{I}/I$, since the elements of R/I whose powers lie in I are exactly the images of elements of $\sqrt{I}$. Applying the nilradical formula in the ring R/I and pulling back to R gives

$$\sqrt{I} = \bigcap_{\substack{\mathfrak{p} \in \mathrm{Spec}(R) \\ \mathfrak{p} \supseteq I}} \mathfrak{p}.$$

This has immediate geometric consequences. For example, this gives another proof that $\mathbb{V}(I) = \mathbb{V}(\sqrt{I})$. Thus the Zariski-closed subsets of $\mathrm{Spec}(R)$ determined by I and $\sqrt{I}$ are identical. This interacts cleanly with quotient rings; in particular, an argument like the one above shows that $\mathrm{Spec}(R/I)$ and $\mathrm{Spec}(R/\sqrt{I})$ are homeomorphic.

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(7) **Nilradicals and Reduced Rings.** Let R be a ring.

(a) Explain why $\mathrm{Nil}(R)$ is an ideal and $\mathrm{Nil}(R) \subset \bigcap_{\mathfrak{p} \in \mathrm{Spec}(R)} \mathfrak{p}$.

(b) Prove that if $f \in R$ is not a nilpotent then $R[\frac{1}{f}] \neq 0$ and hence contains a prime ideal. Hint: If $R[\frac{1}{f}] = 0$, then $1 = 0$ in $R[\frac{1}{f}]$. What does that say about some power of f?

(c) Use the correspondence between prime ideals in $R[\frac{1}{f}]$ and R not containing f to show there is a prime ideal $\mathfrak{p} \subset R$ such that $f \notin \mathfrak{p}$.

(d) Deduce the theorem

$$\mathrm{Nil}(R) = \bigcap_{\mathfrak{p} \in \mathrm{Spec}(R)} \mathfrak{p}.$$

(e) Define R_{red} to be equal to $R/\mathrm{Nil}(R)$. Prove that R_{red} is reduced.

(f) Use the above characterization to prove that $D(f) = \varnothing$ if and only if f is a nilpotent element.

(g) Show that $D(f) = \mathrm{Spec}(R)$ if and only if f is a unit in R.

(h) Let $\pi : R \to R_{\mathrm{red}}$ be the natural quotient map. Prove that the induced map $\pi^{\#} : \mathrm{Spec}(R_{\mathrm{red}}) \to \mathrm{Spec}(R)$ is a homeomorphism. (Hint: We already know it is continuous. Use part 7d to show it is a bijection. Finally, show that the map is closed by proving that for every ideal $I \supseteq \mathrm{Nil}(R)$, the closed set $\mathbb{V}(I/\mathrm{Nil}(R)) \subseteq \mathrm{Spec}(R_{\mathrm{red}})$ maps to $\mathbb{V}(I) \subseteq \mathrm{Spec}(R)$.)

(i) Compute R_{red} for each of the following rings:

(i) $\mathbb{K}[\epsilon]/\langle \epsilon^2 \rangle$,

(ii) $\mathbb{K}[x]/\langle x^3 \rangle$,

(iii) $\mathbb{K}[x,y]/\langle x^2, xy \rangle$,

(iv) $\mathbb{Z}/12\mathbb{Z}$.

Given a ring homomorphism $\phi : R \to S$, we know that it induces a continuous map $\phi^{\#} : \mathrm{Spec}(S) \to \mathrm{Spec}(R)$ of topological spaces. A natural geometric question is: what is the fibre of this map over a point $[\mathfrak{p}] \in \mathrm{Spec}(R)$? Recall that if $f : X \to Y$ is a map of topological spaces, then the fibre over a point $y \in Y$ is just the preimage $f^{-1}(\{y\}) = \{x \in X \mid f(x) = y\} \subset X$, considered with the subspace topology. In algebraic geometry, authors often write X_y for the fibre over y when the map f is clear. Returning to the induced map $\phi^{\#}$, since we know that $\phi^{\#}([\mathfrak{q}]) = [\phi^{-1}(\mathfrak{q})]$, we can directly describe the fibre as

$$(\phi^{\#})^{-1}([\mathfrak{p}]) = \left\{ [\mathfrak{q}] \in \mathrm{Spec}(S) \;\middle|\; \phi^{-1}(\mathfrak{q}) = \mathfrak{p} \right\} \subset \mathrm{Spec}(S).$$

However, we would like a better, more algebraic description of this set. Answering this involves two conceptual steps: localization and reduction to the residue field. It is worth pausing to try to get an intuitive understanding of each step. As a roadmap: to compute the fibre over $[\mathfrak{p}]$, we first localize at $\mathfrak{p}$, then quotient by $\mathfrak{p}R_{\mathfrak{p}}$; this is what leads naturally to the fibre ring $\kappa(\mathfrak{p}) \otimes_R S$.

Let $f : X \to Y$ be any continuous map of topological spaces. The fibre over a point $y \in Y$ is local on Y in the sense that it really only depends on the behavior of f at the point y, or perhaps in an open neighborhood of y. More precisely, if $U \subset Y$ contains y and $V = f^{-1}(U)$, then

$$f^{-1}(\{y\}) = (f|_V)^{-1}(\{y\}),$$

where $f|_V : V \to U$ is the restriction of f. If we wished to be fancy, we might draw a commutative diagram of topological spaces

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ {\scriptstyle\iota}\uparrow & & \uparrow \\ V & \xrightarrow[f|_V]{} & U \end{array}$$

where ι is the inclusion of V into X. Since $V = f^{-1}(U)$ and $y \in U$, every point of the fibre over y already lies in V, so the displayed equality holds. In other words, restricting f to V does not change the fibre over y; more generally, we may replace X by any subset that still contains the fibre over y.

Returning to the algebraic situation $\phi^\# : \mathrm{Spec}(S) \to \mathrm{Spec}(R)$, we happen to know from the previous exercise on localization a very nice subset of $\mathrm{Spec}(R)$ containing the point $[\mathfrak{p}]$, namely

$$[\mathfrak{p}] \in X_U \subset \mathrm{Spec}(R) \qquad \text{where} \qquad U = R \setminus \mathfrak{p}.$$

Even better, the localization map $\eta : R \to R_\mathfrak{p}$ induces a homeomorphism from $\mathrm{Spec}(R_\mathfrak{p})$ onto X_U. We could thus hope to compute the fibre over $[\mathfrak{p}]$ by trying to factor $\phi^\#$ through $\eta^\#$, i.e. to find a commutative diagram of topological spaces as shown on the left below for some subset $V \subset \mathrm{Spec}(S)$.

$$\begin{array}{ccc} \mathrm{Spec}(S) & \xrightarrow{\phi^\#} & \mathrm{Spec}(R) \\ \uparrow & & {\scriptstyle\eta^\#}\uparrow \\ V & \dashrightarrow & \mathrm{Spec}(R_\mathfrak{p}) \end{array} \qquad\qquad \begin{array}{ccc} R & \xrightarrow{\phi} & S \\ {\scriptstyle\eta}\downarrow & & \downarrow \\ R_\mathfrak{p} & \dashrightarrow & ?? \end{array}.$$

It turns out we can find such a subset using the universal property of localization. To see why, let us just hope that V can be realized as the spectrum of some ring and that the dashed maps arise from ring homomorphisms. This means we are trying to complete the diagram of rings above on the right. Recall that the image of a multiplicatively closed set is again multiplicatively closed, so we may localize S at

$$T := \phi(R \setminus \mathfrak{p}) \subset S.$$

Let $\eta_S : S \to T^{-1}S$ be the localization map. We now have a diagram in which the diagonal map is the composition of ϕ and η_S:

$$\begin{array}{ccc} R & \xrightarrow{\phi} & S \\ {\scriptstyle\eta}\downarrow & \searrow & \downarrow{\scriptstyle\eta_S} \\ R_\mathfrak{p} & \xrightarrow{\widetilde{\phi}} & T^{-1}S \end{array}$$

where the diagonal map is just the composition of ϕ and η_S. Every element of $R \setminus \mathfrak{p}$ maps to a unit in $T^{-1}S$ under the diagonal map, so the universal property of localization gives a unique ring map $\widetilde{\phi} : R_\mathfrak{p} \longrightarrow T^{-1}S$

making the diagram commute. Passing to spectra gives a commutative diagram

$$\begin{array}{ccc} \operatorname{Spec}(S) & \xrightarrow{\phi^\#} & \operatorname{Spec}(R) \\ \uparrow{\scriptstyle \eta_S^\#} & & \uparrow{\scriptstyle \eta^\#} \\ \operatorname{Spec}(T^{-1}S) & \xrightarrow{\widetilde{\phi}^\#} & \operatorname{Spec}(R_\mathfrak{p}) \end{array}$$

and hence the fibre of $\phi^\#$ over $[\mathfrak{p}]$ is naturally identified with the fibre of $\widetilde{\phi}^\#$ over the point $[\mathfrak{p}R_\mathfrak{p}]$. Since $\eta^\#$ and $\eta_S^\#$ are homeomorphisms onto their images, this reduces the computation of the fibre of $\phi^\#$ over $[\mathfrak{p}]$ to the computation of the fibre of $\widetilde{\phi}^\#$ over $[\mathfrak{p}R_\mathfrak{p}]$.

Now we would like to understand this latter fibre algebraically. Since $R_\mathfrak{p}$ is a local ring with unique maximal ideal $\mathfrak{p}R_\mathfrak{p}$, a prime ideal $\mathfrak{q} \subset T^{-1}S$ lies over $\mathfrak{p}R_\mathfrak{p}$ if and only if it contains the extended ideal $\mathfrak{p}(T^{-1}S)$. Thus the fibre is exactly the closed subset

$$\mathbb{V}\big(\mathfrak{p}(T^{-1}S)\big) \subset \operatorname{Spec}(T^{-1}S),$$

so

$$(\phi^\#)^{-1}([\mathfrak{p}]) \cong \operatorname{Spec}\big(T^{-1}S/\mathfrak{p}(T^{-1}S)\big).$$

At first glance this quotient ring looks a bit mysterious, but tensor products package it very nicely. In fact there is a natural isomorphism $R_\mathfrak{p} \otimes_R S \cong T^{-1}S$. The *residue field* at $\mathfrak{p}$ is $\kappa(\mathfrak{p}) := R_\mathfrak{p}/\mathfrak{p}R_\mathfrak{p}$, which one checks is isomorphic to $\operatorname{Frac}(R/\mathfrak{p})$. (Recall $R/\mathfrak{p}$ is a domain since $\mathfrak{p}$ is a prime ideal.) There is then an isomorphism

$$\kappa(\mathfrak{p}) \otimes_R S \cong (R_\mathfrak{p} \otimes_R S)/\mathfrak{p}(R_\mathfrak{p} \otimes_R S) \cong T^{-1}S/\mathfrak{p}(T^{-1}S).$$

Therefore the fibre of $\phi^\#$ over $[\mathfrak{p}]$ is naturally homeomorphic to $\operatorname{Spec}(\kappa(\mathfrak{p}) \otimes_R S)$. The ring $\kappa(\mathfrak{p}) \otimes_R S$ is called the *fibre ring* of ϕ at $[\mathfrak{p}]$. The next exercise asks you to prove this carefully.

---

(8) **Residue Field.** Let R be a ring. Given a prime ideal $\mathfrak{p} \subset R$, the residue field of R at $\mathfrak{p}$ is the ring $R_\mathfrak{p}/\mathfrak{p}R_\mathfrak{p}$, which we denote by $\kappa(\mathfrak{p})$.

(a) Explain why $\mathfrak{p}R_\mathfrak{p}$ is the unique maximal ideal of $R_\mathfrak{p}$. Deduce that $\kappa(\mathfrak{p})$ is a field.

(b) Show that the composite map

$$R \longrightarrow R_\mathfrak{p} \longrightarrow \kappa(\mathfrak{p})$$

has kernel $\mathfrak{p}$. Deduce that it induces an injective ring map $R/\mathfrak{p} \hookrightarrow \kappa(\mathfrak{p})$.

(c) Prove that every nonzero element of $R/\mathfrak{p}$ becomes invertible in $\kappa(\mathfrak{p})$. Conclude that $\kappa(\mathfrak{p}) \cong \operatorname{Frac}(R/\mathfrak{p})$.

(d) Use part (c) to compute the residue field $\kappa(\mathfrak{p})$ in each of the following cases. In each case it may be helpful to first identify the quotient ring $R/\mathfrak{p}$:

(i) $R = \mathbb{Z}$ and $\mathfrak{p} = \langle p \rangle$ for a prime p.

(ii) $R = \mathbb{Z}$ and $\mathfrak{p} = \langle 0 \rangle$.

(iii) $R = \mathbb{K}[x]$ and $\mathfrak{p} = \langle x - a \rangle$ for $a \in \mathbb{K}$.

(iv) $R = \mathbb{K}[x]$ and $\mathfrak{p} = \langle 0 \rangle$.

(v) $R = \mathbb{R}[x]$ and $\mathfrak{p} = \langle x^2 + 1 \rangle$.

(vi) $R = \mathbb{K}[x, y]$ and $\mathfrak{p} = \langle y - x^2 \rangle$.

(vii) $R = \mathbb{K}[x, y]$ and $\mathfrak{p} = \langle x, y \rangle$.

(viii) $R = \mathbb{Z}[x]$ and $\mathfrak{p} = \langle p, x \rangle$ for a prime p.

(ix) $R = \mathbb{Z}[i]$ and $\mathfrak{p} = \langle 1 + i \rangle$.

(x) $R = \mathbb{Z}[\sqrt{-5}]$ and $\mathfrak{p} = \langle 2, 1 + \sqrt{-5} \rangle$.

(9) **Fibre Ring.** Let $\phi : R \to S$ be a ring map and fix a point $[\mathfrak{p}] \in \operatorname{Spec}(R)$. Set $U := R \setminus \mathfrak{p}$ and $T := \phi(U) \subset S$. Let $\widetilde{\phi} : R_{\mathfrak{p}} \longrightarrow T^{-1}S$ be the map induced by the universal property of localization discussed above.

(a) Prove that there is a natural isomorphism

$$\begin{aligned} R_{\mathfrak{p}} \otimes_R S &\xrightarrow{\sim} T^{-1}S \\ \tfrac{r}{u} \otimes s &\longmapsto \tfrac{\phi(r)s}{\phi(u)}. \end{aligned}$$

(b) Tensor the exact sequence

$$\mathfrak{p}R_{\mathfrak{p}} \longrightarrow R_{\mathfrak{p}} \longrightarrow \kappa(\mathfrak{p}) \longrightarrow 0$$

with S over R to obtain an exact sequence

$$(\mathfrak{p}R_{\mathfrak{p}}) \otimes_R S \longrightarrow R_{\mathfrak{p}} \otimes_R S \longrightarrow \kappa(\mathfrak{p}) \otimes_R S \longrightarrow 0.$$

Hint: View S as an R-module via ϕ, and use the right exactness of $- \otimes_R S$, which will be proved in Worksheet 5-1.

(c) Show that the image of

$$(\mathfrak{p}R_{\mathfrak{p}}) \otimes_R S \longrightarrow R_{\mathfrak{p}} \otimes_R S$$

is exactly the ideal $\mathfrak{p}(R_{\mathfrak{p}} \otimes_R S)$.

(d) Deduce that there are natural isomorphisms

$$\kappa(\mathfrak{p}) \otimes_R S \cong (R_{\mathfrak{p}} \otimes_R S)/\mathfrak{p}(R_{\mathfrak{p}} \otimes_R S) \cong T^{-1}S/\mathfrak{p}(T^{-1}S).$$

(10) **Fibres of Spec Maps.** Let $\phi : R \to S$ be a ring map and fix a point $[\mathfrak{p}] \in \operatorname{Spec}(R)$. Set $U := R \setminus \mathfrak{p}$ and $T := \phi(U) \subset S$. Let $\widetilde{\phi} : R_{\mathfrak{p}} \longrightarrow T^{-1}S$ be the map induced by the universal property of localization discussed above.

(a) Show that the fibre $(\phi^{\#})^{-1}([\mathfrak{p}])$ is naturally homeomorphic to the fibre of $\widetilde{\phi}^{\#} : \operatorname{Spec}(T^{-1}S) \longrightarrow \operatorname{Spec}(R_{\mathfrak{p}})$ over the point $[\mathfrak{p}R_{\mathfrak{p}}]$.

(b) Let $\mathfrak{q} \in \operatorname{Spec}(T^{-1}S)$. Prove that $\widetilde{\phi}^{-1}(\mathfrak{q}) = \mathfrak{p}R_{\mathfrak{p}}$ if and only if $\mathfrak{p}(T^{-1}S) \subseteq \mathfrak{q}$. Hint: Use that $R_{\mathfrak{p}}$ is a local ring with unique maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$.

(c) Deduce that

$$(\phi^{\#})^{-1}([\mathfrak{p}]) \cong \mathbb{V}\big(\mathfrak{p}(T^{-1}S)\big) \subset \operatorname{Spec}(T^{-1}S),$$

and hence that there is a homeomorphism of topological spaces $(\phi^{\#})^{-1}([\mathfrak{p}]) \cong \operatorname{Spec}\big(T^{-1}S/\mathfrak{p}(T^{-1}S)\big)$.

(d) Use Exercise 9 to conclude that $(\phi^{\#})^{-1}([\mathfrak{p}]) \cong \operatorname{Spec}(\kappa(\mathfrak{p}) \otimes_R S)$.

(e) Deduce that the fibre over $[\mathfrak{p}]$ is empty if and only if $\kappa(\mathfrak{p}) \otimes_R S = 0$.

(11) **Computing Fibres of Spec Maps.** In this problem, we use the previous exercise to compute several examples:

(a) Let $\phi : \mathbb{Z} \to \mathbb{Z}[x]$ be the inclusion. Compute the fibre of the induced map $\phi^{\#} : \operatorname{Spec}(\mathbb{Z}[x]) \to \operatorname{Spec}(\mathbb{Z})$ over the generic point $\langle 0 \rangle$ and over a closed point $\langle p \rangle$ for $p \in \mathbb{Z}$ prime.

(b) Fix an integer $n \geq 2$, and consider the localization map $\eta : \mathbb{Z} \to \mathbb{Z}[\frac{1}{n}]$. For which points of $\operatorname{Spec}(\mathbb{Z})$ is the fibre with respect to $\eta^{\#}$ empty? Compute the nonempty fibres.

(c) Let $\mathbb{K}$ be a field and consider the map

$$\mathbb{K}[t] \longrightarrow \mathbb{K}[x, y], \qquad t \longmapsto x.$$

Compute the fibre of the induced map over $\langle 0 \rangle \in \operatorname{Spec}(\mathbb{K}[t])$ and the fibre over $\langle t - a \rangle$ for $a \in \mathbb{K}$.

(d) Consider the inclusion of $\mathbb{C}$-algebras $\mathbb{C}[t] \to \mathbb{C}[x, y, t]/\langle xy - t \rangle$ given by $t \mapsto t$.

(i) Compute the fibre over the generic point $\langle 0 \rangle$.

(ii) Compute the fibre over $\langle t - a \rangle$ for $a \neq 0$.

(iii) Compute the fibre over $\langle t \rangle$.

(iv) Draw the corresponding map on closed points and compare it with the fibres above.

WORKSHEET 6.1: EXACTNESS PT. I

Throughout this course, "ring" means *commutative* ring with unity. In the previous worksheets we introduced modules, homomorphisms, quotient modules, free modules, Hom, tensor products, and localization. The goal of this worksheet is to organize these constructions around one central idea: *exactness*. Exactness is a way to say that one map accounts for exactly the elements killed by the next map. It packages kernels, images, quotient modules, presentations, and many theorems into one flexible language.

Let R be a ring. A *sequence* of R-modules is $\ldots, M_0, M_1, M_2, \ldots$ together with R-module homomorphisms $\phi_i : M_i \to M_{i-1}$. We often arrange a sequence into a diagram like

$$\cdots \xrightarrow{\phi_{i+1}} M_i \xrightarrow{\phi_i} M_{i-1} \xrightarrow{\phi_{i-1}} M_{i-2} \longrightarrow \cdots.$$

We are most interested in sequences when the maps are related in some nice way that allows us to compare the image of one map with the kernel of the next. A *chain complex*, or simply a *complex*, of R-modules is a sequence in which $\phi_{i-1} \circ \phi_i = 0$ for every i. Equivalently, $\operatorname{img}(\phi_i) \subset \ker(\phi_{i-1})$ for all i. We often write $(M_\bullet, \phi_\bullet)$ for a complex and call the maps ϕ_i *differentials*. A complex is *exact at* M_i if $\operatorname{img}(\phi_{i+1}) = \ker(\phi_i)$. One way to measure the failure, or success, of a complex to be exact at M_i is by looking at the *i-th homology group* of $(M_\bullet, \phi_\bullet)$, which is defined to be the quotient R-module:

$$H_i\left(M_\bullet, \phi_\bullet\right) \coloneqq \frac{\ker\left(\phi_i\right)}{\operatorname{img}\left(\phi_{i+1}\right)}.$$

The definition of a complex guarantees this quotient makes sense, since $\operatorname{img}(\phi_{i+1}) \subset \ker(\phi_i)$. Equivalently, a complex is exact at M_i if and only if $H_i(M_\bullet, \phi_\bullet) = 0$. (Here 0 denotes the zero R-module.) A complex is *exact* if it is exact at every module where this condition makes sense. Thus every exact sequence is a complex, but a complex need not be exact.

Complexes appear throughout commutative algebra and other areas of mathematics. Perhaps the most important special case of a complex is a short exact sequence. A *short sequence* is a complex of the form

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0.$$

It is called a *short exact sequence* if it is exact at A, B, and C. Strictly speaking, a complex extends infinitely in both directions, so the diagram above is shorthand for

$$\cdots \longrightarrow 0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0 \longrightarrow \cdots,$$

where all modules outside the displayed range are zero. We will follow the standard convention of suppressing these trailing and leading zero modules. Saying that this sequence is exact is the same as saying that i is injective, p is surjective, and $\operatorname{img}(i) = \ker(p)$. Thus a short exact sequence says that A is being identified with a submodule of B, and that the quotient of B by this submodule is isomorphic to C.

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(1) **First Encounters with Exactness.** Let A, B, C be R-modules.

(a) Prove that a sequence

$$0 \longrightarrow A \xrightarrow{i} B$$

is exact at A if and only if i is injective.

(b) Prove that a sequence

$$B \xrightarrow{p} C \longrightarrow 0$$

is exact at C if and only if p is surjective.

(c) Prove carefully that

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

is exact if and only if i is injective, p is surjective, and $\operatorname{img}(i) = \ker(p)$.

(d) If

$$A \xrightarrow{f} B \xrightarrow{g} C$$

is exact at B, prove that $g \circ f = 0$.

(e) Give an example of maps of R-modules

$$A \xrightarrow{f} B \xrightarrow{g} C$$

such that $g \circ f = 0$ but the sequence is not exact at B.

(f) Consider the complex below. Compute the homology at B in each of the following cases:

$$A \xrightarrow{f} B \xrightarrow{g} C$$

(i) $f = 0$ and $g = 0$.

(ii) f is injective and $g = 0$.

(iii) g is injective.

(2) **Examples of Short Exact Sequences.** Let R be a ring. In this problem we will see that a number of constructions we are quite familiar with can be framed in terms of short exact sequences.

(a) Let $I \subset R$ be an ideal. Construction a short exact sequence

$$0 \longrightarrow I \xrightarrow{\iota} R \xrightarrow{\pi} R/I \longrightarrow 0$$

where π is the natural quotient map and ι is the inclusion I into R.

(b) Let $N \subset M$ be a submodule. Prove that the sequence

$$0 \longrightarrow N \xrightarrow{\iota} M \xrightarrow{\pi} M/N \longrightarrow 0$$

is exact, where ι is the inclusion of N into M.

(c) Let $I \subset J \subset R$ be ideals. Construct a short exact sequence

$$0 \longrightarrow J/I \longrightarrow R/I \longrightarrow R/J \longrightarrow 0.$$

Hint: Define $J/I \to R/I$ by $j+I \mapsto j+I$ and define $R/I \to R/J$ by $r+I \mapsto r+J$. Check well-definedness before checking exactness.

(d) For a positive integer n, prove that multiplication by n gives

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0$$

which is a short exact sequence of $\mathbb{Z}$-modules.

(3) **Kernels, Cokernels, and Images.** Let $\phi : A \to B$ be a homomorphism of R-modules. The *cokernel* of ϕ is the R-module $\operatorname{coker}(\phi) := B/\operatorname{img}(\phi)$.

(a) Prove that the sequence

$$0 \longrightarrow \ker(\phi) \xrightarrow{\iota} A \xrightarrow{\phi} B$$

where ι is the inclusion of $\ker(\phi)$ into A is exact.

(b) Suppose

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C$$

is an exact sequence of R-modules. Prove that f induces an isomorphism $A \cong \ker(g)$.

(c) Prove that the sequence

$$A \xrightarrow{\phi} B \xrightarrow{\pi} \operatorname{coker}(\phi) \longrightarrow 0$$

where π is the natural quotient map is exact.

(d) Suppose

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

is an exact sequence of R-modules. Prove that g induces an isomorphism $\operatorname{coker}(f) \cong C$.

(e) Deduce there is an exact sequence of R-modules

$$0 \longrightarrow \ker(\phi) \xrightarrow{\iota} A \xrightarrow{\phi} B \xrightarrow{\pi} \operatorname{coker}(\phi) \longrightarrow 0 \ .$$

(f) Prove that ϕ induces an isomorphism $A/\ker(\phi) \cong \operatorname{img}(\phi)$, and deduce that the exact sequence in part (e) factors through $\operatorname{img}(\phi)$ as two short exact sequences

$$0 \longrightarrow \ker(\phi) \xrightarrow{\iota} A \xrightarrow{\bar{\phi}} \operatorname{img}(\phi) \longrightarrow 0 \quad \text{and} \quad 0 \longrightarrow \operatorname{img}(\phi) \xrightarrow{j} B \xrightarrow{\pi} \operatorname{coker}(\phi) \longrightarrow 0$$

where $\bar{\phi}$ is ϕ with codomain restricted to $\operatorname{img}(\phi)$ and j is the inclusion.

(4) **Presentations of Modules.** Let R be a ring. A *presentation* of an R-module M is an exact sequence of the form

$$F_1 \xrightarrow{\varphi} F_0 \longrightarrow M \longrightarrow 0$$

where F_0 and F_1 are free modules.

(a) Explain why a presentation of M as above says that M is obtained from the generators of F_0 by imposing the relations coming from $\varphi(F_1)$. Hint: Use exactness to identify M with $F_0/\mathrm{img}(\varphi)$.

(b) Let $R = \mathbb{K}[x]$ and $M = R/\langle x^3 \rangle$. Write a presentation of M using finite free R-modules.

(c) Let $R = \mathbb{K}[x,y]$ and $M = R/\langle x^2, xy, y^2 \rangle$. Write a presentation of M using one free generator and three relations.

(d) Let A be an $m \times n$ matrix with entries in R, viewed as a map $R^n \to R^m$. Interpret the sequence

$$R^n \xrightarrow{A} R^m \longrightarrow \mathrm{coker}(A) \longrightarrow 0$$

as a presentation of $\mathrm{coker}(A)$.

We just saw that a short exact sequence $0 \to A \xrightarrow{i} B \xrightarrow{p} C \to 0$ of R-modules says that B has a submodule isomorphic to A (namely $\mathrm{img}\, i$) whose quotient is isomorphic to C (namely $B/\mathrm{img}\, i \cong C$). One might hope that knowing A and C determines B up to isomorphism, and in the simplest cases it does: $B \cong A \oplus C$. When this happens – and when the isomorphism is compatible with the maps i and p – the sequence is called split. More precisely, a short exact sequence is *split* (also called *split exact*) if there exists an isomorphism of R-modules $h : B \to A \oplus C$ such that the following diagram commutes

$$\begin{array}{ccccccccc}
0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C & \longrightarrow & 0 \\
 & & \| & & \downarrow{\scriptstyle h} & & \| & & \\
0 & \longrightarrow & A & \xrightarrow{\iota} & A \oplus C & \xrightarrow{\pi} & C & \longrightarrow & 0
\end{array}$$

where $\iota(a) = (a, 0)$ is the natural inclusion and $\pi(a, c) = c$ is the natural projection. Saying that the diagram commutes means that the two squares commute. This is the generic situation for vector spaces: every subspace has a complement, so every short exact sequence of vector spaces splits. Over more general rings, however, splitting is the exception rather than the rule.

We can characterize when a short exact sequence splits in terms of the existence of either a section of p or a retraction of i. A *section* of $p : B \to C$ is an R-module homomorphism $s : C \to B$ such that $p \circ s = \mathrm{Id}_C$. A *retraction* of $i : A \to B$ is an R-module homomorphism $r : B \to A$ such that $r \circ i = \mathrm{Id}_A$. Note that in general s is not an inverse to p, and r is not an inverse to i; the compositions in the other order need not be the identity.

Lemma 1 (Splitting Lemma)**.** *Let R be a ring, and consider a short exact sequence of R-modules*

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0.$$

The following are equivalent:

(i) The sequence splits, i.e. $B \cong A \oplus C$ in a way compatible with i and p.

(ii) The map p admits a section $s : C \to B$.

(iii) The map i admits a retraction $r : B \to A$.

Condition (ii) is often called *right split*, while condition (iii) is called *left split*. Thus, for R-modules, "right split," "left split," and "split" are all equivalent. Warning: This equivalence is special to the setting of R-modules (more generally, abelian categories). For example, in the category of non-abelian groups, a sequence may admit a section without being a direct product: such extensions correspond to semidirect products. Fortunately, none of these issues arise for R-modules.

---

(5) **Split Short Exact Sequences.** Let

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

be a short exact sequence of R-modules.

(a) Suppose there exists a homomorphism $s : C \to B$ such that $p \circ s = \mathrm{Id}_C$. Prove that the map defined below is an isomorphism of R-modules:

$$\Phi : A \oplus C \longrightarrow B, \qquad (a, c) \longmapsto i(a) + s(c).$$

(b) Conversely, suppose there is an isomorphism $B \cong A \oplus C$ under which i becomes the standard inclusion $A \to A \oplus C$ and p becomes the standard projection $A \oplus C \to C$. Construct a homomorphism $s : C \to B$ with $p \circ s = \mathrm{Id}_C$.

(c) Suppose there exists a homomorphism $r : B \to A$ such that $r \circ i = \mathrm{Id}_A$.

(i) Prove that the map

$$B \longrightarrow A \oplus C, \qquad b \longmapsto (r(b), p(b))$$

is injective.

(ii) Prove that it is surjective. Hint: Given $(a, c) \in A \oplus C$, choose $b_0 \in B$ with $p(b_0) = c$ and modify b_0 by an element of $i(A)$.

(iii) Conclude that $B \cong A \oplus C$.

(d) Conversely, suppose there is an isomorphism $B \cong A \oplus C$ under which i becomes the standard inclusion $a \mapsto (a, 0)$ and p becomes the standard projection $(a, c) \mapsto c$. Construct a homomorphism $r : B \to A$ with $r \circ i = \mathrm{Id}_A$.

(6) **Examples of Split & Non-Split Exact Sequences.**

(a) Prove that every short exact sequence of vector spaces over a field $\mathbb{K}$ splits. Hint: Choose a basis for the first term, extend it to a basis for the middle term, and identify the remaining basis vectors with a basis for the third term.

(b) Explain why the Rank-Nullity theorem for finite dimensional vector spaces over $\mathbb{K}$ is a consequence of the splitting theorem.

(c) Prove that the sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

does not split as a sequence of $\mathbb{Z}$-modules.

(d) Let p be a prime number. Prove that the sequence of $\mathbb{Z}$-modules:

$$0 \longrightarrow \mathbb{Z}/p\mathbb{Z} \longrightarrow \mathbb{Z}/p^2\mathbb{Z} \longrightarrow \mathbb{Z}/p\mathbb{Z} \longrightarrow 0,$$

where the first map sends $\bar{1} \mapsto \bar{p}$ and the second is reduction modulo p, does not split. Hint: A section would send $\bar{1} \in \mathbb{Z}/p\mathbb{Z}$ to an element of $\mathbb{Z}/p^2\mathbb{Z}$ whose reduction modulo p is $\bar{1}$. What is the order of such an element?

(e) Let $R = \mathbb{K}[x]$. Prove that the sequence below is exact, but does not split as a sequence of R-modules:

$$0 \longrightarrow \langle x \rangle \longrightarrow \mathbb{K}[x] \longrightarrow \mathbb{K} \longrightarrow 0.$$

Hint: If a section $s : \mathbb{K} \to \mathbb{K}[x]$ existed as a $\mathbb{K}[x]$-module map, what could $s(1)$ be? Use the fact that $x \cdot 1 = 0$ in $\mathbb{K} \cong \mathbb{K}[x]/\langle x \rangle$.

(f) Let R be a ring and let $e \in R$ be an idempotent, meaning $e^2 = e$. Prove that

$$0 \longrightarrow eR \hookrightarrow R \twoheadrightarrow R/eR \longrightarrow 0$$

splits. Then prove that the map $R \to (1-e)R$ given by $r \mapsto (1-e)r$ induces an isomorphism $R/eR \cong (1-e)R$.

We now introduce just enough categorical language to talk cleanly about Hom, tensor product, and localization. A *category* is a collection of objects together with, for each pair of objects, a set of morphisms between them, equipped with an associative composition law and an identity morphism for each object. The quintessential category, both in this class and arguably beyond, is the category of R-modules, denoted $R-\mathbf{Mod}$, whose objects are R-modules and whose morphisms are R-module homomorphisms.

A *functor* between categories $\mathbf{C}$ and $\mathbf{D}$ is a mapping that assigns to each object of $\mathbf{C}$ an object of $\mathbf{D}$ and to each morphism in $\mathbf{C}$ a morphism in $\mathbf{D}$, preserving identity morphisms and composition. A functor is contravariant if it reverses the order of composition and covariant if it preserves the order of composition. More precisely, a *covariant functor* $F : \mathbf{C} \to \mathbf{D}$ is the data of

(1) for every object X of $\mathbf{C}$, $F(X)$ is an object of $\mathbf{D}$,

(2) for every morphism $f : X \to Y$ in $\mathbf{C}$, $F(f) : F(X) \to F(Y)$ is a morphism in $\mathbf{D}$,

which satisfies the following conditions:

(i) $F(\mathrm{Id}_X) = \mathrm{Id}_{F(X)}$ for every object X of $\mathbf{C}$,

(ii) $F(g \circ f) = F(g) \circ F(f)$ for all composable morphisms $f : X \to Y$ and $g : Y \to Z$ in $\mathbf{C}$.

Similarly a *contravariant functor* $F : \mathbf{C} \to \mathbf{D}$ is the data of:

(1) for every object X of $\mathbf{C}$, $F(X)$ is an object of $\mathbf{D}$,

(2) for every morphism $f : X \to Y$ in $\mathbf{C}$, $F(f) : F(Y) \to F(X)$ is a morphism in $\mathbf{D}$,

which satisfies the following conditions:

(i) $F(\mathrm{Id}_X) = \mathrm{Id}_{F(X)}$ for every object X of $\mathbf{C}$,

(ii) $F(g \circ f) = F(f) \circ F(g)$ for all composable morphisms $f : X \to Y$ and $g : Y \to Z$ in $\mathbf{C}$.

We have already constructed several operations that take modules as input and produce new modules. For example, after fixing an R-module T, the assignments

$$M \longmapsto \mathrm{Hom}_R(T, M), \qquad M \longmapsto \mathrm{Hom}_R(M, T), \qquad M \longmapsto M \otimes_R T$$

each send an object of $R-\mathbf{Mod}$ to another object of $R-\mathbf{Mod}$. Similarly, if $U \subset R$ is a multiplicatively closed set, localization gives an assignment

$$M \longmapsto U^{-1}M,$$

which sends an R-module to a module over the localized ring $U^{-1}R$. We would like to view each of these constructions as defining functors:

$$R\text{-}\mathbf{Mod} \xrightarrow{\mathrm{Hom}_R(T,\text{-})} R\text{-}\mathbf{Mod} \qquad R\text{-}\mathbf{Mod} \xrightarrow{\mathrm{Hom}_R(\text{-},T)} R\text{-}\mathbf{Mod}$$

$$R\text{-}\mathbf{Mod} \xrightarrow{\text{-}\otimes_R T} R\text{-}\mathbf{Mod} \qquad R\text{-}\mathbf{Mod} \xrightarrow{U^{-1}\text{-}} U^{-1}R\text{-}\mathbf{Mod}$$

However, a functor is not merely a rule on objects. To view these constructions functorially, we must also understand what they do to morphisms. Thus, given an R-module homomorphism $f : M \to N$, we must define induced maps between the corresponding Hom modules, tensor products, and localizations, and then check that these induced maps respect identities and composition. The following exercise asks you to make these constructions explicit and verify that they really do define functors.

---

(7) **Hom, Tensor, and Localization as Functors.** Throughout let R be a ring and fix an R-module T.

(a) Let T be a fixed R-module. If $f : M \to N$ is a homomorphism, define

$$f_* : \mathrm{Hom}_R(T, M) \to \mathrm{Hom}_R(T, N), \qquad \phi \longmapsto f \circ \phi.$$

Prove that $\mathrm{Hom}_R(T, -)$ is a covariant functor from $R-\mathbf{Mod}$ to $R-\mathbf{Mod}$.

(b) If $f : M \to N$ is a homomorphism, define

$$f^* : \mathrm{Hom}_R(N,T) \to \mathrm{Hom}_R(M,T), \qquad \psi \longmapsto \psi \circ f.$$

Prove that $\mathrm{Hom}_R(-,T)$ is a contravariant functor from $R-\mathbf{Mod}$ to $R-\mathbf{Mod}$.

(c) If $f : M \to N$ is a homomorphism, use the universal property of tensor products to construct

$$f \otimes \mathrm{Id}_T : M \otimes_R T \longrightarrow N \otimes_R T, \qquad m \otimes t \longmapsto f(m) \otimes t.$$

Prove that $- \otimes_R T$ is a covariant functor from $R-\mathbf{Mod}$ to $R-\mathbf{Mod}$.

(d) Let $U \subset R$ be a multiplicatively closed set. If $f : M \to N$ is a homomorphism, define

$$U^{-1}f : U^{-1}M \to U^{-1}N, \qquad \frac{m}{u} \longmapsto \frac{f(m)}{u}.$$

Prove that this is well-defined and that $U^{-1}(-)$ is a covariant functor from $R-\mathbf{Mod}$ to $U^{-1}R-\mathbf{Mod}$.

The functors constructed above let us ask a new version of the exactness question. Instead of asking whether a given sequence of modules is exact, we can ask whether a construction preserves exactness. Roughly speaking, a functor is exact if applying it to an exact sequence does not introduce any new homology: the kernel of one map is still the image of the previous. This is a very strong condition, and many of the most important functors in algebra preserve only part of an exact sequence.

For the functors in this worksheet, applying the functor to a sequence again produces a sequence: these functors are additive, so they send zero maps to zero maps, and they respect composition. Let $F : R\text{-}\mathbf{Mod} \to S\text{-}\mathbf{Mod}$ be one of the covariant functors under consideration, where S is some ring. We say that F is *exact* if, whenever

$$M' \xrightarrow{f} M \xrightarrow{g} M''$$

is exact at M, the transformed sequence

$$F(M') \xrightarrow{F(f)} F(M) \xrightarrow{F(g)} F(M'')$$

is exact at $F(M)$. Equivalently, F sends exact sequences to exact sequences. In particular, an exact covariant functor sends every short exact sequence

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

to a short exact sequence

$$0 \longrightarrow F(A) \xrightarrow{F(i)} F(B) \xrightarrow{F(p)} F(C) \longrightarrow 0.$$

Thus exactness means that $F(i)$ remains injective, $F(p)$ remains surjective, and $\mathrm{img}(F(i)) = \ker(F(p))$. A covariant functor F is called *left exact* if every exact sequence

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C$$

is sent to a sequence

$$0 \longrightarrow F(A) \xrightarrow{F(i)} F(B) \xrightarrow{F(p)} F(C)$$

which is exact at $F(A)$ and $F(B)$. So a left exact functor preserves the injectivity at the left end and preserves the kernel condition in the middle, but it may fail to make $F(p)$ surjective. Dually, F is called *right exact* if every exact sequence

$$A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

is sent to a sequence

$$F(A) \xrightarrow{F(i)} F(B) \xrightarrow{F(p)} F(C) \longrightarrow 0$$

which is exact at $F(B)$ and $F(C)$. Thus a right exact functor preserves the quotient at the right end, but it may fail to preserve injectivity at the left end. A functor that is both left exact and right exact is exact. One can also define exactness, left exactness, and right exactness for contravariant functors; in that case, the arrows are reversed.

The next exercises focus on three fundamental examples. The Hom functors are naturally left exact. For $\mathrm{Hom}_R(T,-)$, the reason is that a map $T \to B$ whose composite with $B \to C$ is zero is exactly a map whose image lands in the submodule $A \subseteq B$. For the contravariant functor $\mathrm{Hom}_R(-,T)$, the same idea appears in quotient form: a map $B \to T$ that kills A factors uniquely through $C = B/A$. In both cases, surjectivity can fail; a map into or out of one end of a short exact sequence need not lift or extend across the middle term.

Tensor product behaves in the opposite direction. The functor $- \otimes_R T$ is always right exact: it preserves cokernels and therefore preserves quotient presentations. This is why tensor product is so useful for computing modules from generators and relations. But tensor product need not be left exact; after tensoring, an injective map can acquire a kernel. The modules T for which tensoring with T is exact are precisely the flat modules, which will be introduced below.

Localization is the especially well-behaved example in commutative algebra. Since $U^{-1}M \cong U^{-1}R \otimes_R M$, localization is right exact for the same reason tensor product is right exact. What makes localization better than an arbitrary tensor product is that it is also left exact: if a fraction becomes zero after applying a localized map, then after clearing denominators its numerator already came from the previous module. Thus localization preserves all exact sequences, and this exactness is one of the main reasons localization is such a powerful tool.

---

(8) **Left Exactness of** $\mathrm{Hom}_R(T,-)$**.** Let T be an R-module and let

$$0 \longrightarrow A \xhookrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

be a short exact sequence.

(a) Define the maps i_* and p_* by postcomposition, and prove that $p_* \circ i_* = 0$.

(b) Prove that $i_* : \mathrm{Hom}_R(T,A) \to \mathrm{Hom}_R(T,B)$ is injective.

(c) Suppose $\phi : T \to B$ satisfies $p \circ \phi = 0$. Prove that ϕ factors uniquely through $i : A \to B$. In other words, prove that there is a unique homomorphism $\widetilde{\phi} : T \to A$ such that $\phi = i \circ \widetilde{\phi}$.

(d) Conclude that the sequence below is exact

$$0 \longrightarrow \mathrm{Hom}_R(T,A) \xhookrightarrow{i_*} \mathrm{Hom}_R(T,B) \xrightarrow{p_*} \mathrm{Hom}_R(T,C).$$

(e) Suppose the original short exact sequence splits. Prove that $p_* : \mathrm{Hom}_R(T,B) \to \mathrm{Hom}_R(T,C)$ is surjective. Thus $\mathrm{Hom}_R(T,-)$ sends split short exact sequences to short exact sequences.

(f) Let $R = \mathbb{Z}$ and apply $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z},-)$ to

$$0 \longrightarrow \mathbb{Z} \xhookrightarrow{\cdot 2} \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

Compute the resulting sequence and explain why $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z},-)$ is not right exact.

(9) **Left Exactness of $\mathrm{Hom}_R(-,T)$.** Let T be an R-module and consider the short exact sequence below:

$$0 \longrightarrow A \xhookrightarrow{i} B \xrightarrow{p} C \longrightarrow 0.$$

(a) Using precomposition define the maps as below. Prove that $i^* \circ p^* = 0$:

$$\mathrm{Hom}_R(C,T) \xrightarrow{p^*} \mathrm{Hom}_R(B,T) \xrightarrow{i^*} \mathrm{Hom}_R(A,T):$$

(b) Prove that p^* is injective.

(c) Suppose $\psi : B \to T$ satisfies $\psi \circ i = 0$. Prove that ψ factors uniquely through $p : B \to C$. In other words, prove that there is a unique homomorphism $\widetilde{\psi} : C \to T$ such that $\psi = \widetilde{\psi} \circ p$.

(d) Conclude that the sequence below is exact:

$$0 \longrightarrow \mathrm{Hom}_R(C,T) \xhookrightarrow{p^*} \mathrm{Hom}_R(B,T) \xrightarrow{i^*} \mathrm{Hom}_R(A,T).$$

(e) Suppose the original short exact sequence splits. Prove that $i^* : \mathrm{Hom}_R(B,T) \to \mathrm{Hom}_R(A,T)$ is surjective. Thus $\mathrm{Hom}_R(-,T)$ sends split short exact sequences to short exact sequences.

(f) Let $R = \mathbb{Z}$ and apply $\mathrm{Hom}_{\mathbb{Z}}(-,\mathbb{Z})$ to

$$0 \longrightarrow \mathbb{Z} \xhookrightarrow{\cdot 2} \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

Compute the resulting sequence and explain why $\mathrm{Hom}_{\mathbb{Z}}(-,\mathbb{Z})$ is not exact.

(10) **Hom Computations from Exactness.**

(a) Let M be an R-module and let $I \subseteq R$ be an ideal. Use the short exact sequence

$$0 \longrightarrow I \xhookrightarrow{} R \twoheadrightarrow R/I \longrightarrow 0$$

together with $\mathrm{Hom}_R(-,M)$ to recover the description

$$\mathrm{Hom}_R(R/I,M) \cong \{m \in M \mid Im = 0\}.$$

Hint: Identify $\mathrm{Hom}_R(R,M)$ with M by evaluating a homomorphism at $1 \in R$. Under this identification, what does the condition that a map $R \to M$ vanish on I become?

(b) Let N be an abelian group and n a positive integer. Apply $\mathrm{Hom}_{\mathbb{Z}}(-,N)$ to

$$0 \longrightarrow \mathbb{Z} \xhookrightarrow{\cdot n} \mathbb{Z} \twoheadrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0$$

Show that there is an exact sequence

$$0 \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z},N) \hookrightarrow N \xrightarrow{\cdot n} N$$

Deduce that $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z},N)$ is naturally isomorphic to the subgroup of n-torsion elements of N.

(c) For positive integers m,n, use the previous part to compute $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z},\mathbb{Z}/n\mathbb{Z})$.

(d) Let $R = \mathbb{K}[x]$ and $M = R/\langle x^b \rangle$. Compute $\mathrm{Hom}_R(R/\langle x^a \rangle, M)$ for positive integers a and b.

(e) Let A be an $m \times n$ matrix with entries in R, let $M = \mathrm{coker}(A)$, and let N be an R-module. Use the presentation

$$R^n \xrightarrow{A} R^m \twoheadrightarrow M \longrightarrow 0$$

to describe $\mathrm{Hom}_R(M,N)$ as the kernel of an explicit map $N^m \to N^n$. Hint: Identify $\mathrm{Hom}_R(R^m,N)$ with N^m and $\mathrm{Hom}_R(R^n,N)$ with N^n. The induced map is precomposition with A; in coordinates it is given by the transpose matrix A^t acting on column vectors, with the usual convention.

(11) **Right Exactness of Tensor Product.** Let T be an R-module and let the below be an exact sequence:

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

(a) Prove that $g \otimes \mathrm{Id}_T : B \otimes_R T \to C \otimes_R T$ is surjective.

(b) Construct a bilinear map $C \times T \to (B \otimes_R T)/\langle f(a) \otimes t \mid a \in A,\ t \in T \rangle$t by choosing a representative $b \in B$ of $c \in C$ and sending (c,t) to the class of $b \otimes t$. Check that this is well-defined.

(c) Since $C \cong B/\mathrm{img}(f)$, use the universal property of quotients and tensor products to show that

$$C \otimes_R T \cong (B \otimes_R T)/\langle f(a) \otimes t \mid a \in A,\ t \in T \rangle.$$

(d) Prove that $\ker(g \otimes \mathrm{Id}_T) = \mathrm{img}(f \otimes \mathrm{Id}_T)$.

(e) Conclude that the sequence below is exact:

$$A \otimes_R T \longrightarrow B \otimes_R T \longrightarrow C \otimes_R T \longrightarrow 0.$$

(f) Suppose the original short exact sequence splits. Prove that $g \times \mathrm{Id}_T$ is injective. Thus $- \otimes_R T$ sends split short exact sequences to short exact sequences.

(12) **Tensor Product Need Not Be Left Exact.** The next example shows that tensor product can turn a nonzero injective map into the zero map.

(a) Tensor the short exact sequence

$$0 \longrightarrow \mathbb{Z} \xhookrightarrow{\cdot 2} \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

with $\mathbb{Z}/2\mathbb{Z}$. Compute the resulting sequence explicitly.

(b) Explain why the map $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$ induced by multiplication by 2 is the zero map.

(c) Conclude that $- \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$ is not left exact.

(d) Let n be a positive integer. Compute $(\mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/2\mathbb{Z})$ and use this to predict what happens when one tensors the exact sequence below with $\mathbb{Z}/2\mathbb{Z}$:

$$0 \longrightarrow \mathbb{Z} \xhookrightarrow{\cdot n} \mathbb{Z} \twoheadrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

(13) **Flat Modules.** An R-module T is called *flat* if the functor $- \otimes_R T$ is exact, or equivalently if tensoring with T preserves injective maps.

(a) Explain why right exactness of tensor product implies that T is flat if and only if, for every injection $A \hookrightarrow B$, the induced map $A \otimes_R T \to B \otimes_R T$ is injective.

(b) Prove that R is flat as an R-module.

(c) Prove that a finite free module R^n is flat.

(d) Prove that every free R-module is flat. Hint: Tensor product commutes with direct sums.

Localization is the most important exact functor in commutative algebra. Let $U \subset R$ be a multiplicatively closed set. The localization construction sends an R-module M to a $U^{-1}R$-module $U^{-1}M$, and sends a homomorphism $f : M \to N$ to the homomorphism

$$U^{-1}M \xrightarrow{U^{-1}f} U^{-1}N \qquad \frac{m}{u} \longmapsto \frac{f(m)}{u}.$$

We already saw that $U^{-1}M \cong U^{-1}R \otimes_R M$. Since tensor product is right exact, localization is automatically right exact. The special feature of localization is that it is also left exact. The reason is the denominator-clearing argument: if a fraction maps to zero after localization, then some element of U already kills the relevant numerator.

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(14) **Exactness of Localization.** Let $U \subset R$ be a multiplicatively closed set and let the sequence below be exact at M:

$$M' \xrightarrow{f} M \xrightarrow{g} M''.$$

(a) Prove that

$$U^{-1}M' \xrightarrow{U^{-1}f} U^{-1}M \xrightarrow{U^{-1}g} U^{-1}M''$$

is a sequence of $U^{-1}R$-modules whose consecutive composition is zero.

(b) Prove that $\operatorname{img}(U^{-1}f) \subseteq \ker(U^{-1}g)$.

(c) Let $m/u \in U^{-1}M$ satisfy $(U^{-1}g)(m/u) = 0$. Unpack the definition of equality to zero in $U^{-1}M''$ to show that there exists $t \in U$ such that $g(tm) = 0$.

(d) Use exactness of the original sequence at M to find $m' \in M'$ with $f(m') = tm$.

(e) Prove that $\frac{m}{u} = \frac{f(m')}{tu}$ in $U^{-1}M$, and conclude that $m/u \in \operatorname{img}(U^{-1}f)$.

(f) Deduce that localization preserves exactness of arbitrary exact sequences.

(15) **Localization Preserves Short Exact Sequences.** Let

$$0 \longrightarrow A \xhookrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

be a short exact sequence and let $U \subset R$ be a multiplicatively closed set.

(a) Prove directly that $U^{-1}i : U^{-1}A \to U^{-1}B$ is injective.

(b) Prove directly that $U^{-1}p : U^{-1}B \to U^{-1}C$ is surjective.

(c) Use the previous exercise to conclude that

$$0 \longrightarrow U^{-1}A \xhookrightarrow{} U^{-1}B \xrightarrow{} U^{-1}C \longrightarrow 0$$

is a short exact sequence of $U^{-1}R$-modules.

(d) Combine this with the isomorphism $U^{-1}M \cong U^{-1}R \otimes_R M$ to prove that $U^{-1}R$ is a flat R-module.

(16) **Computing Localized Exact Sequences.**

(a) Let $I \subseteq R$ be an ideal. Use exactness of localization to prove that $U^{-1}(R/I) \cong U^{-1}R/IU^{-1}R$.

(b) Localize the short exact sequence at $U = \{1, 2, 2^2, 2^3, \ldots\}$. Identify each term explicitly

$$0 \longrightarrow \mathbb{Z} \xhookrightarrow{\cdot 6} \mathbb{Z} \xrightarrow{} \mathbb{Z}/6\mathbb{Z} \longrightarrow 0.$$

(c) Localize the same sequence at $V = \mathbb{Z} \setminus \langle 2 \rangle$. Identify $V^{-1}(\mathbb{Z}/6\mathbb{Z})$.

(d) Let $R = \mathbb{K}[x]$ and $M = R/\langle x^2(x-1) \rangle$. Compute $M[1/x]$ using the localized presentation

$$R[1/x] \xrightarrow{\cdot x^2(x-1)} R[1/x] \xrightarrow{} M[1/x] \longrightarrow 0$$

(e) Explain why localizing $R/\langle x^2(x-1) \rangle$ at the prime ideal $\langle x \rangle$ "keeps" the $x = 0$ component and "removes" the $x = 1$ component.

(f) With R and M as in the previous part, compute $M_{\langle x \rangle}$. Hint: What happens to $x - 1$ in $R_{\langle x \rangle}$?

(g) Let $R = \mathbb{K}[x, y]$ and $M = R/\langle xy \rangle$. Compute $M[1/x]$ and $M[1/y]$.

WORKSHEET 6.2: EXACTNESS PT. II

Throughout this course, "ring" means *commutative* ring with unity, and all modules are assumed to be modules over the relevant ring. Let R be a ring and let M be an R-module. Recall from the previous worksheets that if $U \subset R$ is a multiplicatively closed set, the localization of M at U is a $U^{-1}R$-module, denoted $U^{-1}M$, which intuitively consists of elements of the form $\frac{m}{u}$ for $m \in M$ and $u \in U$ (up to equivalence). Further, localization is a functor, meaning given a map $\phi : M \to N$ of R-modules we get a map $U^{-1}(\phi) : U^{-1}M \to U^{-1}N$ of $U^{-1}R$-modules. On elements this map is given by $\frac{m}{u} \mapsto \frac{\phi(m)}{u}$. Further, we saw that localization is an exact functor, meaning that it preserves exact sequences.

The goal of this worksheet is to use localization in two complementary ways. First, we will prove the slogan: *exactness can be checked locally*. This means that many questions about kernels, images, injections, surjections, and isomorphisms can be answered after localizing at every prime ideal, or even every maximal ideal. Second, we will show that at least for *finitely generated* modules, many of powerful pieces of linear algebra can imported into module theory.

Recall that for a prime ideal $\mathfrak{p} \subset R$ we write $M_{\mathfrak{p}}$ for the localization of M at the multiplicatively closed set $R \setminus \mathfrak{p}$. If $f \in R$, we write $M[\frac{1}{f}]$ for the localization of M at the multiplicative set $U = \{1, f, f^2, \ldots\}$. The first local criterion is the most basic one: an R-module is zero if and only if it becomes zero after localizing at every point of $\mathrm{Spec}(R)$. The intuition behind this is simple: If $m \in M$ is nonzero, then its annihilator

$$\mathrm{Ann}(m) = \{r \in R \mid rm = 0\}$$

is a proper ideal of R. Choosing a maximal ideal containing $\mathrm{Ann}(m)$ gives a point of $\mathrm{Spec}(R)$ where m cannot disappear after localization. This is the algebraic version of saying that a nonzero object must be visible somewhere locally.

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(1) **Local Vanishing for Modules.** Let R be a ring and M an R-module. If $\mathscr{S} \subset M$ is any subset, the *annihilator* of $\mathscr{S}$ is the ideal:

$$\mathrm{Ann}_R(\mathscr{S}) := \left\{ r \in R \;\middle|\; rs = 0 \text{ for all } s \in \mathscr{S} \right\}.$$

If $\mathscr{S} = \{m\}$ we write $\mathrm{Ann}_R(m)$ for $\mathrm{Ann}_R(\{m\})$.

(a) If $\mathscr{S} \subset M$ is any subset, prove that $\mathrm{Ann}_R(\mathscr{S})$ is an ideal of R.

(b) Let $\mathfrak{p} \subset R$ be a prime ideal. Given $m \in M$, prove that $m/1 = 0$ in $M_{\mathfrak{p}}$ if and only if there exists $s \in R \setminus \mathfrak{p}$ such that $sm = 0$ in M.

(c) Let M be the zero R-module. Explain why $U^{-1}M = 0$ for any multiplicatively closed subset $U \subset R$.

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(d) Prove that $M = 0$ if and only if $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m} \subset R$. Hint: One direction should be easy. For the other, suppose $m \neq 0$, choose a maximal ideal containing $\mathrm{Ann}(m)$.

(e) Deduce that $M = 0$ if and only if $M_{\mathfrak{p}} = 0$ for every prime ideal $\mathfrak{p} \in \mathrm{Spec}(R)$.

(f) Let $f_1, \ldots, f_n \in R$ generate the unit ideal. Prove that $M = 0$ if and only if $M[\frac{1}{f_i}] = 0$ for every i. Hint: Again, one direction should be straightforward. For the other direction, given $m \in M$, first find powers $f_i^{t_i}$ killing m, then show that the ideal $\langle f_1^{t_1}, \ldots, f_n^{t_n} \rangle$ is the unit ideal.

(g) Let $\phi : M \to N$ be a homomorphism of R-modules. Prove that $\phi = 0$ if and only if $\phi_{\mathfrak{m}} : M_{\mathfrak{m}} \to N_{\mathfrak{m}}$ is the zero map for every maximal ideal $\mathfrak{m} \subset R$.

(2) **Homology and Localization Commute.** Let R be a ring and consider the complex of R-modules below:

$$M' \xrightarrow{f} M \xrightarrow{g} M''$$

Recall this means that $\mathrm{img}(f) \subset \ker(g)$. Define $H := \ker(g)/\mathrm{img}(f)$, which is the homology of this complex at M. Recall that the complex is exact at M if and only if $H = 0$.

(a) Explain why there is a short exact sequence of R-modules shown below. What are the maps?

$$0 \longrightarrow \mathrm{img}(f) \longrightarrow \ker(g) \longrightarrow H \longrightarrow 0$$

(b) Let $U \subset R$ be a multiplicatively closed set. Use exactness of localization and part (a) to show there is an isomorphism:

$$U^{-1}H \cong \frac{\ker(U^{-1}g)}{\mathrm{img}(U^{-1}f)}.$$

(c) Explain why the previous parts prove the slogan: *The homology of the localization is the localization of the homology.*

(d) Deduce that the complex is exact at M if and only if the complex below is exact at $M_{\mathfrak{p}}$ for every prime ideal $\mathfrak{p} \subset R$

$$M'_{\mathfrak{p}} \xrightarrow{f_{\mathfrak{p}}} M_{\mathfrak{p}} \xrightarrow{g_{\mathfrak{p}}} M''_{\mathfrak{p}}.$$

(e) Prove that it is enough to check exactness after localizing at every maximal ideal.

(f) Suppose $f_1, \ldots, f_n \in R$ generate the unit ideal. Prove that the complex is exact at M if and only if the complex below is exact at $M[\frac{1}{f_i}]$ for every $i = 1, \ldots, n$:

$$M'\left[\tfrac{1}{f_i}\right] \longrightarrow M\left[\tfrac{1}{f_i}\right] \longrightarrow M''\left[\tfrac{1}{f_i}\right].$$

(g) Explain why the previous parts prove the slogan: *exactness can be checked locally.*

(3) **Checking Injectivity & Surjectivity Locally.** Let R be a ring and $\phi : M \to N$ be a homomorphism of R-modules.

(a) Prove that ϕ is injective if and only if $\phi_{\mathfrak{m}} : M_{\mathfrak{m}} \to N_{\mathfrak{m}}$ is injective for every maximal ideal $\mathfrak{m} \subset R$. Hint: Apply local vanishing to $\ker(\phi)$.

(b) Prove that ϕ is surjective if and only if $\phi_{\mathfrak{m}} : M_{\mathfrak{m}} \to N_{\mathfrak{m}}$ is surjective for every maximal ideal $\mathfrak{m} \subset R$. Hint: Apply local vanishing to $\operatorname{coker}(\phi)$.

(c) Deduce ϕ is an isomorphism if and only if $\phi_{\mathfrak{m}}$ is an isomorphism for every maximal ideal $\mathfrak{m} \subset R$.

(d) State and prove the analogous criterion using all prime ideals instead of all maximal ideals.

(e) State and prove the analogous criterion using a finite basic open cover $\operatorname{Spec}(R) = D(f_1) \cup \cdots \cup D(f_n)$.

(f) Let $n \geq 1$ be an integer and consider multiplication by n as a map of $\mathbb{Z}$-modules

$$\mathbb{Z} \xrightarrow{\times n} \mathbb{Z}.$$

For which prime ideals $\langle p \rangle \in \operatorname{Spec}(\mathbb{Z})$ does the localized map become an isomorphism? For which prime ideals is it injective? Surjective?

We now turn to a different technique – seemingly unrelated topic – namely attempting to port-over aspects of linear algebra to the theory of modules. The basic idea is that while a module might not have a basis – recall only free modules do – so long as a module has a finite set of generators many classical identities and facts from linear algebra remain true.

Let us first recall the relevant linear algebra. Let $\mathbb{K}$ be a field, and let A be an $n \times n$ matrix over $\mathbb{K}$. For each pair $1 \leq i, j \leq n$, let $A_{\hat{i},\hat{j}}$ denote the $(n-1) \times (n-1)$ matrix obtained from A by deleting row i and column j. The *cofactor* matrix of A is the $n \times n$ matrix whose (i,j)-th entry is $c_{i,j} := (-1)^{i+j} \det(A_{\hat{i},\hat{j}})$. The *adjugate* of A is defined to be $\operatorname{adj}(A) := C^{\mathsf{T}}$ where C is the cofactor matrix of A. A standard theorem from linear algebra says that the identity holds for every $n \times n$ matrix A over the field $\mathbb{K}$:

$$A \operatorname{adj}(A) = \operatorname{adj}(A) A = \det(A) \operatorname{Id}_{n \times n}.$$

This identity is often proved using Laplace expansion along rows and columns. It is also the identity behind Cramer's rule. Conceptually, it says that even if A is not invertible, multiplying by $\operatorname{adj}(A)$ turns A into the scalar matrix $\det(A) I_n$. This formula has surprisingly important consequences for the structure of endomorphisms of vector spaces.

One clean way to see this is to do the computation once in the *universal* ring

$$\mathbb{Z}[x_{i,j} \mid 1 \leq i, j \leq n],$$

where $X = (x_{i,j})$ is the generic $n \times n$ matrix whose entries are variables. Every particular matrix over every commutative ring is obtained from X by substituting $x_{i,j} \mapsto a_{i,j}$. Thus a polynomial matrix identity over this polynomial ring can be specialized to any commutative ring. The same universal idea gives a first proof of Cayley–Hamilton over arbitrary commutative rings: If $\chi_A(t) = \det(tI_n - A)$ is the characteristic polynomial of A, then $\chi_A(A) = 0$. For matrices over fields this is a theorem from linear algebra. For

matrices over arbitrary rings, one proves the identity for the generic matrix over the polynomial ring over $\mathbb{Z}$, and then specializes.

Before we prove this we need to extend our definition of an algebra slightly. Let R be a commutative ring. An R-algebra is a ring A together with a specified ring homomorphism $\phi : R \to A$. We think of this as defining a scalar multiplication on A by elements of R by defining $r \cdot a := \phi(r)a$. A morphism of R-algebras $\psi : A \to B$ is a ring homomorphism that preserves scalars, i.e. $\psi(r \cdot 1_A) = r \cdot 1_B$ for every $r \in R$.

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(4) **Universal Adjugate Formula.** Fix a positive integer n. Let R be a ring and A be an $n \times n$ matrix with entries in R. Define the adjugate matrix of $A = (a_{ij})$ just as one does for matrices over a field. The goal of this exercise is to prove the following identity:

$$A \operatorname{adj}(A) = \operatorname{adj}(A)A = \det(A)\mathrm{Id}_{n\times n}. \tag{1}$$

In this problem let $S = \mathbb{Z}[x_{i,j} \mid 1 \le i, j \le n]$ be the polynomial ring in n^2 variables. Let $X = (x_{i,j})$ be the generic $n \times n$ matrix with entries in S.

(a) Show that an arbitrary ring has a canonical structure of a $\mathbb{Z}$-algebra.

(b) Given an $n \times n$ matrix $A = (a_{i,j})$ with entries in R show that there is a well-defined morphism of $\mathbb{Z}$-algebras:

$$\begin{array}{ccc} S & \xrightarrow{\mathrm{ev}_A} & R \\ x_{i,j} & \longmapsto & a_{i,j} \end{array}.$$

(c) Let $X = (x_{i,j})$ be the generic $n \times n$ matrix whose entries are the variables in S. Assume that (1) holds true for the matrix X. Use the homomorphism ev_A to show the identity holds true for A.

(d) Since S is a domain it embeds into its field of fractions $S \subset \mathrm{Frac}(S)$. Explain why this allows us to view X as a matrix with entries in $\mathrm{Frac}(S)$.

(e) Viola! Conclude the matrix equation (1) holds for arbitrary rings from the corresponding linear algebra fact (over fields). Be sure to check the field case too, if it's not familiar to you. It's easy, if you know about computing determinants by "expanding along a row or column".

(5) **Universal Cayley-Hamilton Theorem.** Fix a positive integer n. Let R be a ring and A be an $n \times n$ matrix with entries in R. Define the *characteristic polynomial* of A to be $\chi_A(t) = \det(tI_n - A) \in R[t]$. The goal of this is to prove the Cayley-Hamilton theorem for rings, namely: $\chi_A(A) = 0$. In this problem let $S = \mathbb{Z}[x_{i,j} \mid 1 \le i, j \le n]$ be the polynomial ring in n^2 variables. Let $X = (x_{i,j})$ be the generic $n \times n$ matrix with entries in S.

(a) Prove the Cayley-Hamilton theorem for the generic matrix X, i.e., $\chi_X(X) = 0$. Hint: Again use that you may embed S into its field of fractions.

(b) Use the specialization map ev_A from the previous problem to prove Cayley–Hamilton for every $n \times n$ matrix A over every commutative ring R.

We now use our universal adjugate formula to prove what is called the *determinant trick*, which allows us to extend the Cayley–Hamilton to endomorphisms of finitely generated modules. Recall an endomorphism of an R-module M is just an R-module homomorphism $\phi : M \to M$.

Theorem 1 (Cayley-Hamilton)**.** *Let R be a ring and $I \subset R$ an ideal. Let M be an R-module that can be generated as an R-module by n elements. Let $\phi : M \to M$ be an endomorphism of M. If $\phi(M) \subset IM$ then there exists a monic polynomial*

$$p(t) = t^n + p_1 t^{n-1} + \cdots + p_{n-1} t + p_n \in R[t]$$

with $p_k \in I^k$, such that $p(\phi) = 0$ as an endomorphism on M.

Before proving the theorem, let us spell out what the notation means. If $\phi : M \to M$ is an endomorphism, then we write $\phi^2 := \phi \circ \phi$, $\phi^3 := \phi \circ \phi \circ \phi$, and, more generally, ϕ^n means the n-fold composition of ϕ with itself. We also set $\phi^0 = \mathrm{Id}_M$. Thus, given a polynomial $p(t) = t^n + p_1 t^{n-1} + \cdots + p_{n-1} t + p_n \in R[t]$ then $p(\phi) = \phi^n + p_1 \phi^{n-1} + \cdots + p_{n-1}\phi + p_n \mathrm{Id}_M$ is in $\mathrm{End}_R(M)$. The equation $p(\phi) = 0$ means that this endomorphism is the zero map; i.e., $p(\phi)(m) = 0$ for all $m \in M$.

There is another useful way to say the same thing. Giving an R-module M together with an endomorphism $\phi : M \to M$ is the same as giving M the structure of an $R[t]$-module extending its original R-module structure. The variable t acts on M by the rule $t \cdot m := \phi(m)$ and $t^k \cdot m = \phi^k(m)$. Therefore, for a polynomial $p(t) \in R[t]$, the action of $p(t)$ on M is exactly the endomorphism $p(\phi)$. With this language, the conclusion $p(\phi) = 0$ says that the polynomial $p(t)$ annihilates M as an $R[t]$-module $p(t) \cdot M = 0$.

The determinant trick constructs such an annihilating polynomial from a finite set of generators of M. The key point is that, although M need not have a basis, the endomorphism ϕ gives us enough "linear equations" among a chosen generating set to apply the adjugate identity.

Let M be generated by $x_1, \ldots, x_n$, let $I \subset R$ be an ideal, and let $\phi : M \to M$ be an endomorphism such that $\phi(M) \subseteq IM$. Since the x_i generate M, we can write

$$\phi(x_i) = \sum_{j=1}^{n} a_{ij} x_j \qquad \text{with} \qquad a_{ij} \in I.$$

These are n linear equations in the generators x_i, but the coefficients include the operator ϕ. The key observation is that scalar multiplication by elements of R commutes with ϕ. Thus the subring $R[\phi] \subseteq \mathrm{End}_R(M)$ generated by scalar multiplications and by ϕ is commutative; equivalently, it is the image of the evaluation map $R[t] \to \mathrm{End}_R(M)$ sending t to ϕ.

Let $A = (a_{ij})$. The equations above may be rewritten as

$$(\phi \,\mathrm{Id}_{n\times n} - A) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = 0.$$

Now the matrix $\phi \mathrm{Id}_{n\times n} - A$ has entries in the commutative ring $R[\phi]$, so the adjugate identity applies. Multiplying by the adjugate matrix shows that $\det(\phi I_n - A)$ kills each generator x_i, and hence kills all of M. Expanding this determinant gives a monic polynomial in ϕ whose non-leading coefficients lie in powers of I. This is the determinant trick.

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(6) **The Determinant Trick.** Let M be an R-module generated by $x_1, \ldots, x_n$, let $I \subset R$ be an ideal, and let $\phi \in \mathrm{End}_R(M)$ satisfy $\phi(M) \subseteq IM$.

(a) Prove that there exist elements $a_{ij} \in I$ such that

$$\phi(x_i) = \sum_{j=1}^{n} a_{ij} x_j \qquad \text{for every } i = 1, \ldots, n.$$

(b) Let $A = (a_{ij})$ and let $B = \phi I_n - A$, viewed as a matrix with entries in the commutative ring $R[\phi] \subseteq \mathrm{End}_R(M)$. Prove that

$$B \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = 0.$$

(c) Use the identity $\mathrm{adj}(B)B = \det(B) I_n$ to prove that $\det(\phi I_n - A) x_i = 0$ for every i.

(d) Deduce the Cayley-Hamilton theorem stated above.

(7) **Applications of the Cayley–Hamilton Theorem.** Let $A = (a_{ij})$ be an $n \times n$ matrix with entries in R, and let $\phi_A : R^n \to R^n$ be the corresponding R-linear map.

(a) Apply the determinant trick to $M = R^n$, the standard basis $e_1, \ldots, e_n$, $I = R$, and $\phi = \phi_A$. Prove that $\chi_A(\phi_A) = 0$.

(b) Translate the previous statement into the matrix identity $\chi_A(A) = 0$.

(c) Prove that if $A \in \mathrm{Mat}_{n\times n}(R)$ and $\det(A)$ is a unit, then A is invertible. Hint: Use the adjugate matrix.

(d) Prove the converse: if A is invertible, then $\det(A)$ is a unit.

(e) Combine this with the local criterion for isomorphisms to prove that $A : R^n \to R^n$ is an isomorphism if and only if $A_{\mathfrak{p}} : R^n_{\mathfrak{p}} \to R^n_{\mathfrak{p}}$ is an isomorphism for every prime ideal $\mathfrak{p} \subset R$.

(f) Show that the condition in part (e) is equivalent to $\det(A) \notin \mathfrak{p}$ for every prime ideal $\mathfrak{p} \subset R$, and hence to $\det(A)$ being a unit.

(8) **Cayley–Hamilton Computations.** Compute the characteristic polynomial in each case and verify the Cayley–Hamilton identity directly.

(a) $R = \mathbb{Z}$ and $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

(b) R is any ring and $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

(c) $R = \mathbb{Z}/6\mathbb{Z}$ and $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$.

(d) R is any ring, $a, b \in R$, and $A = \begin{pmatrix} 0 & -b \\ 1 & -a \end{pmatrix}$.

(e) Let A be a 2×2 matrix over R with trace τ and determinant δ. Prove directly that

$$A^2 - \tau A + \delta I_2 = 0.$$

We now get to a series of amazingly powerful results, all of which can roughly be called some version of Nakayama's Lemma. Somewhat annoyingly each of these results will have its own hypothesis and conclusions, however, they all fit into a broad framework: The slogan is: *for finitely generated modules, if nothing remains after reducing modulo a sufficiently small ideal, then there was nothing there to begin with*. We will begin with the most used versions of this lemma before stating it in its most general form.

Lemma 2 (Nakayama's Lemma I)**.** *Let R be a ring and let M be a finitely generated R-module. If $I \subset R$ is an ideal such that $IM = M$ then there exists an element $r \in I$ that acts on M as the identity, i.e. $(1-r)M = 0$.*

Recall that a ring R is local if it has a unique maximal ideal $\mathfrak{m} \subset R$, in which case we denote it by $(R, \mathfrak{m})$. (Note Mel Hochster's notes adopt the somewhat non-standard terminology that a local ring be required to be Noetherian. We, and most commutative algebraists, do not do this.) In this setting, since $\mathfrak{m} \subset R$ is maximal the quotient $\kappa(R) := R/\mathfrak{m}$ is a field, which we call the *residue field* of R. For any ideal $I \subset R$, the quotient module $M/IM \cong M \otimes_R (R/I)$, and so has the structure of an R/I-module. This means that when we specialize to the case when $I = \mathfrak{m}$ the module $M/\mathfrak{m}M$ is isomorphic to $M \otimes_R (R/\mathfrak{m}) = M \otimes_R \kappa(R)$, meaning $M/\mathfrak{m}M$ is a $\kappa(R)$-module, i.e. it is just a vector space over the field $\kappa(R)$!

Lemma 3 ((Local) Nakayama's Lemma II)**.** *Let $(R, \mathfrak{m})$ be a local ring and let M be a finitely generated R-module. If $\mathfrak{m}M = M$ then $M = 0$.*

Lemma 4 ((Local) Nakayama's Lemma III)**.** *Let $(R, \mathfrak{m})$ be a local ring and let M be a finitely generated R-module. A finite set $\{x_1, \ldots, x_n\} \subset M$ is a generating set for M if and only if their images $\{\overline{x}_1, \ldots, \overline{x}_n\}$ span $M/\mathfrak{m}M$ as a vector space over the field $R/\mathfrak{m} = \kappa(R)$.*

While the above lemmas are the most common versions of Nakayama's Lemma one might use in commutative algebra and algebraic geometry. They all are, essentially special cases or corollaries, of a more general version. The *Jacobson radical* of R is

$$\mathrm{Jac}(R) := \bigcap_{\mathfrak{m} \in \mathrm{mSpec}(R)} \mathfrak{m},$$

the intersection of all maximal ideals of R. Note this is not the same thing in general as the nilradical $\sqrt{\langle 0\rangle}$, which is the set of nilpotent elements and is equal to the intersection of all prime ideals of R. If $a \in \mathrm{Jac}(R)$, then $1+a$ is a unit. Indeed, if $1+a$ were not a unit, it would be contained in some maximal ideal $\mathfrak{m}$; since $a \in \mathfrak{m}$ as well, this would force $1 \in \mathfrak{m}$, a contradiction. This observation will lead us to the following general version of Nakayama's Lemma.

Lemma 5 (Nakayama's Lemma IV)**.** *Let R be a ring and let M be a finitely generated R-module. If $I \subset R$ is an ideal such that I is contained in the Jacobson radical of R and $IM = M$ then $M = 0$.*

A brief historical aside: Like most results, a number of versions of Nakayama's lemma appeared before Tadashi Nakayama stated it in its now-familiar form in 1951. Krull proved the case when M was an ideal, and Goro Azumaya published a substantially generalized version (for all associative rings) in 1951. According to Matsumura, "Priority is obscure, and although it is usually called the Lemma of Nakayama, late Prof. Nakayama did not like the name."

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(9) **Examples and Hypotheses for Nakayama's Lemma.**

(a) Let $R = \mathbb{Z}$, $I = \langle 2\rangle$, and $M = \mathbb{Z}/3\mathbb{Z}$. Verify that $IM = M$ and find $a \in I$ such that $(1-a)M = 0$.

(b) In the previous example, explain why your computation does not imply that $M = 0$. Which hypothesis from Nakayama's Lemma is missing?

(c) Let $R = \mathbb{K}[x]_{\langle x\rangle}$ and let $M = R[1/x]/R$. Prove that $M \neq 0$ but $xM = M$. Hint: Show that $1/x + R$ is nonzero and that multiplication by x is surjective on M.

(d) Explain why part (c) shows that finite generation cannot be removed from Nakayama's Lemma.

(10) **The Jacobson Radical.** Let R be a ring.

(a) Prove that if $a \in \mathrm{Jac}(R)$, then $1+a$ is a unit in R.

(b) More generally, prove that if $a \in \mathrm{Jac}(R)$ and $r \in R$, then $1+ra$ is a unit.

(c) Prove that if $(R, \mathfrak{m})$ is a local ring, then $\mathrm{Jac}(R) = \mathfrak{m}$.

(d) Compute $\mathrm{Jac}(R)$ for the following rings: $R = \mathbb{K}$ a field, $R = \mathbb{Z}$, and $R = \mathbb{Z}/12\mathbb{Z}$.

(e) Give an example of a ring R for which $\mathrm{Jac}(R)$ and $\sqrt{\langle 0\rangle}$ are not equal.

(11) **Proving Nakayama's Lemma.** Let R be a ring, $I \subset R$ an ideal, and M a finitely generated R-module.

(a) Use the determinant trick with $\phi = \mathrm{Id}_M$ to prove Nakayama's Lemma I: if $IM = M$, then there exists $r \in I$ such that $(1-r)M = 0$.

(b) Assume that $I \subset \mathrm{Jac}(R)$. Use part (a) to deduce Nakayama's Lemma IV: if $IM = M$, then $M = 0$.

(c) Explain why the local version of Nakayama's Lemma II follows from Nakayama's Lemma IV.

(12) **Applications of Nakayama's Lemma.** Let R be a ring and M a finitely generated R-module. Let $I \subset R$ be an ideal, and assume that $I \subset \mathrm{Jac}(R)$.

(a) Let $N \subseteq M$ be a submodule. Prove that if $M = N + IM$, then $M = N$. Hint: Apply Nakayama's Lemma to M/N.

(b) Let $\phi : M \to N$ be a homomorphism of finitely generated R-modules. Prove that ϕ is surjective if and only if the induced map

$$M/IM \longrightarrow N/IN$$

is surjective. Hint: Apply Nakayama's Lemma to $\mathrm{coker}(\phi)$.

(c) Let $(R, \mathfrak{m})$ be local and let M be finitely generated. Prove that $M = 0$ if and only if $M/\mathfrak{m}M = 0$.

(d) Let $(R, \mathfrak{m})$ be local and let $\phi : M \to N$ be a homomorphism of finitely generated R-modules. Prove that if the induced map

$$M/\mathfrak{m}M \longrightarrow N/\mathfrak{m}N$$

is an isomorphism, then ϕ is surjective. Give an example showing that ϕ need not be injective.

(e) Give an example showing that an injective map of finitely generated modules can become non-injective after reducing modulo $\mathfrak{m}$. Hint: Use multiplication by x on $\mathbb{K}[x]_{\langle x \rangle}$.

(13) **Surjective Endomorphisms of Finitely Generated Modules.** Let M be a finitely generated R-module and let $\phi : M \to M$ be a surjective endomorphism.

(a) Explain how we can view M as an $R[t]$-module by letting t act as ϕ. Why is M finitely generated as an $R[t]$-module?

(b) Use that ϕ is surjective to prove that $tM = M$ as an $R[t]$-module.

(c) Apply Nakayama's Lemma I over the ring $R[t]$ with the ideal $\langle t \rangle$ to prove that there exists a polynomial $q(t) \in R[t]$ such that

$$(1 - tq(t))M = 0.$$

(d) Deduce that ϕ is an automorphism of M. Hint: Translate the identity in part (c) into an identity involving ϕ and $q(\phi)$.

(e) Now suppose $(R, \mathfrak{m})$ is local and M is finitely generated. Prove that if the induced map $M/\mathfrak{m}M \to M/\mathfrak{m}M$ is an isomorphism, then ϕ is an automorphism of M.

(14) **More Applications of Nakayama's Lemma.** Let R be a nonzero ring.

(a) Prove that $R^n \cong R^m$ as R-modules if and only if $n = m$. Hint: Choose a maximal ideal $\mathfrak{m}$ and reduce modulo $\mathfrak{m}$.

(b) Let M be a free R-module such that $M \cong R^n$. Prove that any set of n elements that generate M is a basis for M as an R-module. Hint: Use the previous exercise on surjective endomorphisms.

(c) Define the *rank* of a finitely generated free R-module M to be the number of elements in a basis for M. Prove that the rank of M is well-defined.

(15) **Minimal Generating Sets over Local Rings.** Let $(R, \mathfrak{m})$ be a local ring with residue field $\kappa = R/\mathfrak{m}$. Let M be a finitely generated R-module and let $x_1, \ldots, x_r$ be elements of M.

(a) Prove that if $x_1, \ldots, x_r$ generate M as an R-module, then their images generate $M/\mathfrak{m}M$ as a κ-vector space.

(b) Prove the converse: if the images of $x_1, \ldots, x_r$ generate the κ-vector space $M/\mathfrak{m}M$, then $x_1, \ldots, x_r$ generate M as an R-module.

(c) Deduce Nakayama's Lemma III.

(d) Let $\mu_R(M)$ denote the minimal number of generators of M as an R-module. Prove that $\mu_R(M) = \dim_\kappa(M/\mathfrak{m}M)$.

(e) We say that $x_1, \ldots, x_r$ are *minimal generators* for M if they generate M and $r = \mu_R(M)$. Prove that $x_1, \ldots, x_t$ are minimal generators for M if and only if their images form a basis of $M/\mathfrak{m}M$.

(16) **Computing Minimal Numbers of Generators.** Let $R = \mathbb{K}[x, y]_{\langle x,y \rangle}$ with maximal ideal $\mathfrak{m} = \langle x, y \rangle R$ and residue field $\mathbb{K}$.

(a) Compute the minimal number of generators of $R/\langle x^2, xy, y^2 \rangle$ as an R-module.

(b) Compute the minimal number of generators of $\mathfrak{m}$ as an R-module.

(c) Compute the minimal number of generators of $\mathfrak{m}^2$ as an R-module.

(d) Let $S = \mathbb{Z}_{\langle p \rangle}$ and $M = S/\langle p^4 \rangle \oplus S/\langle p^2 \rangle$. Compute the minimal number of generators of M as an S-module.

(17) **Local Generators and Local Surjectivity.** Let R be a ring. Let M be a finitely generated R-module and let $x_1, \ldots, x_r \in M$.

(a) Prove that $x_1, \ldots, x_r$ generate M if and only if, for every maximal ideal $\mathfrak{m} \subset R$, their images generate $M_\mathfrak{m}/\mathfrak{m}M_\mathfrak{m}$ as a vector space over $\kappa(\mathfrak{m}) = R_\mathfrak{m}/\mathfrak{m}R_\mathfrak{m}$.

(b) Let $\phi : M \to N$ be a homomorphism of finitely generated R-modules. Prove that ϕ is surjective if and only if, for every maximal ideal $\mathfrak{m} \subset R$, the induced map below is surjective:

$$M_\mathfrak{m}/\mathfrak{m}M_\mathfrak{m} \longrightarrow N_\mathfrak{m}/\mathfrak{m}N_\mathfrak{m}$$

(c) Explain why the analogous statement for injectivity is false by revisiting the multiplication-by-x example over $\mathbb{K}[x]_{\langle x \rangle}$.

(d) Let $f_1, \ldots, f_n \in R$ generate the unit ideal. State and prove a version of part (b) where one first localizes at each f_i.

WORKSHEET 6.3: GRÖBNER BASES PT. III

Fix a field $\mathbb{K}$ and set $S = \mathbb{K}[x_1, \ldots, x_n]$. Unless otherwise specified, all ideals live in S and all monomial orders are denoted by $<$. In the previous Gröbner basis worksheets we introduced monomial orders, multivariable division, monomial ideals, initial ideals, and the definition of a Gröbner basis. We also proved that Gröbner bases exist and that they solve the ideal membership problem. This worksheet is the last Gröbner basis worksheet of the course. Its goal is to consolidate the definitions, introduce the standard monomial basis of a quotient, and give the computational theorem which makes Gröbner bases effective: Buchberger's criterion and algorithm. The guiding principle is that a Gröbner basis lets us replace questions about a complicated ideal I by questions about the monomial ideal $\text{in}_<(I)$.

Recall a *monomial ordering* on $\mathbb{K}[x_1, \ldots, x_n]$ is a total well-ordering $<$ on $\mathbb{Z}^n_{\geq 0}$ such that if $\alpha < \beta$ then $\alpha + \gamma < \beta + \gamma$ for all $\gamma \in \mathbb{Z}^n_{\geq 0}$. Using the bijection between monomials in $\mathbb{K}[x_1, \ldots, x_n]$ and elements of $\mathbb{Z}^n_{\geq 0}$ we think of a monomial ordering as giving us a way to compare monomials by saying $x^\alpha < x^\beta$ if and only if $\alpha < \beta$. Recall that for a nonzero polynomial $f \in S$ we write $\text{LT}(f) = \text{LC}(f)\text{LM}(f)$ where $\text{LM}(f)$ is the largest monomial appearing in f, $\text{LC}(f) \in \mathbb{K}^\times$ is its coefficient, and $\text{LT}(f)$ is the corresponding term.

Definition 1. Let $<$ be a monomial order on $\mathbb{K}[x_1, \ldots, x_n]$. The *initial ideal* of a nonzero ideal $I \subset \mathbb{K}[x_1, \ldots, x_n]$ with respect to $<$ is:

$$\text{in}_<(I) := \langle \text{LT}(f) \mid f \in I \rangle.$$

A finite subset $\mathcal{G} = \{g_1, \ldots, g_s\} \subset I$ is a *Gröbner basis* for I with respect to $<$ if $\text{in}_<(I) = \langle \text{LT}(g_1), \ldots, \text{LT}(g_s) \rangle$.

It is standard to set $\text{in}_<(\langle 0 \rangle) = \langle 0 \rangle$.

---

(1) **Reviewing leading terms and initial ideals.** Let $S = \mathbb{K}[x, y, z]$.

(a) For

$$f = 3x^2z - 5xy^3 + 7y^4z + xz^5$$

compute $\text{LM}(f)$, $\text{LC}(f)$, and $\text{LT}(f)$ using each of the following orders: lex with $x > y > z$, grlex with $x > y > z$, and grevlex with $x > y > z$.

(b) Prove that if $f, g \in S$ are nonzero, then

$$\text{LT}(fg) = \text{LT}(f)\text{LT}(g), \qquad \text{LM}(fg) = \text{LM}(f)\text{LM}(g), \qquad \text{LC}(fg) = \text{LC}(f)\text{LC}(g).$$

Where do you use the defining property of a monomial order?

(c) Let $I = \langle xy - 1, y^2 - 1 \rangle \subset \mathbb{K}[x, y]$ and use lex order with $x > y$. Show that

$$\langle \text{LT}(xy - 1), \text{LT}(y^2 - 1) \rangle = \langle xy, y^2 \rangle.$$

Then show that $x - y \in I$ and explain why this proves $\{xy - 1, y^2 - 1\}$ is not a Gröbner basis.

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(d) Give an example of an ideal I and a generating set $F = \{f_1, \ldots, f_s\}$ such that

$$\langle \mathrm{LT}(f_1), \ldots, \mathrm{LT}(f_s) \rangle \subsetneq \mathrm{in}_<(I).$$

Your example may be the one from part (c), but make sure to identify an explicit element of $\mathrm{in}_<(I)$ missing from the smaller ideal.

(2) **Division and normal forms.** Let $G = (g_1, \ldots, g_t)$ be an ordered list of nonzero polynomials in S. Dividing f by G produces an expression

$$f = q_1 g_1 + \cdots + q_t g_t + r$$

where no term of r is divisible by any of $\mathrm{LT}(g_1), \ldots, \mathrm{LT}(g_t)$. We call such an r a *remainder* or *normal form* of f with respect to G and sometimes write $r = \overline{f}^G$ or $r = \mathrm{NF}_G(f)$.

(a) Explain why the remainder can depend on the order of the list G if G is not a Gröbner basis.

(b) Prove that if G is a Gröbner basis for $I = \langle G \rangle$, then $f \in I$ if and only if the remainder on division by G is zero.

(c) Prove that if G is a Gröbner basis for I, then the remainder of f on division by G is independent of all choices made during the division algorithm. Hint: If two remainders r and r' occur, compare $r - r'$.

(d) Let $I = \langle xy - 1, y^2 - 1 \rangle \subset \mathbb{K}[x, y]$ with lex order $x > y$. Divide $x - y$ by the ordered list $(xy - 1, y^2 - 1)$. Explain why the answer is consistent with part (b).

(3) **Minimal and reduced Gröbner bases.** A Gröbner basis $G = \{g_1, \ldots, g_t\}$ is *minimal* if every g_i is monic and no $\mathrm{LM}(g_i)$ divides $\mathrm{LM}(g_j)$ for $i \neq j$. It is *reduced* if every g_i is monic and no monomial appearing in $g_i - \mathrm{LT}(g_i)$ is divisible by any $\mathrm{LM}(g_j)$.

(a) Prove that every reduced Gröbner basis is minimal.

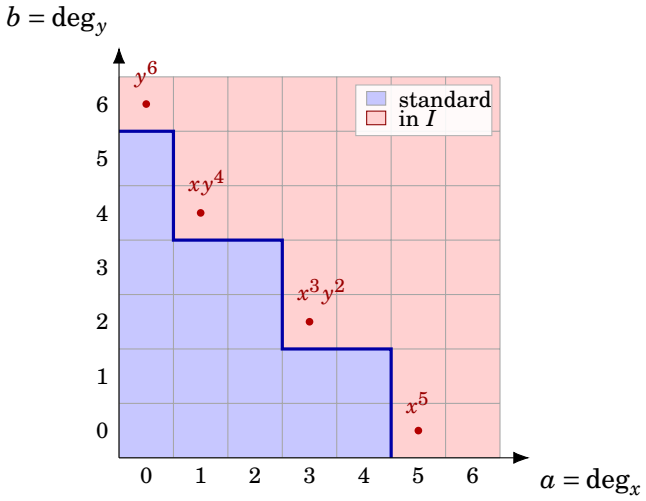
(b) Starting from any Gröbner basis G, explain how to produce a minimal Gröbner basis by rescaling and deleting redundant elements.

(c) Starting from a minimal Gröbner basis, explain how to reduce the non-leading terms of each g_i by the other basis elements. Why does this process produce a reduced Gröbner basis?

(d) Prove uniqueness of reduced Gröbner bases for a fixed ideal and fixed monomial order. Hint: If G and H are reduced Gröbner bases for I, first compare their leading monomials using the uniqueness of minimal monomial generators of $\mathrm{in}_<(I)$. Then compare the two basis elements with the same leading monomial.

For monomial ideals, the monomials outside the ideal form a "staircase." For example if we consider the ideal $I = \langle x^5, x^3y^2, xy^4, y^6 \rangle$ in $\mathbb{K}[x, y]$ we can picture it as via the diagram below. The monomials in I are

shown in red while those that are non-zero in the quotient S/I are shown in blue. For a general ideal, the same staircase is obtained from the initial ideal. These monomials give canonical basis for S/I.



Definition 2. Let $I \subset S$ be an ideal. A monomial x^α is called a *standard monomial* for I with respect to $<$ if $x^\alpha \notin \text{in}_<(I)$.

We denote the set of standard monomials by $\text{std}_<(I)$. If $G = \{g_1, \ldots, g_t\}$ is a finite set of nonzero polynomials, we say x^α is *standard for* G if no $\text{LM}(g_i)$ divides x^α. If G is a Gröbner basis for I, then being standard for I is the same as being standard for G. Thus the standard monomials can be read off from the leading monomials of any Gröbner basis.

Theorem 3 (Standard Monomial Basis Theorem)**.** *Let $I \subset S$ be an ideal and fix a monomial order. The residue classes of the standard monomials form a $\mathbb{K}$-basis of S/I.*

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(4) **Proving the Standard Monomial Basis Theorem.** Let $G = \{g_1, \ldots, g_t\}$ be a Gröbner basis for I.

- (a) Show that the remainder obtained by dividing any $f \in S$ by G is a $\mathbb{K}$-linear combination of standard monomials.
- (b) Deduce that the residue classes of the standard monomials span S/I.
- (c) Suppose h is a nonzero $\mathbb{K}$-linear combination of standard monomials. Prove that $h \notin I$. Hint: What can you say about $\text{LT}(h)$ if $h \in I$?
- (d) Deduce linear independence and conclude Theorem **??**.
- (e) Explain why Theorem **??** gives another proof that normal forms with respect to a Gröbner basis are unique.

(5) **Computing standard monomials.** In each part, list the standard monomials and draw the staircase picture when there are only two variables.

(a) $M = \langle x^3, x^2y, y^4 \rangle \subset \mathbb{K}[x,y]$.

(b) $M = \langle x^2, xy, y^3 \rangle \subset \mathbb{K}[x,y]$.

(c) $M = \langle x^2z, xyz, y^2z, z^3 \rangle \subset \mathbb{K}[x,y,z]$.

(d) Let $I = \langle x - y^2, y^3 - 1 \rangle \subset \mathbb{K}[x,y]$ with lex order $x > y$. Verify that the displayed generators are a reduced Gröbner basis. List $\mathrm{std}_<(I)$ and compute $\dim_{\mathbb{K}}(S/I)$.

(e) With I as in part (d), compute the normal forms of x^5, $x^2y + y^7$, and $xy^2 + x + y$.

(6) **Finite quotients and powers of variables.** Let $I \subset S$ be an ideal and fix a monomial order.

(a) Prove that S/I is finite-dimensional as a $\mathbb{K}$-vector space if and only if $\mathrm{std}_<(I)$ is finite. In this case,

$$\dim_{\mathbb{K}}(S/I) = \#\,\mathrm{std}_<(I).$$

(b) Let $M \subset S$ be a monomial ideal. Prove that S/M is finite-dimensional over $\mathbb{K}$ if and only if, for every variable x_i, there exists an integer $a_i \geq 1$ such that $x_i^{a_i} \in M$.

(c) Deduce that S/I is finite-dimensional over $\mathbb{K}$ if and only if, for every variable x_i, there exists $a_i \geq 1$ such that $x_i^{a_i} \in \mathrm{in}_<(I)$.

(d) Apply part (c) to the ideals

$$\langle x - y^2, y^3 - 1 \rangle, \qquad \langle xy - 1 \rangle, \qquad \langle x^2, y^2, z^2, xy - yz \rangle.$$

We now turn to the computational heart of the subject. The obstruction to a generating set being a Gröbner basis is cancellation of leading terms. Buchberger's insight was that it is enough to check the simplest possible cancellations, namely pairwise cancellations.

Definition 4. Let $f, g \in S$ be nonzero. Write $\mathrm{LT}(f) = c_f x^\alpha$ and $\mathrm{LT}(g) = c_g x^\beta$, and let $x^\gamma = \mathrm{lcm}(x^\alpha, x^\beta)$. The *$S$-polynomial* of f and g is

$$S(f,g) = \frac{x^{\gamma-\alpha}}{c_f} f - \frac{x^{\gamma-\beta}}{c_g} g.$$

Thus the leading terms of the two summands cancel.

Theorem 5 (Buchberger's Criterion)**.** *Let $G = \{g_1, \ldots, g_t\} \subset S$ and set $I = \langle G \rangle$. Then G is a Gröbner basis for I if and only if every $S(g_i, g_j)$ has remainder zero on division by G.*

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(7) **S-polynomials by hand.** Work in $\mathbb{K}[x,y]$ with lex order $x > y$.

(a) Let $g_1 = xy - 1$ and $g_2 = y^2 - 1$. Compute $S(g_1, g_2)$ and reduce it by the ordered list (g_1, g_2).

(b) Add the nonzero remainder from part (a) to the list. Call it g_3. Compute $S(g_1, g_3)$ and $S(g_2, g_3)$ and reduce both by (g_1, g_2, g_3).

(c) Conclude that $\{g_1, g_2, g_3\}$ is a Gröbner basis for $\langle xy-1, y^2-1\rangle$.

(d) Reduce this Gröbner basis to the unique reduced Gröbner basis.

(8) **Another full Buchberger computation.** Work in $\mathbb{K}[x,y]$ with lex order $x > y$. Let

$$f_1 = x^2 - y, \qquad f_2 = xy - 1,$$

and set $I = \langle f_1, f_2\rangle$.

(a) Compute $S(f_1, f_2)$ and reduce it by (f_1, f_2).

(b) Continue Buchberger's algorithm until every pair reduces to zero.

(c) Show that the reduced Gröbner basis is $\{x - y^2,\ y^3 - 1\}$.

(d) Find the standard monomials of S/I and to solve the system $x^2 - y = xy - 1 = 0$ over $\mathbb{C}$.

(9) **Proving Buchberger's Criterion.** We prove Theorem **??** in pieces. Let $G = \{g_1, \ldots, g_t\}$ and $I = \langle G\rangle$.

(a) Prove the easy direction: if G is a Gröbner basis, then every $S(g_i, g_j)$ reduces to zero modulo G.

(b) For the converse, let $f \in I$ and write $f = q_1 g_1 + \cdots + q_t g_t$. For such a representation, define

$$\delta(q_1, \ldots, q_t) = \max_i \mathrm{LM}(q_i g_i).$$

Choose a representation of f for which δ is as small as possible. Explain why this makes sense.

(c) Show that if the terms of monomial δ in the sum $\sum q_i g_i$ do not cancel, then $\mathrm{LT}(f)$ is divisible by some $\mathrm{LT}(g_i)$.

(d) Suppose instead that the terms of monomial δ cancel. Show that the cancellation gives a nontrivial relation among leading terms of some g_i's.

(e) Use the assumption that every $S(g_i, g_j)$ reduces to zero to replace this relation by a combination of the g_i's whose terms all have monomial strictly smaller than δ.

(f) Explain why part (e) contradicts the minimal choice of the representation unless $\mathrm{LT}(f)$ is divisible by some $\mathrm{LT}(g_i)$.

(g) Conclude that $\mathrm{in}_<(I) = \langle \mathrm{LT}(g_1), \ldots, \mathrm{LT}(g_t)\rangle$.

(10) **Buchberger's Algorithm.** Buchberger's criterion leads directly to an algorithm.

Algorithm 1 Buchberger's Algorithm

Input: nonzero polynomials $F = \{f_1, \dots, f_s\} \subset S$ and a monomial order $<$
$G \leftarrow F$
$P \leftarrow \{\{g,h\} \mid g,h \in G,\ g \neq h\}$
while $P \neq \varnothing$ **do**
    choose a pair $\{g,h\} \in P$ and remove it from P
    divide $S(g,h)$ by G and let r be the remainder
    if $r \neq 0$ **then**
        $P \leftarrow P \cup \{\{r,k\} \mid k \in G\}$
        $G \leftarrow G \cup \{r\}$
    end if
end while
return G

(a) Prove that whenever a nonzero remainder r is added, the monomial ideal $\langle \mathrm{LT}(g) \mid g \in G \rangle$ strictly increases.

(b) Use the Noetherian property of S (or Dickson's Lemma for monomial ideals) to prove that the algorithm terminates.

(c) Prove that the output is a Gröbner basis for $\langle F \rangle$.

(11) **Ideal membership, inclusion, and equality.** Let $I = \langle f_1, \dots, f_s \rangle$ and $J = \langle h_1, \dots, h_r \rangle$ be ideals of S.

(a) Suppose G_J is a Gröbner basis for J. Prove that $I \subseteq J$ if and only if every $\mathrm{NF}_{G_J}(f_i) = 0$.

(b) Explain how to test whether $I = J$ using two Gröbner basis computations.

(c) Use part (a) to decide whether

$$\langle x^2 - y, xy - 1 \rangle \subseteq \langle x - y^2, y^3 - 1 \rangle$$

in $\mathbb{K}[x,y]$ with lex order $x > y$.

(d) Let G be a Gröbner basis for I. Prove that f and g have the same image in S/I if and only if $\mathrm{NF}_G(f) = \mathrm{NF}_G(g)$.

(12) **Radical membership: the Rabinowitsch trick.** Let $I = \langle f_1, \dots, f_s \rangle \subset S$ and let $f \in S$. Introduce a new variable u.

(a) Prove that

$$f \in \sqrt{I} \quad \Longleftrightarrow \quad 1 \in \langle f_1, \dots, f_s, 1 - uf \rangle \subset S[u].$$

Hint: Interpret $S[u]/\langle 1 - uf \rangle$ as the localization $S[f^{-1}]$.

(b) Explain how this turns radical membership into an ideal membership test using a Gröbner basis.

(c) Use this criterion to show that $x \in \sqrt{\langle x^2, xy \rangle} \subset \mathbb{K}[x,y]$.

(d) Is $y \in \sqrt{\langle x^2, xy \rangle}$? Justify your answer algebraically and with the Gröbner basis criterion.

(13) **Computer check.** Use *Macaulay2* to verify at least two computations from this worksheet.

(a) Compute a Gröbner basis for $\langle xy - 1, y^2 - 1 \rangle$ with lex order $x > y$.

(b) Compute a Gröbner basis for $\langle x^2 - y, xy - 1 \rangle$ with lex order $x > y$.

(c) Compute the standard monomials for one zero-dimensional ideal above.

(d) Compute one intersection ideal from Problems 15 and 16.

(e) Compare the reduced Gröbner basis returned by the computer with your hand computation. Which intermediate choices differed, and why did the reduced answer not depend on those choices?

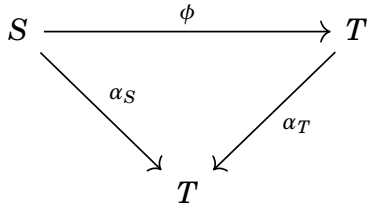
WORKSHEET 7.1: INTEGRAL EXTENSION PT. I

Throughout this course, "ring" means *commutative* ring with unity and all ring homomorphisms preserve unity. In the previous worksheets we have repeatedly moved between rings, modules, quotients, localization, and spectra. The goal of this worksheet is to introduce a language that lets us treat many ring maps as geometric families: the language of *R-algebras*. Once this language is in place, we will distinguish two finiteness notions that sound similar but behave very differently: finite generation as an algebra and finite generation as a module.

The main bridge between these two notions is ntegrality. Integral elements are elements of an extension ring that satisfy monic polynomial equations over the base. The monic condition is what lets one convert algebraic equations into module-theoretic finiteness. This is one of the basic mechanisms behind many finiteness theorems in commutative algebra and algebraic geometry.

Let R be a ring. An *R-algebra* is a ring S together with a specified ring homomorphism $\alpha : R \longrightarrow S$. Note we do not require that α be injective, however, the case when α is an inclusion, i.e., $\mathbb{K}$ into $\mathbb{K}[x]$, is generally the motivating example. We often think of an R-algebra structure on a ring S as giving us a way to multiply elements of S by elements of R, i.e., $r \cdot s := \alpha(r)s$. Thus every R-algebra S is naturally an R-module. The key difference between an R-module M and an R-algebra S is that for module we may only multiple elements of M by elements of R, but for an R-algebra we every element of S may be multiplied together. The key analogy might be: R-modules are to vector spaces, as R-algebras are to rings.

An R-algebra homomorphism $\phi : S \to T$ is a ring homomorphism compatible with the structure maps from R, meaning that the diagram



commutes. Equivalently, $\phi(\alpha_S(r)) = \alpha_T(r)$ for every $r \in R$. An *R-subalgebra* of S is a subring $S' \subset S$, which contains $\alpha(R)$. Given any set $\mathscr{S} \subset S$ the *R-algebra generated by $\mathscr{S}$* is the smallest R-subalgebra of S containing $\mathscr{S}$. We often write $R[\mathscr{S}]$ to denote R-subalgebra generated by $\mathscr{S}$, because it can explicitly be written down as the set of polynomials in the elements in $\mathscr{S}$ with coefficients in R.

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(1) **First Encounters with R-algebras.** Let $\alpha : R \to S$ be an R-algebra.

(a) Verify that the rule $r \cdot s = \alpha(r)s$ makes S into an R-module.

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(b) Give an example where α is not injective. In your example, identify a nonzero element of R that acts as zero on S.

(c) Show that every quotient R/I is naturally an R-algebra.

(d) Show that if S and T are R-algebras and $\phi : S \to T$ is an R-algebra homomorphism, then ϕ is R-linear for the underlying R-module structures.

(e) Let $\mathbb{K} \subseteq \mathbb{E}$ be a field extension. Explain why $\mathbb{E}$ is a $\mathbb{K}$-algebra. What is the underlying $\mathbb{K}$-module structure?

(2) **Polynomial Rings and Generation of R-algebras.** Let S be an R-algebra and let $s_1, \ldots, s_n \in S$.

(a) Construct an R-algebra homomorphism

$$\mathrm{ev}_{s_1,\ldots,s_n} : R[x_1, \ldots, x_n] \longrightarrow S, \qquad f(x_1, \ldots, x_n) \longmapsto f(s_1, \ldots, s_n).$$

(b) Show that the image of this map is the smallest R-subalgebra of S containing $s_1, \ldots, s_n$. We denote this image by $R[s_1, \ldots, s_n]$.

(c) Prove that $S = R[s_1, \ldots, s_n]$ if and only if $\mathrm{ev}_{s_1,\ldots,s_n}$ is surjective.

(d) Deduce that S is generated as an R-algebra by n elements if and only if $S \cong R[x_1, \ldots, x_n]/I$ as an R-algebra for some ideal $I \subset R[x_1, \ldots, x_n]$.

(3) **Examples of R-algebras.** In each case, identify a finite set of algebra generators over the indicated base ring. Try to choose a minimal-looking set, though you do not need to prove minimality.

(a) $R[x]/\langle x^2 \rangle$ over R.

(b) $R[x, y]/\langle xy \rangle$ over R.

(c) R/I over R.

(d) $\mathbb{K}[t, t^{-1}]$ over $\mathbb{K}$.

(e) $\mathbb{Z}[i]$ over $\mathbb{Z}$.

(f) $\mathbb{K}[x, y]/\langle y^2 - x^3 \rangle$ over $\mathbb{K}[x]$.

(g) $\mathbb{K}(t)[[t]]$ over the $\mathbb{K}(t)$.

(h) $\mathbb{Z}[\sqrt{n}\, n \in \mathbb{Z}_{\geq 1}]$ over $\mathbb{Z}$.

As discussed above an R-algebra S is also always an R-module with the same R-multiplication. This leads to two different finiteness conditions depending on whether we wish to view S as an R-module or as an R-algebra. Let S be an R-algebra.

Definition 1. We say that S is *algebra-finite over* R if S is finitely generated s as an R-algebra.

Definition 2. We say that S is *module-finite over* R if S is finitely generated as an R-module.

The condition S be algebra-finite over R is sometimes called being *finite type* over R, i.e. we would say S is finite type over R if and only if S is algebra-finite over R. Similarly, many authors will use the phrase S

is *finite* over R or S is a *finite R-algebra* to mean S if a R-algebra that is module-finite,. In this worksheet we will attempt to restrict to using *algebra-finite* and *module-finite* to hopefully minimize confusion.

The key point is that module-finite is much stronger than algebra-finite. A module-finite algebra is controlled by finitely many R-linear combinations. An algebra-finite algebra is controlled by finitely many elements and all polynomial expressions in them.

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(4) **Module-Finite Implies Algebra-Finite.** Let S be an R-algebra and suppose that S is generated as an R-module by $m_1, \ldots, m_n \in S$.

(a) Show that the R-subalgebra $R[m_1, \ldots, m_n]$ contains every R-linear combination of $m_1, \ldots, m_n$.

(b) Deduce that $S = R[m_1, \ldots, m_n]$.

(c) Conclude that every module-finite R-algebra is algebra-finite over R.

(5) **Algebra-Finite Need Not Imply Module-Finite.** Assume R is nonzero.

(a) Show that $R[x]$ is algebra-finite over R.

(b) Suppose $p_1, \ldots, p_n \in R[x]$. Let d be the maximum of their degrees. Show that every R-linear combination of the p_i has degree at most d, unless it is zero.

(c) Deduce that $R[x]$ is not module-finite over R. Hint: Is x^{d+1} in the R-span of $p_1, \ldots, p_n$?

(d) Specialize to $R = \mathbb{K}$ a field and restate the result in the language of vector spaces.

(6) **Permanence of Algebra-Finiteness.** The goal of this exercise is to explore how being algebra-finite plays with many natural algebraic operations: quotienting, tensoring, localizing, and composing the underlying morphisms giving the algebra structure. Consider maps of rings:

$$R \xrightarrow{\alpha} S \xrightarrow{\beta} T \ .$$

The map α gives S an R-algebra structure. On the other hand T has two different algebra structures: we may think of T as an S-algebra via the map β or think of T as an R-algebra via $\beta \circ \alpha$.

(a) If S is algebra-finite over R and $J \subseteq S$ is an ideal, prove that S/J is algebra-finite over R.

(b) If S is algebra-finite over R and T is algebra-finite over S, prove that T is algebra-finite over R.

(c) Give an example where α and β are injective and T is algebra-finite over R, but S is not algebra-finite over R. Hint: Let $R = \mathbb{K}$, and $T = \mathbb{K}[x, y]$, what happens if $S = \mathbb{K}[x, xy, xy^2, xy^3, \ldots]$?

(d) If S is algebra-finite over R, and $R \to T$ is a ring map, prove that $T \otimes_R S$ is algebra-finite over T. What are algebra generators?

(e) If $U \subset R$ is multiplicatively closed and S is algebra-finite over R, prove that $U^{-1}S$ is algebra-finite over $U^{-1}R$.

(7) **Permanence of Module-Finiteness.** The goal of this exercise is to explore how being module-finite plays with many natural algebraic operations: quotienting, tensoring, localizing, and composing the underlying morphisms giving the algebra structure. Consider maps of rings:

$$R \xrightarrow{\alpha} S \xrightarrow{\beta} T \ .$$

The map α gives S an R-module structure. On the other hand T has two different module structures: we may think of T as an S-module via the map β or think of T as an R-module via $\beta \circ \alpha$.

(a) If S is module-finite over R and $J \subseteq S$ is an ideal, prove that S/J is module-finite over R.

(b) If S is module-finite over R and T is module-finite over S, prove that T is module-finite over R.

(c) If S is module-finite over R, and $R \to T$ is a ring map, prove that $T \otimes_R S$ is module-finite over T.

(d) If $U \subset R$ is multiplicatively closed and S is module-finite over R, prove that $U^{-1}S$ is module-finite over $U^{-1}R$.

(8) **Comparing the Two Notions in Examples.** Decide whether each algebra is algebra-finite and whether it is module-finite over the indicated base ring. Justify your answers.

(a) $\mathbb{K}[x]$ over $\mathbb{K}$.

(b) $\mathbb{K}[x]/\langle x^2 \rangle$ over $\mathbb{K}$.

(c) $\mathbb{Z}[i]$ over $\mathbb{Z}$.

(d) $\mathbb{Z}[1/2]$ over $\mathbb{Z}$.

(e) $\mathbb{K}[t]$ over $\mathbb{K}[t^2]$.

(f) $\mathbb{K}[t, t^{-1}]$ over $\mathbb{K}[t]$.

Let S be an R-algebra and let $s \in S$. We say that s is *integral over* R if there exists a monic polynomial with coefficients in R that s satisfies. In other words, s is integral over R if there are elements $a_1, \ldots, a_n \in R$ such that

$$s^n + a_1 s^{n-1} + a_2 s^{n-2} + \cdots + a_{n-1} s + a_n = 0$$

in S, where each a_i acts through the structure map $R \to S$. The word *monic* is essential. For example, every rational number satisfies some nonzero polynomial over $\mathbb{Z}$, but only the integers satisfy monic polynomial equations over $\mathbb{Z}$. The monic condition is exactly what allows us to solve for high powers of s in terms of lower powers.

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(9) **First Examples of Integral Elements.** Let S be an R-algebra.

(a) Show that every element in the image of $R \to S$ is integral over R.

(b) Show that every nilpotent element of S is integral over R.

(c) Let $S = R[x]/\langle x^2 - r \rangle$ for some $r \in R$. Show that the image of x in S is integral over R.

(d) Show that $i \in \mathbb{Z}[i]$ is integral over $\mathbb{Z}$.

(e) Show that $t \in \mathbb{K}[t]$ is integral over $\mathbb{K}[t^2]$.

(f) Show that $\frac{1}{2}(\sqrt{-3}+1) \in \mathbb{Q}(\sqrt{-3})$ is integral over $\mathbb{Z}[\sqrt{-3}]$.

(g) Show directly that $1/2 \in \mathbb{Q}$ is not integral over $\mathbb{Z}$. Hint: Suppose $(1/2)^n + a_1(1/2)^{n-1} + \cdots + a_n = 0$ and clear denominators.

(10) **Integral Elements Give Finite Modules.** Let S be an R-algebra and let $s \in S$.

(a) Suppose that the element $s \in S$ is integral over R, and

$$s^n + a_1 s^{n-1} + \cdots + a_n = 0$$

with $a_i \in R$. Show that every power s^m with $m \geq n$ is an R-linear combination of $1, s, \ldots, s^{n-1}$.

(b) Deduce that $R[s]$ is generated as an R-module by $1, s, \ldots, s^{n-1}$.

(c) Find a finite set of R-module generators for $S = \mathbb{K}[t]$ over $R = \mathbb{K}[t^2]$

(d) Find a finite set of R-module generators for $S = \mathbb{Z}[i]$ over $R = \mathbb{Z}$.

(11) **A Fundamental Criterion.** Let S be an R-algebra and let $s \in S$. Prove that the following are equivalent.

(a) s is integral over R.

(b) $R[s]$ is module-finite over R.

(c) There exists a finitely generated R-submodule $M \subseteq S$ such that $1 \in M$ and $sM \subseteq M$.

(12) **Integral Elements Form a Subring.** Let $R \to S$ be an R-algebra. Let $\overline{R}^S$ denote the set of elements of S that are integral over R. When R is an integral domain and S is a field we call $\overline{R}^S$ the *integral closure of R in S*. The *integral closure* or *normalization* of a domain R is the integral closure of R in its field of fractions $S = \mathrm{Frac}(R)$. We denote the integral closure of R by $\overline{R}$.

(a) If $x \in S$ is integral over R, prove that $R[x]$ is module-finite over R.

(b) If $x, y \in S$ are integral over R, prove that $R[x, y]$ is module-finite over R. Hint: First show $R[x]$ is finite over R, then show $R[x, y]$ is finite over $R[x]$.

(c) Let $s \in R[x, y]$. Show that multiplication by s defines an R-linear map

$$\mu_s : R[x, y] \longrightarrow R[x, y], \qquad f \longmapsto sf.$$

Use Cayley–Hamilton, or the determinant trick, to prove that s is integral over R.

(d) Conclude that $x + y$, xy, and $-x$ are integral over R.

(e) Conclude that $\overline{R}^S$ is a subring of S containingnthe image of R.

(13) **Computations with Integrality.** Let $\mathbb{K}$ be a field.

(a) Prove that the elements of $\mathbb{Q}$ integral over $\mathbb{Z}$ are exactly the integers. Hint: Write a rational number in lowest terms.

(b) Show that $\mathbb{K}[t]$ is module-finite over $\mathbb{K}[t^2, t^3]$. Find explicit module generators.

(c) Show that $\mathbb{K}[x,y]/\langle y^2 - x^3\rangle$ is module-finite over $\mathbb{K}[x]$. Find explicit module generators.

(d) Let $S = \mathbb{K}[x, x^{-1}]$, viewed as a $\mathbb{K}[x]$-algebra. Show that x^{-1} is not integral over $\mathbb{K}[x]$.

(e) Deduce that $\mathbb{K}[x, x^{-1}]$ is algebra-finite but not module-finite over $\mathbb{K}[x]$.

(14) **UFDs Are Integrally Closed.** Let R be a UFD with fraction field $S = \mathrm{Frac}(R)$.

(a) Explain how we can think of S as an R-algebra?

(b) Let $\alpha \in S$. Explain why we can write $\alpha = \frac{a}{b}$ with $a, b \in R$, $b \neq 0$, and a and b having no common irreducible factors.

(c) Give an example of a domain R that is *not* a UFD for which part (b) is false.

(d) Suppose $\alpha = a/b \in S$ is integral over R and

$$\left(\frac{a}{b}\right)^n + r_1\left(\frac{a}{b}\right)^{n-1} + \cdots + r_{n-1}\left(\frac{a}{b}\right) + r_n = 0$$

for $r_1, \ldots, r_n \in R$. By clearing denominators show that b divides a^n.

(e) Show that if a and b have no common irreducible factor and $b \mid a^n$, then b is a unit. Hint: Use unique factorization.

(f) Conclude that $\alpha = a/b \in R$. Therefore every element of K that is integral over R already lies in R.

(g) Conclude that every UFD is integrally closed.

(h) Show that $R = \mathbb{K}[t^2, t^3]$ is not integrally closed by finding an explicitly integral element in $S = \mathrm{Frac}(R) = \mathbb{K}(t)$ not contained in R.

(i) Show that $\mathbb{Z}[\sqrt{-3}]$ is not integrally closed. What does this mean about it being a UFD?

An R-algebra S is called an *integral extension* of R if every element of S is integral over R. Said differently, the structure map $R \to S$ is integral if all elements of the target satisfy monic polynomial equations over the source. Integral extensions are the precise setting where algebra generation and module generation interact especially well. The exercises below prove the fundamental comparison theorem:

$$\boxed{\text{module-finite}} \iff \boxed{\text{algebra-finite and integral}}.$$

More explicitly, the exercises that follow will guide us through the proof of the following theorem.

Theorem 3. *Let $R \to S$ be an R-algebra. The R-algebra S is module-finite over R if and only if it S is algebra-finite as an R-algebra.*

Note you will often, especially in algebraic geometry, see this expressed as finite is equivalent to finite type and integral. However, again we try to avoid this language for now.

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(15) **Module-Finite Extensions Are Integral.** Let S be a module-finite R-algebra.

(a) Fix $s \in S$. Show that multiplication by s defines an R-linear map

$$\mu_s : S \longrightarrow S \qquad \text{given by} \qquad x \longmapsto sx.$$

(b) Apply the Cayley-Hamilton theorem (explain why you can apply it) to μ_s to deduce that s is integral over R.

(c) Conclude that every module-finite R-algebra is integral over R.

(16) **Finite Type Integral Extensions Are Module-Finite.** Suppose $S = R[s_1, \ldots, s_n]$ and each s_i is integral over R.

(a) Prove that $R[s_1]$ is module-finite over R.

(b) Prove that $R[s_1, s_2]$ is module-finite over $R[s_1]$, and hence module-finite over R.

(c) Continue by induction to prove that S is module-finite over R.

(d) Deduce that if S is algebra-finite and integral over R, then S is module-finite over R.

(17) **The Comparison Theorem.** Let S be an R-algebra. Prove that the following are equivalent.

(a) S is module-finite over R.

(b) S is algebra-finite over R and integral over R.

Then revisit Problem 9 and explain how the theorem classifies each example.

(18) **Permanence of Integral Extensions.** Let $R \to S \to T$ be maps of rings.

(a) Prove that every quotient R/I is integral over R.

(b) Suppose S is integral over R and T is integral over S. Prove that T is integral over R. Hint: If $t \in T$ satisfies a monic equation over S, first adjoin to R the finitely many coefficients appearing in that equation.

(c) If S is integral over R and $J \subseteq S$ is an ideal, prove that S/J is integral over $R/(J \cap R)$.

(d) If S is integral over R and $U \subset R$ is multiplicatively closed, prove that $U^{-1}S$ is integral over $U^{-1}R$.

(e) More generally, if S is integral over R and $R \to R'$ is any ring map, prove that $R' \otimes_R S$ is integral over R'.

(19) **A First Consequence for Maximal Ideals.** Let $A \subseteq B$ be an integral extension of domains.

(a) Suppose B is a field. Prove that A is a field. Hint: If $0 \neq a \in A$, then $a^{-1} \in B$ is integral over A; write a monic equation for a^{-1} and multiply by a suitable power of a.

(b) Let $R \to S$ be an integral ring map and let $\mathfrak{q} \in \operatorname{Spec}(S)$ with contraction $\mathfrak{p} = \mathfrak{q} \cap R$. Show that $S/\mathfrak{q}$ is integral over $R/\mathfrak{p}$.

(c) Deduce that if $\mathfrak{q}$ is maximal, then $\mathfrak{p}$ is maximal.

(d) Prove the converse: if $\mathfrak{p}$ is maximal, then $\mathfrak{q}$ is maximal. Hint: $S/\mathfrak{q}$ is an integral domain integral over the field $R/\mathfrak{p}$.

(e) Conclude that under an integral ring map, a prime ideal upstairs is maximal if and only if its contraction downstairs is maximal.

(20) **More Examples of Integral Extensions.** Decide whether each map is integral. If it is integral and algebra-finite, give module generators.

(a) $\mathbb{Z} \to \mathbb{Z}[i]$.

(b) $\mathbb{K}[t^2] \to \mathbb{K}[t]$.

(c) $\mathbb{K}[t^2, t^3] \to \mathbb{K}[t]$.

(d) $\mathbb{K}[x] \to \mathbb{K}[x, x^{-1}]$.

(e) $\mathbb{K}[x] \to \mathbb{K}[x, y]/\langle y^2 - x^3 \rangle$.

(f) $\mathbb{K}[x^2, xy, y^2] \to \mathbb{K}[x, y]$. Hint: Show that x and y are integral over the source.

We now introduce a numerical invariant of a ring. A *chain of prime ideals of length* d in a ring R is a strictly increasing chain

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_d$$

of prime ideals. Thus the length counts the number of strict inclusions, not the number of prime ideals appearing in the chain. For a nonzero ring R, the *Krull dimension* of R is

$$\dim(R) := \sup\{d \in \mathbb{Z}_{\geq 0} \mid \text{there exists a chain } \mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_d \text{ in } \operatorname{Spec}(R)\}.$$

If there are chains of arbitrarily large finite length, we write $\dim(R) = \infty$. The zero ring has no prime ideals; different authors use different conventions for its dimension, so we will avoid using the zero ring in basic dimension computations unless a convention is specified. We will eventually prove the following theorem, which will turn out to be key to computing the dimension of most rings.

Theorem 4. *If* $\mathbb{K}$ *is a field then* $\dim \mathbb{K}[x_1, x_2, \ldots, x_n] = n$.

Proving that the dimension of the polynomial ring $\mathbb{K}[x_1, x_2, \ldots, x_n]$ is at least n is relatively easy, all one must do is find an explicit chain of prime ideals of length n. You will do this in the exercises below. Proving the equality in the theorem requires us to prove the other inequality, i.e. we must show there is no chain

of prime ideals of length more than n. This turns out to be surprisingly difficult. For the remainder of the worksheet you may assume this theorem unless it trivializes an exercise.

You will also prove that if R is any ring then $\dim R[x] \geq \dim R + 1$, again by exhibiting an explicit chain of prime ideals. Intuition tells us that this should be an equality, however, this turns out to be false unless we put additional hypotheses on R, namely that R be Noetherian. This highlights a somewhat tedious fact about dimension theory in commutative algebra: unless one assumes some form of finiteness condition (Noetherian, finitely generated $\mathbb{K}$-algebra, integral, etc.) there are many pathologies.

Using the correspondence between prime ideals and irreducible closed subsets of $\mathrm{Spec}(R)$, thee intuition is that longer chains of prime ideals correspond to longer chains of irreducible closed subsets in $\mathrm{Spec}(R)$, ordered in the opposite direction.

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(21) **Unpacking the Definition.** Let R be a nonzero ring.

(a) Show that $\dim(R) = 0$ if and only if every prime ideal of R is maximal.

(b) Show that every field has Krull dimension 0.

(c) Show that if R is an integral domain that is not a field, then $\dim(R) \geq 1$.

(d) Let $I \subseteq R$ be an ideal. Use the correspondence between prime ideals of R/I and prime ideals of R containing I to describe $\dim(R/I)$ in terms of chains of prime ideals in R.

(e) Prove that $\dim(R) = \dim(R_{\mathrm{red}})$, where $R_{\mathrm{red}} = R/\mathrm{Nil}(R)$.

(22) **Dimension Zero Examples.** Compute the Krull dimension of each ring.

(a) $\mathbb{K}$.

(b) $\mathbb{K} \times \mathbb{K}$.

(c) $\mathbb{K}[\epsilon]/\langle \epsilon^2 \rangle$.

(d) $\mathbb{Z}/12\mathbb{Z}$.

(e) $\mathbb{K}[x]/\langle x^n \rangle$ for $n \geq 1$.

(f) $\mathbb{K}[x,y]/\langle x, y \rangle$.

(23) **Dimension One Examples.**

(a) Prove that $\dim(\mathbb{Z}) = 1$.

(b) More generally, prove that every principal ideal domain that is not a field has Krull dimension 1.

(c) Prove that $\dim(\mathbb{K}[x]) = 1$.

(d) Compute $\dim(\mathbb{K}[x]/\langle f \rangle)$ for a nonzero nonunit polynomial $f \in \mathbb{K}[x]$.

(e) Compute $\dim(\mathbb{Z}[1/n])$ for an integer $n \geq 2$.

(24) **Building Longer Chains.** Let $\mathbb{K}$ be a field.

(a) Prove that

$$\langle 0\rangle \subsetneq \langle x_1\rangle \subsetneq \langle x_1, x_2\rangle \subsetneq \cdots \subsetneq \langle x_1, \ldots, x_n\rangle$$

is a chain of prime ideals in $\mathbb{K}[x_1, \ldots, x_n]$.

(b) Deduce that $\dim(\mathbb{K}[x_1, \ldots, x_n]) \geq n$.

(c) Let R be any nonzero ring. Show that $\dim(R[x]) \geq \dim(R) + 1$ whenever $\dim(R)$ is finite. Hint: Extend a chain in R to $R[x]$, then add the ideal generated by the final prime together with x.

(25) **Dimension and Quotients.** Let $\mathbb{K}$ be a field.

(a) Compute $\dim(\mathbb{K}[x, y]/\langle xy\rangle)$. Hint: Every prime ideal containing xy contains x or y.

(b) Compute $\dim(\mathbb{K}[x, y]/\langle x^2, xy\rangle)$. Hint: First pass to the reduced ring.

(c) Compute $\dim(\mathbb{K}[x, y]/\langle y - x^2\rangle)$.

(d) Find a chain of prime ideals of length 2 in $\mathbb{K}[x, y, z]/\langle xy\rangle$.

(e) Assuming $\dim(\mathbb{K}[x_1, \ldots, x_n]) = n$, compute the dimension of $\mathbb{K}[x, y, z]/\langle z - y^2 + x^3\rangle$.

(26) **Local Dimension.** Let $\mathfrak{p} \in \operatorname{Spec}(R)$.

(a) Recall that prime ideals of $R_{\mathfrak{p}}$ correspond to prime ideals $\mathfrak{q} \subseteq R$ with $\mathfrak{q} \subseteq \mathfrak{p}$. Use this to describe $\dim(R_{\mathfrak{p}})$ in terms of chains of prime ideals in R ending inside $\mathfrak{p}$.

(b) Compute $\dim(\mathbb{K}[x]_{\langle x\rangle})$.

(c) Compute $\dim(\mathbb{Z}_{\langle p\rangle})$ for a prime number p.

(d) Find a chain showing that $\dim(\mathbb{K}[x, y]_{\langle x,y\rangle}) \geq 2$.

(e) Assuming $\dim(\mathbb{K}[x, y]) = 2$, prove that $\dim(\mathbb{K}[x, y]_{\langle x,y\rangle}) = 2$.

WORKSHEET 7.2: INTEGRAL EXTENSION PT. II

Throughout this course, "ring" means *commutative* ring with unity and all ring homomorphisms preserve unity. In the previous worksheet we introduced integral elements and integral extensions. We saw that integrality is the mechanism by which monic polynomial equations turn algebraic generation into module generation. The goal of this worksheet is to see what integrality does to prime ideals and to use this to make dimension computable for finitely generated algebras over a field.

Let $\phi : R \to S$ be a ring map. Recall that ϕ induces a continuous map

$$\mathrm{Spec}(S) \xrightarrow{\phi^\#} \mathrm{Spec}(R), \qquad \mathfrak{q} \longmapsto \phi^{-1}(\mathfrak{q}).$$

The central question of the first half of this worksheet is: *how much of* $\mathrm{Spec}(R)$ *is seen by* $\mathrm{Spec}(S)$*, and how do chains of prime ideals compare?* For integral extensions the answer is surprisingly rigid. Prime ideals downstairs have primes upstairs, chains can be lifted upward, and no two comparable upstairs primes can lie over the same downstairs prime. These facts are called *Lying Over*, *Going Up*, and *Incomparability*. A complementary statement, *Going Down*, is more delicate: it is true for flat maps, and for integral extensions of domains when the base is integrally closed.

In the second half of the worksheet we use these theorems to prove and apply *Noether Normalization*. The slogan is that every finitely generated algebra over a field is finite over a polynomial ring. Thus, after a finite map, an affine algebra behaves like affine space of some dimension. This is one of the main reasons polynomial rings control dimension theory.

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(1) **A First Consequence for Maximal Ideals.** Let $\phi : R \to S$ be an integral ring map.

(a) Suppose S is a field and R is a domain. Prove that R is a field. Hint: If $r \neq 0 \in R$, then $\phi(r)^{-1} \in S$ is integral over R; write a monic equation for $\phi(r)^{-1}$ and multiply by a suitable power of $\phi(r)$.

(b) Suppose S is a domain and R is a field. Prove that S is a field.

(c) Let $\phi : R \to S$ be an integral ring map and let $\mathfrak{q} \in \mathrm{Spec}(S)$ with contraction $\mathfrak{p} = \phi^{-1}(\mathfrak{q})$. Show that $S/\mathfrak{q}$ is integral over $R/\mathfrak{p}$.

(d) Deduce that if $\mathfrak{q}$ is maximal, then $\mathfrak{p}$ is maximal.

(e) Prove the converse: If $\mathfrak{p}$ is maximal, then $\mathfrak{q}$ is maximal. Hint: $S/\mathfrak{q}$ is an integral domain integral over the field $R/\mathfrak{p}$.

(f) Conclude that under an integral ring map, a prime ideal upstairs is maximal if and only if its contraction downstairs is maximal.

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(2) **Prime Ideals Near a Point.** Let $\phi : R \to S$ be a ring map and let $\mathfrak{p} \in \operatorname{Spec}(R)$. Write $U = R \setminus \mathfrak{p}$ and $S_{\mathfrak{p}} = U^{-1}S$.

(a) Prove that if $\mathfrak{q} \in \operatorname{Spec}(S)$, then $\phi^{-1}(\mathfrak{q})$ is a prime ideal of R.

(b) Prove that prime ideals of $S_{\mathfrak{p}}$ are in bijection with prime ideals $\mathfrak{q} \in \operatorname{Spec}(S)$ such that $\mathfrak{q} \cap \phi(U) = \varnothing$. Equivalently, these are the primes $\mathfrak{q}$ such that $\phi^{-1}(\mathfrak{q}) \subseteq \mathfrak{p}$.

(c) Suppose that $R \subseteq S$ is an injective ring map. Show that $S_{\mathfrak{p}} \neq 0$. Hint: If $1 = 0$ in $S_{\mathfrak{p}}$, then some element of $R \setminus \mathfrak{p}$ becomes zero in S.

(d) Suppose that S is integral over R. Prove that $S_{\mathfrak{p}}$ is integral over $R_{\mathfrak{p}}$.

(e) Let $\mathfrak{Q} \in \operatorname{Spec}(S_{\mathfrak{p}})$ and let $\mathfrak{q} \subset S$ be the corresponding prime ideal. Show that if $\phi_{\mathfrak{p}}^{-1}(\mathfrak{Q}) = \mathfrak{p}R_{\mathfrak{p}}$, then $\phi^{-1}(\mathfrak{q}) = \mathfrak{p}$.

(3) **Lying Over.** Let $R \subseteq S$ be an integral extension of rings. The goal of this exercise is to prove the *Lying Over Theorem*: for every $\mathfrak{p} \in \operatorname{Spec}(R)$, there exists $\mathfrak{q} \in \operatorname{Spec}(S)$ such that $\mathfrak{q} \cap R = \mathfrak{p}$.

(a) Fix $\mathfrak{p} \in \operatorname{Spec}(R)$. Explain why $S_{\mathfrak{p}}$ is a nonzero integral extension of $R_{\mathfrak{p}}$.

(b) Choose a maximal ideal $\mathfrak{N} \subset S_{\mathfrak{p}}$. Use the theorem from the previous worksheet saying that maximal ideals contract to maximal ideals under integral extensions to prove

$$\mathfrak{N} \cap R_{\mathfrak{p}} = \mathfrak{p}R_{\mathfrak{p}}.$$

(c) Pull $\mathfrak{N}$ back to a prime ideal $\mathfrak{q} \subset S$ and prove that $\mathfrak{q} \cap R = \mathfrak{p}$.

(d) State the correct version of Lying Over for a not-necessarily-injective integral map $\phi : R \to S$. Hint: Which prime ideals of R can possibly be contractions of prime ideals of S?

(e) Translate Lying Over into a statement about the image of the map $\phi^{\#} : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$.

(4) **First Examples of Lying Over.** Let $\mathbb{K}$ be a field.

(a) Show that $\mathbb{K}[t]$ is integral over $\mathbb{K}[t^2]$. For a prime ideal $\mathfrak{q} \subset \mathbb{K}[t]$, compute $\mathfrak{q} \cap \mathbb{K}[t^2]$ when $\mathfrak{q} = \langle t - a \rangle$.

(b) Assume $\mathbb{K}$ is algebraically closed and $\operatorname{char}(\mathbb{K}) \neq 2$. For $u = t^2$, describe the primes of $\mathbb{K}[t]$ lying over the maximal ideal $\langle u - a \rangle \subset \mathbb{K}[u]$ when $a = 0$ and when $a \neq 0$.

(c) Show that $\mathbb{K}[t]$ is integral over $\mathbb{K}[t^2, t^3]$. Identify the prime of $\mathbb{K}[t]$ lying over the cusp point $\langle t^2, t^3 \rangle \subset \mathbb{K}[t^2, t^3]$.

(d) Show that $\mathbb{Z}[i]$ is integral over $\mathbb{Z}$. Compute the primes of $\mathbb{Z}[i]$ lying over $\langle 2 \rangle$, $\langle 3 \rangle$, and $\langle 5 \rangle$ by factoring $x^2 + 1$ modulo 2, 3, and 5.

(e) Let $R \subseteq S$ be integral and let $I \subset R$ be an ideal. Prove that the induced map

$$\{\mathfrak{q} \in \operatorname{Spec}(S) \mid \mathfrak{q} \supseteq IS\} \longrightarrow \{\mathfrak{p} \in \operatorname{Spec}(R) \mid \mathfrak{p} \supseteq I\}$$

is surjective.

The next theorem explains how integral extensions interact with inclusions of prime ideals. Geometrically, if $\mathfrak{p}_0 \subseteq \mathfrak{p}_1$ in $\mathrm{Spec}(R)$, then $\mathfrak{p}_1$ is a specialization of $\mathfrak{p}_0$. The theorem says that specializations can be lifted through an integral extension once the starting point upstairs has been chosen.

Theorem 1 (Going Up)**.** *Let $R \subseteq S$ be an integral extension. Suppose*

$$\mathfrak{p}_0 \subseteq \mathfrak{p}_1$$

are prime ideals of R, and suppose $\mathfrak{q}_0 \in \mathrm{Spec}(S)$ satisfies $\mathfrak{q}_0 \cap R = \mathfrak{p}_0$. Then there exists $\mathfrak{q}_1 \in \mathrm{Spec}(S)$ such that

$$\mathfrak{q}_0 \subseteq \mathfrak{q}_1 \qquad \textit{and} \qquad \mathfrak{q}_1 \cap R = \mathfrak{p}_1.$$

More generally, every chain of prime ideals downstairs can be lifted to a chain of prime ideals upstairs, after choosing a prime upstairs over the first prime in the chain.

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(5) **Proof of Going Up.** Let $R \subseteq S$ be integral, let $\mathfrak{p}_0 \subseteq \mathfrak{p}_1$ be prime ideals of R, and let $\mathfrak{q}_0 \in \mathrm{Spec}(S)$ lie over $\mathfrak{p}_0$.

(a) Prove that $S/\mathfrak{q}_0$ is integral over $R/\mathfrak{p}_0$.

(b) Show that $\mathfrak{p}_1/\mathfrak{p}_0$ is a prime ideal of $R/\mathfrak{p}_0$.

(c) Apply Lying Over to the integral extension $R/\mathfrak{p}_0 \subseteq S/\mathfrak{q}_0$ to produce a prime ideal $\mathfrak{q}_1/\mathfrak{q}_0$ of $S/\mathfrak{q}_0$ lying over $\mathfrak{p}_1/\mathfrak{p}_0$.

(d) Check that the corresponding prime ideal $\mathfrak{q}_1 \subset S$ satisfies $\mathfrak{q}_0 \subseteq \mathfrak{q}_1$ and $\mathfrak{q}_1 \cap R = \mathfrak{p}_1$.

(e) Prove the chain version of Going Up by induction: if

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

is a chain in R and $\mathfrak{q}_0$ lies over $\mathfrak{p}_0$, then there is a chain

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_n$$

in S with $\mathfrak{q}_i \cap R = \mathfrak{p}_i$ for all i.

(6) **Incomparability.** Let $R \subseteq S$ be an integral extension. The goal of this exercise is to prove: if $\mathfrak{q} \subseteq \mathfrak{q}'$ are prime ideals of S and $\mathfrak{q} \cap R = \mathfrak{q}' \cap R$, then $\mathfrak{q} = \mathfrak{q}'$.

(a) Set $\mathfrak{p} = \mathfrak{q} \cap R = \mathfrak{q}' \cap R$. Explain why $R/\mathfrak{p} \subseteq S/\mathfrak{q}$ is an integral extension of domains.

(b) Let $\mathfrak{P} = \mathfrak{q}'/\mathfrak{q} \subset S/\mathfrak{q}$. Show that $\mathfrak{P} \cap (R/\mathfrak{p}) = \langle 0 \rangle$.

(c) Suppose $0 \neq b \in \mathfrak{P}$. Choose a monic equation of minimal degree

$$b^n + a_1 b^{n-1} + \cdots + a_{n-1} b + a_n = 0$$

with $a_i \in R/\mathfrak{p}$. Prove that $a_n = 0$.

(d) Use the fact that $S/\mathfrak{q}$ is a domain to get a monic equation for b of smaller degree, a contradiction.

(e) Conclude Incomparability.

(7) **Dimension is Preserved by Integral Extensions.** Let $R \subseteq S$ be an integral extension of nonzero rings.

(a) Let

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_n$$

be a chain of prime ideals in S. Use Incomparability to show that

$$\mathfrak{q}_0 \cap R \subsetneq \mathfrak{q}_1 \cap R \subsetneq \cdots \subsetneq \mathfrak{q}_n \cap R$$

is a chain of prime ideals in R. Deduce that $\dim(S) \leq \dim(R)$.

(b) Let

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

be a chain of prime ideals in R. Use Lying Over and Going Up to lift it to a chain of prime ideals in S. Deduce that $\dim(R) \leq \dim(S)$.

(c) Conclude that $\dim(R) = \dim(S)$.

(d) Use this theorem to compute the Krull dimension of each ring below.

(i) $\mathbb{K}[t^2, t^3]$.

(ii) $\mathbb{K}[x, y]/\langle y^2 - x^3 \rangle$.

(iii) $\mathbb{K}[x^2, xy, y^2]$.

(iv) $\mathbb{Z}[i]$.

(8) **Closed Sets Under Integral Maps.** Let $R \subseteq S$ be integral and let $\phi^{\#} : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ be the induced map.

(a) Let $J \subseteq S$ be an ideal. Show that

$$\phi^{\#}(\mathbb{V}(J)) = \mathbb{V}(J \cap R).$$

Hint: One inclusion is direct. For the other, apply Lying Over to $R/(J \cap R) \subseteq S/J$ after checking the kernel carefully.

(b) Deduce that $\phi^{\#}$ is a closed map.

(c) If $R \subseteq S$ is finite and $\mathbb{K}$ is algebraically closed, explain why the induced map on affine algebraic sets should be thought of as a finite-to-one map. Make this precise by showing that for each $\mathfrak{p} \in \operatorname{Spec}(R)$, the fibre ring $\kappa(\mathfrak{p}) \otimes_R S$ is finite-dimensional over $\kappa(\mathfrak{p})$.

We now turn to *Going Down*. For a ring map $R \to S$, Going Down means that chains can be lifted in the opposite direction: if $\mathfrak{p}_0 \subseteq \mathfrak{p}_1$ in R and a prime $\mathfrak{q}_1$ upstairs has already been chosen over $\mathfrak{p}_1$, then there should exist a prime $\mathfrak{q}_0 \subseteq \mathfrak{q}_1$ lying over $\mathfrak{p}_0$.

Going Down is not automatic for integral extensions. There are two especially important settings where it holds. The first is flatness, which is a module-theoretic condition. The second is the classical integral theorem: an integral extension of domains satisfies Going Down when the base is integrally closed. We will not prove these now, but will explore their consequences.

Theorem 2 (Flat Going Down)**.** *Let $\phi : R \to S$ be a flat ring map. Suppose $\mathfrak{p}_0 \subseteq \mathfrak{p}_1$ are prime ideals of R and $\mathfrak{q}_1 \in \operatorname{Spec}(S)$ satisfies $\phi^{-1}(\mathfrak{q}_1) = \mathfrak{p}_1$. Then there exists $\mathfrak{q}_0 \in \operatorname{Spec}(S)$ such that*

$$\mathfrak{q}_0 \subseteq \mathfrak{q}_1 \qquad \textit{and} \qquad \phi^{-1}(\mathfrak{q}_0) = \mathfrak{p}_0.$$

Theorem 3 (Going Down for Normal Domains)**.** *Let $A \subseteq B$ be an integral extension of domains. If A is integrally closed in its fraction field, then $A \subseteq B$ satisfies Going Down.*

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(9) **Unpacking Going Down.** Let $\phi : R \to S$ be a ring map.

(a) Write the definition of Going Up for ϕ using a diagram of prime ideals.

(b) Write the definition of Going Down for ϕ using a diagram of prime ideals.

(c) Explain why Going Up is about lifting specializations and Going Down is about lifting generalizations.

(d) Prove directly that a localization map $R \to U^{-1}R$ satisfies Going Down. Hint: Use the correspondence between primes of $U^{-1}R$ and primes of R disjoint from U.

(e) Give an example of a flat map that is not integral and an integral map that is not flat.

(10) **Consequences of Going Down.**

(a) Let $R \to S$ be faithfully flat. Prove that $\operatorname{Spec}(S) \to \operatorname{Spec}(R)$ is surjective. Hint: For $\mathfrak{p} \in \operatorname{Spec}(R)$, show that $\kappa(\mathfrak{p}) \otimes_R S \neq 0$.

(b) Deduce that if $R \to S$ is faithfully flat, then every finite chain of primes in R can be lifted to a chain of primes in S. Conclude that $\dim(S) \geq \dim(R)$.

(c) Apply this to $R \to R[x]$ to recover the inequality $\dim(R[x]) \geq \dim(R) + 1$ for nonzero R. Hint: if $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n$ is a chain in R, use the explicit chain $\mathfrak{p}_0 R[x] \subsetneq \cdots \subsetneq \mathfrak{p}_n R[x] \subsetneq \mathfrak{p}_n R[x] + \langle x \rangle$.

(d) Let $A \subseteq B$ be an integral extension of domains with A integrally closed. Let $\mathfrak{q} \in \operatorname{Spec}(B)$ and $\mathfrak{p} = \mathfrak{q} \cap A$. Use Going Up, Going Down for Normal Domains, and Incomparability to prove

$$\operatorname{ht}(\mathfrak{q}) = \operatorname{ht}(\mathfrak{p}).$$

(e) Specialize the previous part to a finite extension $\mathbb{K}[y_1, \ldots, y_d] \subseteq B$ where B is a domain. What does it say about heights of primes in B?

We now move from arbitrary integral extensions to finitely generated algebras over a field. The next theorem is one of the main structural results in commutative algebra. It says that a finitely generated algebra over a field is finite over a polynomial subring.

Theorem 4 (Noether Normalization)**.** *Let $\mathbb{K}$ be a field and let A be an algebra-finite $\mathbb{K}$-algebra. Then there exist algebraically independent elements $y_1, \ldots, y_d \in A$ such that A is module-finite over the polynomial subring*

$$\mathbb{K}[y_1, \ldots, y_d] \subseteq A.$$

If A is a domain, then $d = \mathrm{tr.deg}_{\mathbb{K}} \mathrm{Frac}(A)$.

The proof is an induction on the number of algebra generators. The key step is a change of variables trick: given a nonzero polynomial relation, we choose new variables so that one old variable satisfies a monic equation over the others.

---

(11) **The Monic Change of Variables Trick.** Let $0 \neq f \in \mathbb{K}[x_1, \ldots, x_n]$. The goal of this exercise is to make a change of variables so that f becomes monic in one variable. Write

$$f = \sum_{\alpha} c_\alpha x_1^{\alpha_1} \cdots x_n^{\alpha_n},$$

where $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{Z}^n_{\geq 0}$, only finitely many c_α are nonzero, and $c_\alpha \in \mathbb{K}$. Choose an integer N larger than every exponent appearing in every monomial of f; that is, $N > \alpha_i$ for every exponent vector α appearing in f and every i.

(a) Define

$$w(\alpha) = \alpha_n + \alpha_1 N + \alpha_2 N^2 + \cdots + \alpha_{n-1} N^{n-1}.$$

Explain why this is the base-N expansion of an integer whose "digits" are $\alpha_n, \alpha_1, \alpha_2, \ldots, \alpha_{n-1}$

(b) Since all these digits are strictly smaller than N, prove that if $w(\alpha) = w(\beta)$ then then $\alpha = \beta$. Thus the weights $w(\alpha)$ are distinct for distinct monomials of f.

(c) Introduce new variables

$$y_i = x_i - x_n^{N^i} \qquad \text{for } 1 \leq i \leq n-1.$$

Equivalently, $x_i = y_i + x_n^{N^i}$. Substitute these expressions into a monomial

$$x_1^{\alpha_1} \cdots x_{n-1}^{\alpha_{n-1}} x_n^{\alpha_n}.$$

Show that the unique term obtained by choosing $x_n^{N^i}$ from every factor $(y_i + x_n^{N^i})^{\alpha_i}$ is $x_n^{w(\alpha)}$. All other terms contain at least one y_i and have strictly smaller degree in x_n.

(d) Since the weights $w(\alpha)$ are distinct, there is a unique exponent vector $\alpha^{\max}$ appearing in f for which $w(\alpha)$ is largest. Show that, after the substitution, the highest power of x_n appearing in f is

$$c_{\alpha^{\max}} x_n^{w(\alpha^{\max})}.$$

(e) Conclude that, after multiplying f by the nonzero scalar $c_{\alpha^{\max}}^{-1}$ we may regard f as a monic polynomial in x_n with coefficients in

$$\mathbb{K}[y_1, \ldots, y_{n-1}].$$

(f) Let $A = \mathbb{K}[x_1, \ldots, x_n]/\langle f \rangle$. Prove that the image of x_n in A satisfies a monic polynomial with coefficients in the subalgebra generated by the images of

$$y_1, \ldots, y_{n-1}.$$

Therefore the image of x_n is integral over this subalgebra.

(g) Show A is generated as an algebra over $\mathbb{K}[y_1, \ldots, y_{n-1}]$ by the single integral element x_n. Deduce that S is module-finite over $\mathbb{K}[y_1, \ldots, y_{n-1}]$.

(12) **Proof of Noether Normalization.** Let $A = \mathbb{K}[x_1, \ldots, x_n]/I$ be an algebra-finite $\mathbb{K}$-algebra.

(a) If $I = \langle 0 \rangle$, prove the theorem.

(b) If $I \neq \langle 0 \rangle$, choose $0 \neq f \in I$. Use the previous exercise to make a change of variables so that A is module-finite over the image of a polynomial ring in $n-1$ variables.

(c) Let A_1 denote this image subalgebra. Explain why A_1 is algebra-finite over $\mathbb{K}$ using at most $n-1$ algebra generators.

(d) Apply induction to A_1 to find a polynomial subring $\mathbb{K}[y_1, \ldots, y_d] \subseteq A_1$ such that A_1 is module-finite over $\mathbb{K}[y_1, \ldots, y_d]$.

(e) Prove that A is module-finite over $\mathbb{K}[y_1, \ldots, y_d]$.

(f) Explain why the elements $y_1, \ldots, y_d$ are algebraically independent over $\mathbb{K}$.

(13) **Finding Normalizations by Hand.** For each algebra below, find a polynomial subring over which the algebra is module-finite. When possible, give explicit module generators.

(a) $\mathbb{K}[t^2, t^3]$.

(b) $\mathbb{K}[x, y]/\langle y^2 - x^3 \rangle$.

(c) $\mathbb{K}[x, y]/\langle xy \rangle$. Hint: Try $x + y$.

(d) $\mathbb{K}[x, y, z]/\langle z^2 - xy \rangle$.

(e) $\mathbb{K}[x^2, xy, y^2] \subseteq \mathbb{K}[x, y]$.

(f) $\mathbb{K}[s^2, s^3, t]$.

(14) **Zariski's Lemma from Noether Normalization.** Let A be an algebra-finite $\mathbb{K}$-algebra that is also a field.

(a) By Noether Normalization, choose $\mathbb{K}[y_1, \ldots, y_d] \subseteq A$ such that A is module-finite over $\mathbb{K}[y_1, \ldots, y_d]$. Explain why this extension is integral.

(b) Use the earlier fact that if B is a field integral over a domain A_0, then A_0 is a field, to prove that $\mathbb{K}[y_1, \ldots, y_d]$ is a field.

(c) Prove that $d = 0$.

(d) Conclude that A is a finite-dimensional vector space over $\mathbb{K}$. Equivalently, every field that is algebra-finite over $\mathbb{K}$ is algebraic and finite over $\mathbb{K}$.

(e) Deduce the weak Nullstellensatz: if $\mathbb{K}$ is algebraically closed and $\mathfrak{m} \subset \mathbb{K}[x_1, \ldots, x_n]$ is maximal, then

$$\mathfrak{m} = \langle x_1 - a_1, \ldots, x_n - a_n \rangle$$

for some $a_1, \ldots, a_n \in \mathbb{K}$.

We finish by extracting consequences for dimension. Noether Normalization reduces the dimension theory of affine algebras to the dimension theory of polynomial rings, while Lying Over, Going Up, and Incomparability tell us that finite integral extensions preserve lengths of prime chains.

Theorem 5 (Dimension of Affine Domains)**.** *Let A be an algebra-finite $\mathbb{K}$-algebra that is a domain. Then*

$$\dim(A) = \operatorname{tr.deg}_{\mathbb{K}} \operatorname{Frac}(A).$$

Equivalently, if $\mathbb{K}[y_1, \ldots, y_d] \subseteq A$ is a Noether normalization, then $\dim(A) = d$.

---

(15) **Dimension of Affine Domains.** Let A be an algebra-finite $\mathbb{K}$-algebra that is a domain.

(a) Let $\mathbb{K}[y_1, \ldots, y_d] \subseteq A$ be a Noether normalization. Use dimension preservation for integral extensions to prove $\dim(A) = d$.

(b) Show that $\operatorname{Frac}(A)$ is algebraic over $\mathbb{K}(y_1, \ldots, y_d)$. Hint: First prove that every element of A is algebraic over $\mathbb{K}(y_1, \ldots, y_d)$.

(c) Deduce that $\operatorname{tr.deg}_{\mathbb{K}} \operatorname{Frac}(A) = d$.

(d) Conversely, explain why d is determined by A and not by the choice of Noether normalization.

(16) **Computations Using Noether Normalization.** Compute the Krull dimension of each ring. In each case, try to exhibit a finite extension from a polynomial subring or reduce to irreducible components.

(a) $\mathbb{K}[t^2, t^3]$.

(b) $\mathbb{K}[s^2, s^3, t]$.

(c) $\mathbb{K}[x, y, z]/\langle z^2 - xy \rangle$.

(d) $\mathbb{K}[x, y, z]/\langle xy, z \rangle$.

(e) $\mathbb{K}[x, y, z]/\langle xz, yz \rangle$.

(f) $\mathbb{K}[x_1, \ldots, x_n]/\langle x_1, \ldots, x_r \rangle$.

(g) $\mathbb{Z}[x]/\langle x^2 + 1 \rangle$.

(h) $\mathbb{K}[x^2, xy, y^2]$.

(17) **Finite Maps to Affine Space.** Let A be an algebra-finite $\mathbb{K}$-algebra and let $P = \mathbb{K}[y_1, \ldots, y_d] \subseteq A$ be a Noether normalization.

(a) Show that the induced map $\operatorname{Spec}(A) \longrightarrow \operatorname{Spec}(P) \cong \mathbb{A}^d_{\mathbb{K}}$ is surjective.

(b) For $\mathfrak{p} \in \mathrm{Spec}(P)$, prove that the fibre ring $\kappa(\mathfrak{p}) \otimes_P A$ is finite-dimensional over $\kappa(\mathfrak{p})$.

(c) Assume $\mathbb{K}$ is algebraically closed and $\mathfrak{p} = \langle y_1 - a_1, \ldots, y_d - a_d \rangle$ is a closed point of $\mathbb{A}^d_{\mathbb{K}}$. Show that the closed points in the fibre over $\mathfrak{p}$ correspond to maximal ideals of the finite-dimensional $\mathbb{K}$-algebra $A/\mathfrak{p}A$.

(d) Prove that a finite-dimensional algebra over a field has only finitely many maximal ideals. Hint: Distinct maximal ideals are comaximal, and the Chinese Remainder Theorem gives a surjection onto a product of fields.

(e) Explain the geometric slogan: Noether normalization gives a finite-to-one projection from an affine algebraic set to affine space of the same dimension.

WORKSHEET 8.1: DEPTH I

Throughout this course, "ring" means *commutative* ring with unity and all ring homomorphisms preserve unity. In the previous worksheets we introduced Krull dimension and height, and saw that finitely generated algebras over a field can often be understood by cutting them down to polynomial rings. The goal of this worksheet is to study sequences of elements that behave like independent equations. The algebraic word for this independence is *regularity*.

A regular sequence is a list of equations such that each new equation remains a nonzerodivisor after imposing the previous ones. This condition is strong enough to force dimension-theoretic consequences: by Krull's Principal Ideal Theorem, one equation cuts height by at most one, and a regular equation really does cut height by exactly one. The same idea leads to *depth*, which measures how long a regular sequence can be found inside the maximal ideal of a local ring.

The final part of the worksheet introduces the *Koszul complex*. This is a small, explicit chain complex attached to a sequence of ring elements. When the sequence is regular, the Koszul complex is exact except in degree zero. Thus the Koszul complex packages regular sequences, depth computations, and explicit syzygies into one object.

Let R be a ring and let M be an R-module. An element $x \in R$ is a *nonzerodivisor on M* if the multiplication map

$$M \longrightarrow M, \qquad m \longmapsto xm$$

is injective. We say that x is *M-regular* if x is a nonzerodivisor on M and $M/xM \neq 0$. A sequence $x_1, \dots, x_r \in R$ is an *M-regular sequence* if x_1 is M-regular and, for each $i \geq 2$, the image of x_i is regular on

$$M/\langle x_1, \dots, x_{i-1}\rangle M.$$

When $M = R$, we simply say that $x_1, \dots, x_r$ is an *R-regular sequence*, or just a *regular sequence* if the ring is clear.

---

(1) **First Encounters with Regular Elements.** Let M be an R-module and let $x \in R$.

(a) Show that x is a nonzerodivisor on M if and only if

$$(0 :_M x) := \{m \in M \mid xm = 0\} = 0.$$

(b) Show that if R is a domain, then every nonzero element of R is a nonzerodivisor on R. Which of these elements are R-regular?

(c) Let $(R, \mathfrak{m})$ be a local ring and let M be a finitely generated nonzero R-module. If $x \in \mathfrak{m}$ is a nonzerodivisor on M, prove that $M/xM \neq 0$. Hint: Use Nakayama's lemma.

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(d) Let $R = \mathbb{K}[x,y]$. Prove that x, y, and $x+y$ are nonzerodivisors on R.

(e) Let $R = \mathbb{K}[x,y]/\langle xy\rangle$. Show that the images of x and y are zerodivisors on R, but the image of $x+y$ is a nonzerodivisor. Hint: Write every element uniquely as

$$c + xf(x) + yg(y) \qquad \text{with } c \in \mathbb{K},\ f \in \mathbb{K}[x],\ g \in \mathbb{K}[y].$$

(f) Let $R = \mathbb{K}[x,y]/\langle x^2, xy\rangle$. Show that the image of x is killed by the maximal ideal $\langle x, y\rangle$.

(2) **Building Regular Sequences.**

(a) Prove that $x_1,\ldots,x_r$ is a regular sequence on $\mathbb{K}[x_1,\ldots,x_n]$ for every $r \le n$.

(b) More generally, let A be a nonzero ring. Prove that $t_1,\ldots,t_r$ is a regular sequence on $A[t_1,\ldots,t_n]$ for every $r \le n$.

(c) In $\mathbb{K}[x,y]$, show that x, y is a regular sequence but x, xy is not a regular sequence.

(d) In $\mathbb{K}[x,y]/\langle xy\rangle$, show that $x+y$ is a regular element but $x+y, x$ is not a regular sequence.

(e) Suppose $x_1,\ldots,x_r$ is an M-regular sequence. Prove that every initial segment $x_1,\ldots,x_i$ is an M-regular sequence.

(f) With notation as in the previous part, prove that $x_{i+1},\ldots,x_r$ is regular on $M/\langle x_1,\ldots,x_i\rangle M$.

(g) Let $U \subset R$ be multiplicatively closed. Prove that the image of an M-regular sequence $x_1,\ldots,x_r$ is a sequence of nonzerodivisors on $U^{-1}M$, after successively quotienting by the previous elements. What extra condition is needed to ensure that it is a regular sequence in the sense that the final quotients are nonzero?

(3) **Regular Sequences and Permutations.** In a general ring, the order of a regular sequence is part of the data. In a Noetherian local ring, however, regular sequences inside the maximal ideal can be permuted.

Theorem 1 (Permuting Regular Sequences)**.** *Let $(R,\mathfrak{m})$ be a Noetherian local ring and let M be a finitely generated R-module. If $x_1,\ldots,x_r \in \mathfrak{m}$ is an M-regular sequence, then every permutation of $x_1,\ldots,x_r$ is an M-regular sequence.*

(a) Use the theorem to prove that if x, y is a regular sequence in a Noetherian local ring, then y, x is a regular sequence.

(b) Let $R = \mathbb{K}[x,y]_{\langle x,y\rangle}$. Prove that x, y and y, x are both regular sequences.

(c) Let $R = \mathbb{K}[x,y]_{\langle x,y\rangle}/\langle xy\rangle$. Explain why x, y is not a regular sequence, even though both x and y lie in the maximal ideal.

(d) Let $R = \mathbb{K}[x,y,z]_{\langle x,y,z\rangle}$. Prove that $x, y+z, z$ is a regular sequence. Then use the theorem to produce five more regular sequences from the same three elements.

Regular sequences are meant to model independent equations. Krull's Principal Ideal Theorem is the first formal statement that one equation cannot cut out something of arbitrarily large codimension.

Theorem 2 (Krull's Principal Ideal Theorem)**.** *Let R be a Noetherian ring and let $x \in R$. If $\mathfrak{p} \in \operatorname{Spec}(R)$ is minimal over the principal ideal $\langle x \rangle$, then*

$$\operatorname{ht}(\mathfrak{p}) \leq 1.$$

More generally, if an ideal can be generated by r elements, then every prime minimal over that ideal has height at most r. This is often called *Krull's Height Theorem*. Thus the number of equations gives an upper bound on the height of every irreducible component they cut out. Regular sequences will give the matching lower bound.

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(4) **Using the Principal Ideal Theorem.** Let R be a Noetherian ring.

(a) Let $x \in R$ and let $\mathfrak{p}$ be minimal over $\langle x \rangle$. Explain why localizing at $\mathfrak{p}$ reduces the Principal Ideal Theorem to the case of a local ring $(R, \mathfrak{m})$ in which $\mathfrak{m}$ is minimal over $\langle x \rangle$.

(b) Let $R = \mathbb{K}[x_1, \ldots, x_n]$ and let $0 \neq f \in R$. If $\mathfrak{p}$ is minimal over $\langle f \rangle$, use the Principal Ideal Theorem to show that $\operatorname{ht}(\mathfrak{p}) = 1$.

(c) Suppose x is a nonzerodivisor on R. Prove that no minimal prime of R contains x. Hint: If $\mathfrak{q}$ is minimal and $x \in \mathfrak{q}$, then $x/1$ is nilpotent in $R_{\mathfrak{q}}$.

(d) Deduce that if x is a nonzerodivisor and $\mathfrak{p}$ is minimal over $\langle x \rangle$, then $\operatorname{ht}(\mathfrak{p}) = 1$.

(e) Give an example showing that the nonzerodivisor hypothesis in the previous part is necessary. Hint: Use $\mathbb{K}[x, y]/\langle xy \rangle$ and the element x.

(5) **From Principal Ideals to Finitely Generated Ideals.** The goal of this exercise is to prove Krull's Height Theorem from the Principal Ideal Theorem. Let $I = \langle x_1, \ldots, x_r \rangle$ and let $\mathfrak{p}$ be minimal over I.

(a) Prove the theorem when $r = 1$.

(b) Assume $r > 1$. Choose a prime ideal $\mathfrak{q} \subseteq \mathfrak{p}$ minimal over $\langle x_1, \ldots, x_{r-1} \rangle$. Explain why such a $\mathfrak{q}$ exists.

(c) By induction, show that $\operatorname{ht}(\mathfrak{q}) \leq r - 1$.

(d) Show that $\mathfrak{p}/\mathfrak{q}$ is minimal over the principal ideal generated by the image of x_r in $R/\mathfrak{q}$.

(e) Apply the Principal Ideal Theorem in $R/\mathfrak{q}$ to show that the largest length of a chain of primes from $\mathfrak{q}$ to $\mathfrak{p}$ is at most 1.

(f) Conclude that $\operatorname{ht}(\mathfrak{p}) \leq r$.

(g) Use the theorem to compute the heights of the minimal primes over the following ideals:

(i) $\langle x, y\rangle \subset \mathbb{K}[x,y,z]$.

(ii) $\langle xy, z\rangle \subset \mathbb{K}[x,y,z]$.

(iii) $\langle xz, yz\rangle \subset \mathbb{K}[x,y,z]$.

(iv) $\langle x^2, xy\rangle \subset \mathbb{K}[x,y]$.

(6) **Regular Sequences Force Height.** Let $(R,\mathfrak{m})$ be a Noetherian local ring.

(a) Suppose $x \in \mathfrak{m}$ is a nonzerodivisor. Let

$$\mathfrak{p}_0/\langle x\rangle \subsetneq \mathfrak{p}_1/\langle x\rangle \subsetneq \cdots \subsetneq \mathfrak{p}_s/\langle x\rangle$$

be a chain of prime ideals in $R/\langle x\rangle$. Choose a minimal prime $\mathfrak{q}$ of R contained in $\mathfrak{p}_0$. Use the previous exercise to show $x \notin \mathfrak{q}$, and deduce that

$$\mathfrak{q} \subsetneq \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_s$$

is a chain in R.

(b) Deduce that if $x \in \mathfrak{m}$ is a nonzerodivisor, then

$$\dim(R/\langle x\rangle) \leq \dim(R) - 1.$$

(c) Use induction to prove that every regular sequence in $\mathfrak{m}$ has length at most $\dim(R)$.

(d) Now let R be any Noetherian ring and let $x_1,\ldots,x_r$ be an R-regular sequence. If $\mathfrak{p}$ is minimal over $I = \langle x_1,\ldots,x_r\rangle$, localize at $\mathfrak{p}$ and prove that $\mathrm{ht}(\mathfrak{p}) \geq r$.

(e) Combine the previous part with Krull's Height Theorem to prove that every prime minimal over I has height exactly r.

(7) **Complete Intersection Examples.** Let $R = \mathbb{K}[x,y,z]$.

(a) Show that x, z is a regular sequence. Compute the height of $\langle x, z\rangle$.

(b) Show that x, yz is a regular sequence. Find the minimal primes over $\langle x, yz\rangle$ and compute their heights.

(c) Show that xy, xz is not a regular sequence. Find a minimal prime over $\langle xy, xz\rangle$ of height 1.

(d) Explain how the previous parts illustrate the difference between being generated by two elements and being generated by a regular sequence of length two.

Let $(R,\mathfrak{m})$ be a Noetherian local ring and let M be a finitely generated nonzero R-module. The *depth* of M is

$$\mathrm{depth}_R(M) := \sup\{r \in \mathbb{Z}_{\geq 0} \mid \text{there exists an } M\text{-regular sequence } x_1,\ldots,x_r \in \mathfrak{m}\}.$$

Equivalently, depth measures how many independent equations from the maximal ideal can act nontrivially on M in succession. We write $\mathrm{depth}(R)$ for $\mathrm{depth}_R(R)$. One should think of depth as a measure of how far M is from having embedded infinitesimal pieces supported only at the closed point.

The dimension bound from the previous section implies

$$\operatorname{depth}_R(M) \leq \dim_R(M),$$

where $\dim_R(M)$ denotes the dimension of the support of M. In particular, for $M = R$ one has $\operatorname{depth}(R) \leq \dim(R)$. A Noetherian local ring is called *Cohen–Macaulay* if

$$\operatorname{depth}(R) = \dim(R).$$

Thus Cohen–Macaulay rings are rings in which the largest possible regular sequences actually exist.

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(8) **First Depth Computations.** Let $(R, \mathfrak{m}, \Bbbk)$ be a Noetherian local ring.

(a) Prove that $\operatorname{depth}_R(\Bbbk) = 0$.

(b) If R is a domain and not a field, prove that $\operatorname{depth}(R) \geq 1$.

(c) Let $S = \Bbbk[x_1, \ldots, x_n]_{\langle x_1, \ldots, x_n \rangle}$. Prove that $\operatorname{depth}(S) = n$.

(d) For $0 \leq r \leq n$, compute

$$\operatorname{depth}_S(S/\langle x_1, \ldots, x_r \rangle).$$

(e) Let $A = \Bbbk[x, y]_{\langle x, y \rangle}/\langle xy \rangle$. Show that $\operatorname{depth}(A) = 1$. Hint: Use the image of $x + y$.

(f) Let $B = \Bbbk[x, y]_{\langle x, y \rangle}/\langle x^2, xy \rangle$. Show that $\operatorname{depth}(B) = 0$. Hint: Find a nonzero element killed by the maximal ideal.

(9) **Depth and Complete Intersections.** The following theorem is one of the basic reasons regular sequences are useful.

Theorem 3 (Depth Drops Along Regular Elements)**.** *Let $(R, \mathfrak{m})$ be a Noetherian local ring and let M be a finitely generated R-module. If $x \in \mathfrak{m}$ is M-regular, then*

$$\operatorname{depth}_R(M/xM) = \operatorname{depth}_R(M) - 1.$$

More generally, if $x_1, \ldots, x_r \in \mathfrak{m}$ is an M-regular sequence, then

$$\operatorname{depth}_R\big(M/\langle x_1, \ldots, x_r \rangle M\big) = \operatorname{depth}_R(M) - r.$$

(a) Use the theorem to compute the depth of $S/\langle f \rangle$, where $S = \Bbbk[x_1, \ldots, x_n]_{\langle x_1, \ldots, x_n \rangle}$ and $0 \neq f \in \langle x_1, \ldots, x_n \rangle$.

(b) Suppose $x_1, \ldots, x_r$ is a regular sequence in $S = \Bbbk[x_1, \ldots, x_n]_{\langle x_1, \ldots, x_n \rangle}$. Prove that $S/\langle x_1, \ldots, x_r \rangle$ is Cohen–Macaulay.

(c) Use this to prove that $\Bbbk[x, y]_{\langle x, y \rangle}/\langle xy \rangle$ is Cohen–Macaulay.

(d) Explain why $\Bbbk[x, y]_{\langle x, y \rangle}/\langle x^2, xy \rangle$ is not Cohen–Macaulay.

(e) Let $S = \Bbbk[x, y, z]_{\langle x, y, z \rangle}$. Show that $S/\langle x, yz \rangle$ is Cohen–Macaulay and compute its dimension.

(10) **Depth Zero.** Let $(R, \mathfrak{m})$ be Noetherian local and let M be a finitely generated nonzero R-module.

(a) Show that if there exists $0 \neq m \in M$ such that $\mathfrak{m}m = 0$, then $\operatorname{depth}_R(M) = 0$.

(b) The converse is also true, but requires the theorem that the zerodivisors on a finite module are the union of its associated primes. Assuming this theorem, prove that if $\operatorname{depth}_R(M) = 0$, then there exists $0 \neq m \in M$ such that $\mathfrak{m}m = 0$.

(c) Interpret the element $\overline{x} \in \mathbb{K}[x,y]_{\langle x,y\rangle}/\langle x^2, xy\rangle$ using part (a).

(d) Give an example of a local ring of depth zero and positive dimension.

We now introduce a complex that detects regular sequences. Start with one element $x \in R$. The *Koszul complex on x with coefficients in M* is

$$\operatorname{Kos}(x;M): \qquad 0 \longrightarrow M \xrightarrow{x} M \longrightarrow 0,$$

where the left copy of M is in homological degree 1 and the right copy is in degree 0. Its homology is easy to read:

$$H_1(\operatorname{Kos}(x;M)) = (0 :_M x), \qquad H_0(\operatorname{Kos}(x;M)) = M/xM.$$

Thus the one-element Koszul complex is exact in positive degree exactly when x is a nonzerodivisor on M.

For a sequence $\mathbf{x} = x_1, \ldots, x_n$, let $F = R^n$ with basis $e_1, \ldots, e_n$. The *Koszul complex* $\operatorname{Kos}(\mathbf{x};M)$ is the complex with

$$\operatorname{Kos}_i(\mathbf{x};M) = \bigwedge^i F \otimes_R M$$

and differential determined by

$$d(e_{j_1} \wedge \cdots \wedge e_{j_i} \otimes m) = \sum_{\ell=1}^{i} (-1)^{\ell-1} e_{j_1} \wedge \cdots \wedge \widehat{e_{j_\ell}} \wedge \cdots \wedge e_{j_i} \otimes x_{j_\ell} m.$$

One checks that $d^2 = 0$, so this is indeed a chain complex. We write $H_i(\mathbf{x};M)$ for its ith homology module.

---

(11) **The One-Element Koszul Complex.** Let $x \in R$.

(a) Compute $H_1(x;M)$ and $H_0(x;M)$ directly from the definition.

(b) Show that $H_1(x;M) = 0$ if and only if x is a nonzerodivisor on M.

(c) Let $R = \mathbb{Z}$ and $M = \mathbb{Z}$. Compute $H_i(n;\mathbb{Z})$ for $n \in \mathbb{Z}$.

(d) Let $R = \mathbb{K}[\epsilon]/\langle \epsilon^2 \rangle$. Compute $H_i(\epsilon;R)$.

(e) Let $R = \mathbb{K}[x]$ and $M = R/\langle x^m \rangle$. Compute $H_i(x;M)$.

(12) **The Two-Element Koszul Complex.** Let $x, y \in R$. The Koszul complex $\operatorname{Kos}(x,y;M)$ has the form

$$0 \longrightarrow M \xrightarrow{d_2} M^2 \xrightarrow{d_1} M \longrightarrow 0,$$

where

$$d_2(m) = (-ym, xm), \qquad d_1(a,b) = xa + yb.$$

(a) Verify that $d_1 d_2 = 0$.

(b) Prove that $H_0(x,y;M) \cong M/\langle x,y\rangle M$.

(c) Assume x is a nonzerodivisor on M and y is a nonzerodivisor on M/xM. Prove that $H_2(x,y;M) = 0$.

(d) Under the same assumptions, prove that $H_1(x,y;M) = 0$. Hint: If $xa + yb = 0$, reduce modulo xM to show that $b \in xM$.

(e) Conclude that if x,y is an M-regular sequence, then $\mathrm{Kos}(x,y;M)$ has no positive-degree homology.

(f) Let $A = \mathbb{K}[x,y]/\langle xy\rangle$. Compute $H_2(x,y;A)$ and find a nonzero class in $H_1(x,y;A)$.

(13) **The General Koszul Complex.** Let $\mathbf{x} = x_1,\dots,x_n$.

(a) Write out the Koszul complex explicitly when $n = 3$.

(b) Verify directly when $n = 3$ that $d^2 = 0$.

(c) Show that $H_0(\mathbf{x};M) \cong M/\langle x_1,\dots,x_n\rangle M$.

(d) Prove that permuting $x_1,\dots,x_n$ changes the Koszul complex only up to a natural isomorphism of complexes. Hint: Permuting the basis of F changes wedge products by signs.

(e) Show that $\mathrm{Kos}(x_1,\dots,x_n;M)$ is naturally isomorphic to the tensor product of the one-element complexes

$$\mathrm{Kos}(x_1;R) \otimes_R \cdots \otimes_R \mathrm{Kos}(x_n;R) \otimes_R M.$$

(14) **Koszul Acyclicity for Regular Sequences.** The goal of this exercise is to prove the following theorem.

Theorem 4 (Koszul Acyclicity)**.** *If $x_1,\dots,x_n$ is an M-regular sequence, then*

$$H_i(x_1,\dots,x_n;M) = 0 \qquad \textit{for all } i > 0,$$

and

$$H_0(x_1,\dots,x_n;M) \cong M/\langle x_1,\dots,x_n\rangle M.$$

Let $\mathbf{x}' = x_1,\dots,x_{n-1}$ and write $K' = \mathrm{Kos}(\mathbf{x}';M)$.

(a) Explain why $\mathrm{Kos}(\mathbf{x};M)$ is obtained from K' by tensoring with the two-term complex $\mathrm{Kos}(x_n;R)$.

(b) Equivalently, show that $\mathrm{Kos}(\mathbf{x};M)$ is the mapping cone of the chain map

$$K' \xrightarrow{x_n} K'$$

given by multiplication by x_n, up to a harmless sign convention.

(c) Use the long exact sequence in homology of a mapping cone to obtain exact sequences

$$H_i(K') \xrightarrow{x_n} H_i(K') \longrightarrow H_i(\mathbf{x}; M) \longrightarrow H_{i-1}(K') \xrightarrow{x_n} H_{i-1}(K').$$

(d) Prove the acyclicity theorem by induction on n.

(e) Apply the theorem to compute the Koszul homology of $x_1, \ldots, x_n$ on $\mathbb{K}[x_1, \ldots, x_n]$.

The Koszul complex also computes depth. Let $(R, \mathfrak{m}, \mathbb{K})$ be a Noetherian local ring, let M be a finitely generated R-module, and suppose $\mathfrak{m} = \langle x_1, \ldots, x_n \rangle$. Then the Koszul homology modules $H_i(x_1, \ldots, x_n; M)$ measure how far the chosen generators are from being an M-regular sequence.

Theorem 5 (Depth from Koszul Homology)**.** *Let $(R, \mathfrak{m})$ be a Noetherian local ring, let M be a finitely generated nonzero R-module, and let $x_1, \ldots, x_n$ generate $\mathfrak{m}$. Then*

$$\operatorname{depth}_R(M) = n - \max\{i \mid H_i(x_1, \ldots, x_n; M) \neq 0\}.$$

In particular, the number on the right is independent of the chosen generating set for $\mathfrak{m}$.

This theorem should be read as follows: the later the Koszul homology survives, the smaller the depth is. If all positive Koszul homology vanishes, then the generators themselves form a regular sequence and the depth is as large as possible.

---

(15) **Computing Depth with Koszul Homology.** Let $(R, \mathfrak{m}, \mathbb{K})$ be Noetherian local and suppose $\mathfrak{m} = \langle x_1, \ldots, x_n \rangle$.

(a) Compute $\operatorname{Kos}(x_1, \ldots, x_n; \mathbb{K})$. What are the differentials?

(b) Show that

$$H_i(x_1, \ldots, x_n; \mathbb{K}) \cong \bigwedge^i \mathbb{K}^n.$$

Use the depth theorem to recover $\operatorname{depth}_R(\mathbb{K}) = 0$.

(c) Let $R = \mathbb{K}[x_1, \ldots, x_n]_{\langle x_1, \ldots, x_n \rangle}$. Use Koszul acyclicity to compute $\operatorname{depth}(R)$.

(d) Let $A = \mathbb{K}[x, y]_{\langle x, y \rangle} / \langle xy \rangle$. Show that $H_2(x, y; A) = 0$ and that the class of $(0, \overline{x}) \in A^2$ is a nonzero class in $H_1(x, y; A)$. Use the depth theorem to compute $\operatorname{depth}(A)$.

(e) Let $B = \mathbb{K}[x, y]_{\langle x, y \rangle} / \langle x^2, xy \rangle$. Show that $\overline{x} \in B$ determines a nonzero class in $H_2(x, y; B)$. Use the depth theorem to compute $\operatorname{depth}(B)$.

(16) **Top Koszul Homology and Socles.** Let $(R, \mathfrak{m})$ be local and suppose $\mathfrak{m} = \langle x_1, \ldots, x_n \rangle$.

(a) Show that the top Koszul differential has kernel

$$H_n(x_1, \ldots, x_n; M) \cong (0 :_M \mathfrak{m}).$$

Hint: The top differential sends m to the tuple whose entries are, up to signs, $x_1 m, \ldots, x_n m$.

(b) The module $(0 :_M \mathfrak{m})$ is called the *socle* of M. Compute the socle of $\mathbb{K}[x,y]_{\langle x,y\rangle}/\langle x^2, xy\rangle$.

(c) Explain why nonzero top Koszul homology forces depth zero.

(d) If M has finite length, prove that its socle is nonzero. Deduce that every nonzero finite-length module over a local ring has depth zero.

(17) **A Final Synthesis.** Let $S = \mathbb{K}[x,y,z]_{\langle x,y,z\rangle}$.

(a) Show that x, yz is an S-regular sequence.

(b) Let $A = S/\langle x, yz\rangle$. Find the minimal primes of A and compute $\dim(A)$.

(c) Compute $\operatorname{depth}(A)$ using the theorem that depth drops along regular sequences.

(d) Now compute the same depth using the Koszul complex on a generating set of the maximal ideal of A.

(e) Compare this example with $C = S/\langle xy, xz\rangle$: find a minimal prime of height one over $\langle xy, xz\rangle$ and explain why xy, xz cannot be a regular sequence.